Group D: Late progression

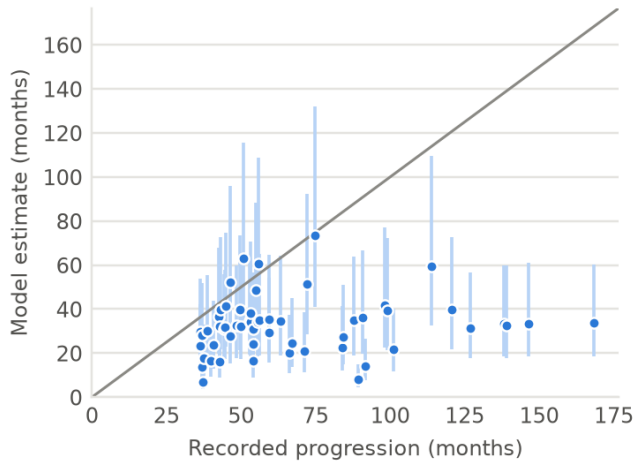
PrimeKG Patient Reasoning · Precision Oncology Research

Group D · 56 patients

Report date 31 Oct 2026

Group D · Late progression recorded progression-free survival: > 36 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
56	**51**	**54.5 mo**	**29%**
5.2% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 36.2–168.2 mo	estimate in this window · median |error| 23 mo



Most frequent path end-points in this group	Patients
hereditary breast ovarian cancer syndrome	4
breast neoplasm	4
Synthesis of PA	4
RAC1 GTPase cycle	3
PKMTs methylate histone lysines	3
breast cancer	3
Formation of the cornified envelope	2
Pregnenolone biosynthesis	2

Logic formulas firing most often in this group	Mean weight	Patients
NStage(p, N0) ⇒ Progress(p)	-0.202	16
Stage(p, II) ⇒ Progress(p)	-0.066	14
Subtype(p, LumA) ⇒ Progress(p)	-0.011	7
Age(p) < 40 ⇒ Progress(p)	+0.055	6
TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	-0.060	6
Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p)	-0.050	4

Most common driver alterations

PIK3CA mutated (29%), TP53 mutated (27%), CCND1 amplified (20%), MYC amplified (18%), GATA3 mutated (12%), FGFR1 amplified (12%), CDH1 mutated (12%), ERBB2 amplified (11%), MAP2K4 mutated (7%), CBFB mutated (7%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-B6-A0RI	44	III	LumA	36.2 mo	30 mo	✗	74	D_progression_after_36m/TCGA-B6-A0RI.pdf
TCGA-B6-A0X7	62	X	LumA	36.3 mo	23 mo	✗	91	D_progression_after_36m/TCGA-B6-A0X7.pdf
TCGA-B6-A0I1	73	II	Basal	36.8 mo	14 mo	✗	8	D_progression_after_36m/TCGA-B6-A0I1.pdf
TCGA-BH-A0C1	61	III	LumA	36.9 mo	28 mo	✗	77	D_progression_after_36m/TCGA-BH-A0C1.pdf
TCGA-A8-A08J	52	IV	Her2	37.1 mo	7 mo	✗	100	D_progression_after_36m/TCGA-A8-A08J.pdf
TCGA-AR-A1AH	51	II	Basal	37.3 mo	18 mo	✗	71	D_progression_after_36m/TCGA-AR-A1AH.pdf
TCGA-BH-A203	78	II	Her2	38.6 mo	30 mo	✗	66	D_progression_after_36m/TCGA-BH-A203.pdf
TCGA-E2-A15O	89	I	LumA	39.8 mo	16 mo	✗	5	D_progression_after_36m/TCGA-E2-A15O.pdf
TCGA-A2-A0SX	48	I	Basal	40.7 mo	24 mo	✗	17	D_progression_after_36m/TCGA-A2-A0SX.pdf
TCGA-A2-A25B	39	II	LumB	42.4 mo	37 mo	✓	24	D_progression_after_36m/TCGA-A2-A25B.pdf
TCGA-E2-A1LL	73	III	Basal	42.8 mo	16 mo	✗	93	D_progression_after_36m/TCGA-E2-A1LL.pdf
TCGA-AR-A2LK	62	III	LumB	42.9 mo	32 mo	✗	82	D_progression_after_36m/TCGA-AR-A2LK.pdf
TCGA-AR-A0TZ	43	III	LumA	43.2 mo	40 mo	✓	98	D_progression_after_36m/TCGA-AR-A0TZ.pdf
TCGA-B6-A0X5	61	II	LumB	44.2 mo	31 mo	✗	41	D_progression_after_36m/TCGA-B6-A0X5.pdf
TCGA-A2-A0EW	53	III	LumA	44.5 mo	32 mo	✗	76	D_progression_after_36m/TCGA-A2-A0EW.pdf
TCGA-A2-A0CK	60	III	LumA	44.8 mo	42 mo	✓	72	D_progression_after_36m/TCGA-A2-A0CK.pdf
TCGA-B6-A0IN	45	X	LumB	46.4 mo	28 mo	✗	91	D_progression_after_36m/TCGA-B6-A0IN.pdf
TCGA-AR-A24L	26	II	LumA	46.5 mo	52 mo	✓	49	D_progression_after_36m/TCGA-AR-A24L.pdf
TCGA-A2-A04V	39	II	LumA	48.3 mo	33 mo	✗	10	D_progression_after_36m/TCGA-A2-A04V.pdf
TCGA-BH-A1EX	67	II	LumA	49.6 mo	40 mo	✓	45	D_progression_after_36m/TCGA-BH-A1EX.pdf
TCGA-B6-A402	47	I	missing	49.8 mo	32 mo	✗	14	D_progression_after_36m/TCGA-B6-A402.pdf
TCGA-HN-A2OB	45	II	LumA	50.7 mo	63 mo	✓	21	D_progression_after_36m/TCGA-HN-A2OB.pdf
TCGA-AR-A0U2	47	III	LumB	52.9 mo	34 mo	✗	79	D_progression_after_36m/TCGA-AR-A0U2.pdf
TCGA-AO-A03U	31	I	Normal	53.1 mo	34 mo	✗	4	D_progression_after_36m/TCGA-AO-A03U.pdf
TCGA-E2-A1L7	40	III	LumB	53.2 mo	38 mo	✓	77	D_progression_after_36m/TCGA-E2-A1L7.pdf
TCGA-AR-A256	45	II	Basal	53.9 mo	16 mo	✗	61	D_progression_after_36m/TCGA-AR-A256.pdf
TCGA-BH-A1FR	73	III	LumA	54.0 mo	24 mo	✗	74	D_progression_after_36m/TCGA-BH-A1FR.pdf
TCGA-AO-A03N	59	II	LumB	54.1 mo	31 mo	✗	33	D_progression_after_36m/TCGA-AO-A03N.pdf
TCGA-BH-A1FL	69	II	LumA	55.0 mo	49 mo	✓	46	D_progression_after_36m/TCGA-BH-A1FL.pdf
TCGA-BH-A1FU	44	I	Normal	55.5 mo	34 mo	✗	5	D_progression_after_36m/TCGA-BH-A1FU.pdf
TCGA-BH-A1EW	38	II	LumA	55.7 mo	61 mo	✓	14	D_progression_after_36m/TCGA-BH-A1EW.pdf
TCGA-BH-A0BQ	39	I	LumA	56.2 mo	35 mo	✗	1	D_progression_after_36m/TCGA-BH-A0BQ.pdf
TCGA-3C-AAAU	55	X	LumA	59.4 mo	29 mo	✗	76	D_progression_after_36m/TCGA-3C-AAAU.pdf
TCGA-E2-A1IE	61	II	LumA	59.5 mo	35 mo	✗	19	D_progression_after_36m/TCGA-E2-A1IE.pdf
TCGA-AO-A03P	54	II	LumB	63.2 mo	35 mo	✗	32	D_progression_after_36m/TCGA-AO-A03P.pdf
TCGA-B6-A0WY	40	III	LumA	66.1 mo	20 mo	✗	75	D_progression_after_36m/TCGA-B6-A0WY.pdf
TCGA-B6-A0RS	38	II	Her2	67.1 mo	25 mo	✗	62	D_progression_after_36m/TCGA-B6-A0RS.pdf
TCGA-B6-A0RH	51	II	Her2	71.2 mo	21 mo	✗	69	D_progression_after_36m/TCGA-B6-A0RH.pdf
TCGA-BH-A1FN	34	II	LumB	72.1 mo	51 mo	✓	38	D_progression_after_36m/TCGA-BH-A1FN.pdf
TCGA-BH-A1FE	31	II	LumA	74.7 mo	74 mo	✓	13	D_progression_after_36m/TCGA-BH-A1FE.pdf
TCGA-A8-A08T	64	IV	LumA	84.1 mo	22 mo	✗	99	D_progression_after_36m/TCGA-A8-A08T.pdf
TCGA-AR-A24O	43	III	LumA	84.2 mo	27 mo	✗	81	D_progression_after_36m/TCGA-AR-A24O.pdf
TCGA-BH-A0C3	47	I	LumB	87.9 mo	35 mo	✗	31	D_progression_after_36m/TCGA-BH-A0C3.pdf
TCGA-B6-A0IB	64	IV	LumB	89.3 mo	8 mo	✗	99	D_progression_after_36m/TCGA-B6-A0IB.pdf
TCGA-BH-A18K	46	I	LumA	90.8 mo	36 mo	✓	5	D_progression_after_36m/TCGA-BH-A18K.pdf
TCGA-B6-A0RP	73	III	LumA	91.6 mo	14 mo	✗	89	D_progression_after_36m/TCGA-B6-A0RP.pdf
TCGA-A2-A25E	34	III	LumB	98.3 mo	42 mo	✓	75	D_progression_after_36m/TCGA-A2-A25E.pdf
TCGA-GM-A2DA	46	II	LumA	99.1 mo	39 mo	✓	16	D_progression_after_36m/TCGA-GM-A2DA.pdf
TCGA-B6-A0RU	40	I	Basal	101.1 mo	22 mo	✗	52	D_progression_after_36m/TCGA-B6-A0RU.pdf

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-BH-A1ES	35	II	missing	113.8 mo	60 mo	✓	73	D_progression_after_36m/TCGA-BH-A1ES.pdf
TCGA-BH-A1FB	60	II	missing	120.6 mo	40 mo	✓	67	D_progression_after_36m/TCGA-BH-A1FB.pdf
TCGA-B6-A0WT	61	II	LumA	127.0 mo	32 mo	✗	21	D_progression_after_36m/TCGA-B6-A0WT.pdf
TCGA-AR-A2LE	69	I	LumA	138.2 mo	33 mo	✗	3	D_progression_after_36m/TCGA-AR-A2LE.pdf
TCGA-B6-A2IU	62	II	LumA	138.9 mo	32 mo	✗	8	D_progression_after_36m/TCGA-B6-A2IU.pdf
TCGA-B6-A0IG	50	II	LumA	146.5 mo	34 mo	✗	34	D_progression_after_36m/TCGA-B6-A0IG.pdf
TCGA-B6-A0WZ	50	II	LumA	168.2 mo	34 mo	✗	17	D_progression_after_36m/TCGA-B6-A0WZ.pdf

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**44**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

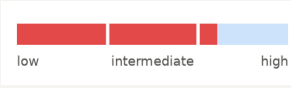
Sample: **TCGA-B6-A0RI-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **0.02** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

74th

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 8% · 60 mo 24%

Model-estimated progression time

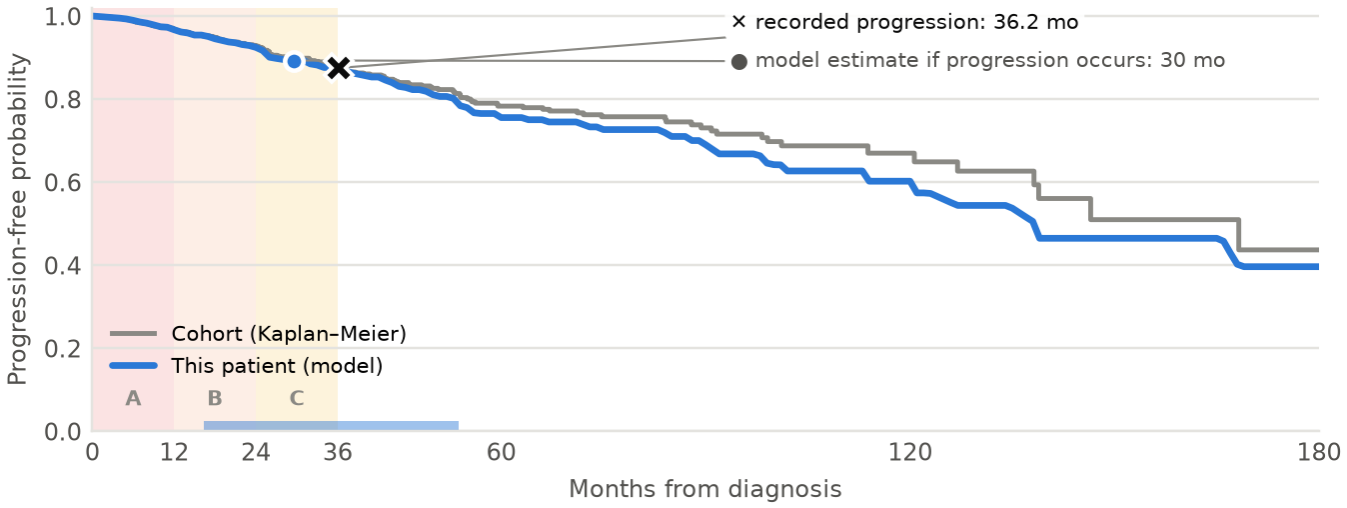
≈ 30 months

Window 24–36 months · IQR 16–54 mo
if progression occurs

Recorded progression

36.2 months

Group D · Late progression
✗ recorded window differs · |error| 7 mo



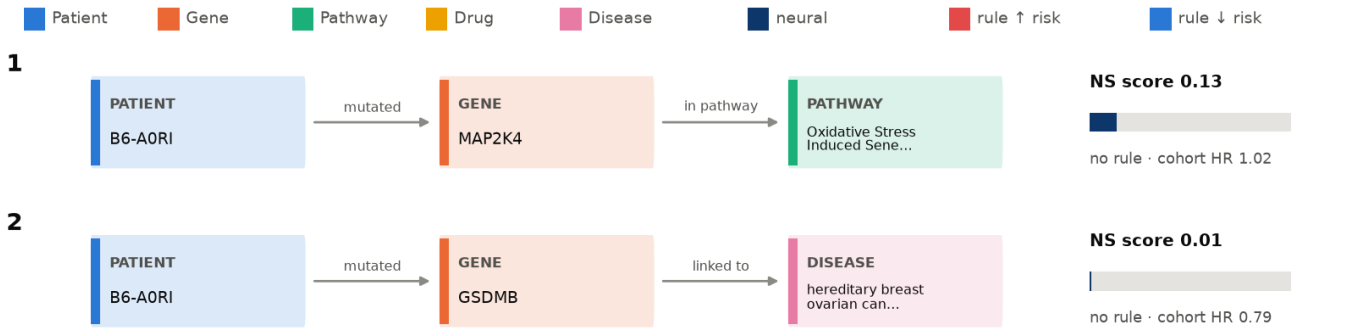
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 44-year-old patient has a LumA tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **74th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.096 per SD). The highest-scoring knowledge-graph path runs from **MAP2K4** to **Oxidative Stress Induced Senescence**. Progression was recorded at 36.2 months; the model's estimate given progression was 30 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

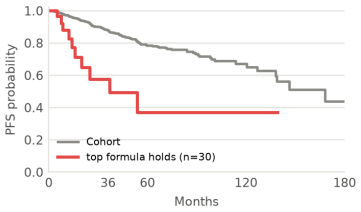


None of the 16 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0RI** mutated → **MAP2K4** in pathway → **Oxidative Stress Induced Senescence**	22	No learned formula fires on this path; its score is neural only. cohort: HR 1.02, n = 326 carriers	**0.13**
2	**TCGA-B6-A0RI** mutated → **GSDMB** linked to → **hereditary breast ovarian cancer syndrome**	1	No learned formula fires on this path; its score is neural only. cohort: HR 0.79, n = 799 carriers	**0.01**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

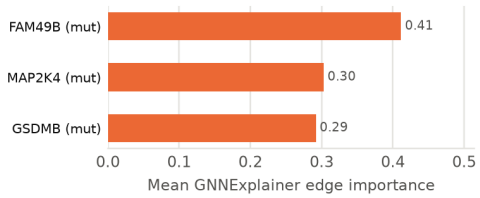
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.096 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	28 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP2K4 · driver	Mutation	p.C266Hfs*8	12	0.30
FAM49B	Mutation	p.L4R	0	0.41
GSDMB	Mutation	p.Q85*, p.S107L	4	0.29



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A1-A0SJ	39	III	LumA	0.82	no progression, followed 13.7 mo
TCGA-AR-A1AU	39	III	LumA	0.71	no progression, followed 94.3 mo
TCGA-BH-A1FR	73	III	LumA	0.70	progressed at 54.0 mo (group D)
TCGA-BH-A0AZ	47	III	LumA	0.70	no progression, followed 63.1 mo
TCGA-B6-A0RO	71	III	LumA	0.68	no progression, followed 162.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical

block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0X7 · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0X7-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.53** · MSIsensor score: **0.16** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 5% · 24 mo 11% · 60 mo 33%

Model-estimated progression time

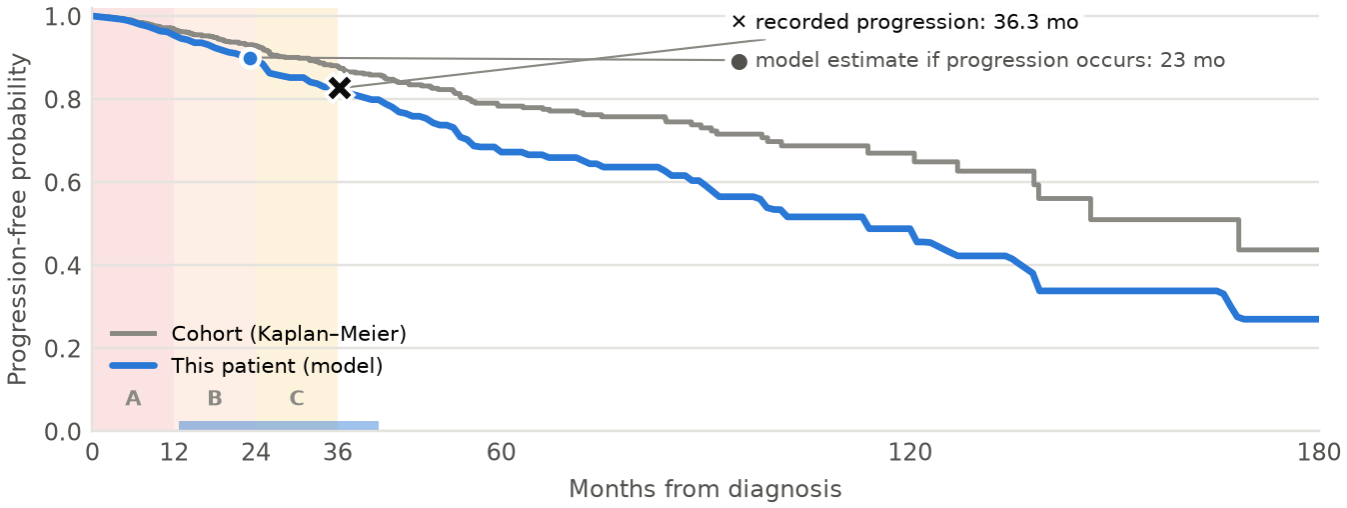
≈ 23 months

Window 12–24 months · IQR 13–42 mo
if progression occurs

Recorded progression

36.3 months

Group D · Late progression
✗ recorded window differs · |error| 13 mo



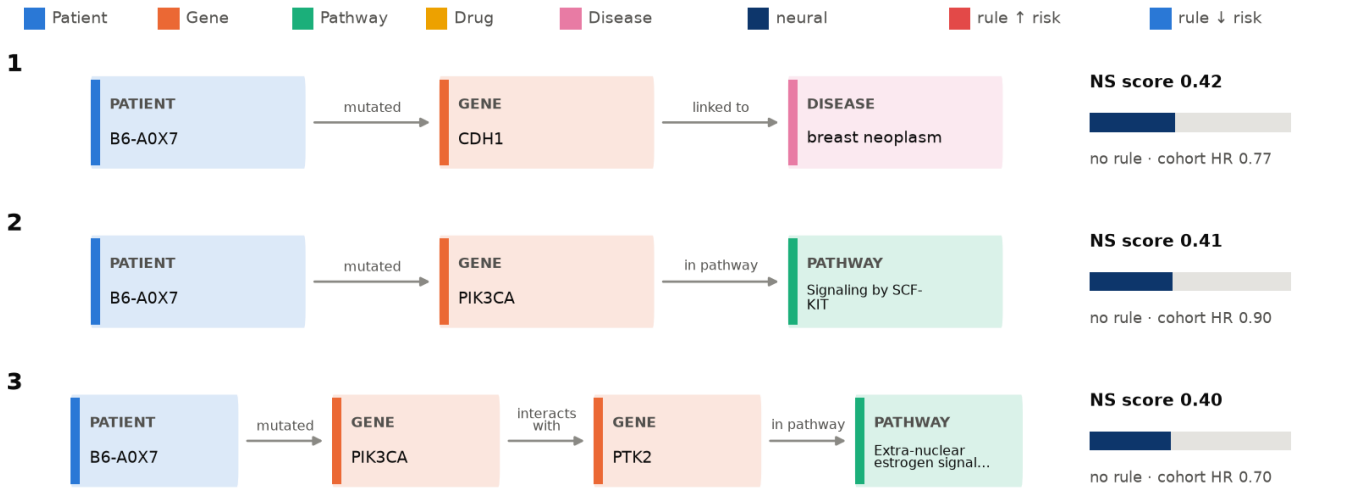
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage X (T1, missing, MX). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T1)* ∧ *Subtype(p, LumA)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.036 per SD). The highest-scoring knowledge-graph path runs from **CDH1** to **breast neoplasm**. Progression was recorded at 36.3 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

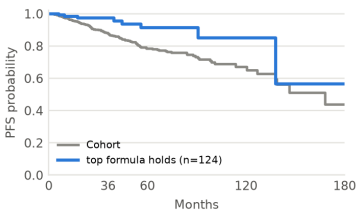


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0X7** mutated → **CDH1** linked to → **breast neoplasm**	71	No learned formula fires on this path; its score is neural only. cohort: HR 0.77, n = 794 carriers	**0.42**
2	**TCGA-B6-A0X7** mutated → **PIK3CA** in pathway → **Signaling by SCF-KIT**	69	No learned formula fires on this path; its score is neural only. cohort: HR 0.90, n = 395 carriers	**0.41**
3	**TCGA-B6-A0X7** mutated → **PIK3CA** interacts with → **PTK2** in pathway → **Extra-nuclear estrogen signaling**	67	No learned formula fires on this path; its score is neural only. cohort: HR 0.70, n = 453 carriers	**0.40**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

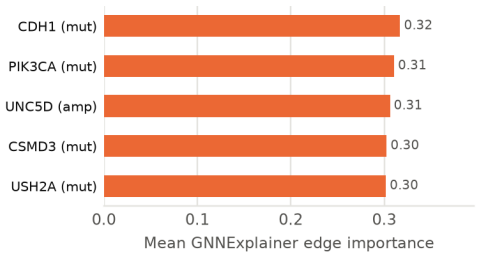
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.036 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	124 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.K675Nfs*2	17	0.32
PIK3CA · driver	Mutation	p.H1047R	70	0.31
UNC5D	High-level amplification	GISTIC ±2	1	0.31
CSMD3	Mutation	p.R615=	0	0.30
USH2A	Mutation	p.T4888M	0	0.30



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they

are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0IC	90	X	LumA	0.72	no progression, followed 50.7 mo
TCGA-XX-A899	46	III	LumA	0.65	no progression, followed 15.4 mo
TCGA-E2-A576	69	I	LumA	0.60	no progression, followed 34.3 mo
TCGA-B6-A0WS	58	X	LumA	0.60	no progression, followed 97.5 mo
TCGA-EW-A1J2	50	II	LumA	0.59	no progression, followed 13.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

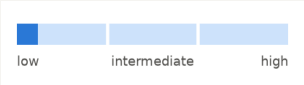
Sample: **TCGA-B6-A0I1-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.7** · MSIsensor score: **0.82** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

8th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 14%

Model-estimated progression time

≈ 14 months

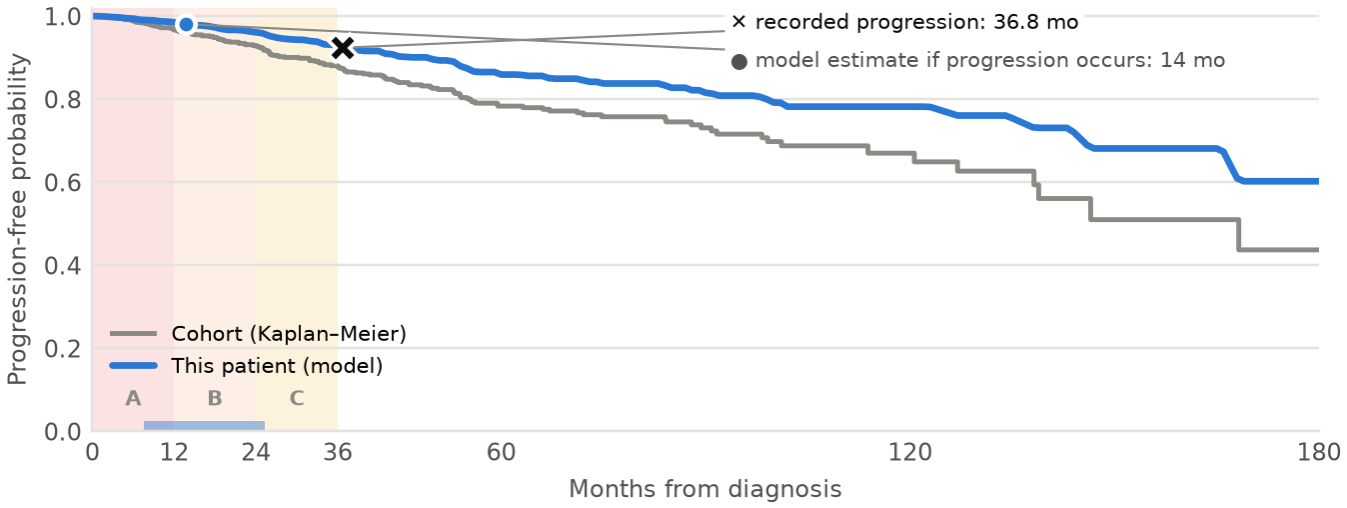
Window 12–24 months · IQR 8–25 mo if progression occurs

Recorded progression

36.8 months

Group D · Late progression

✗ **recorded window differs** · |error| 23 mo



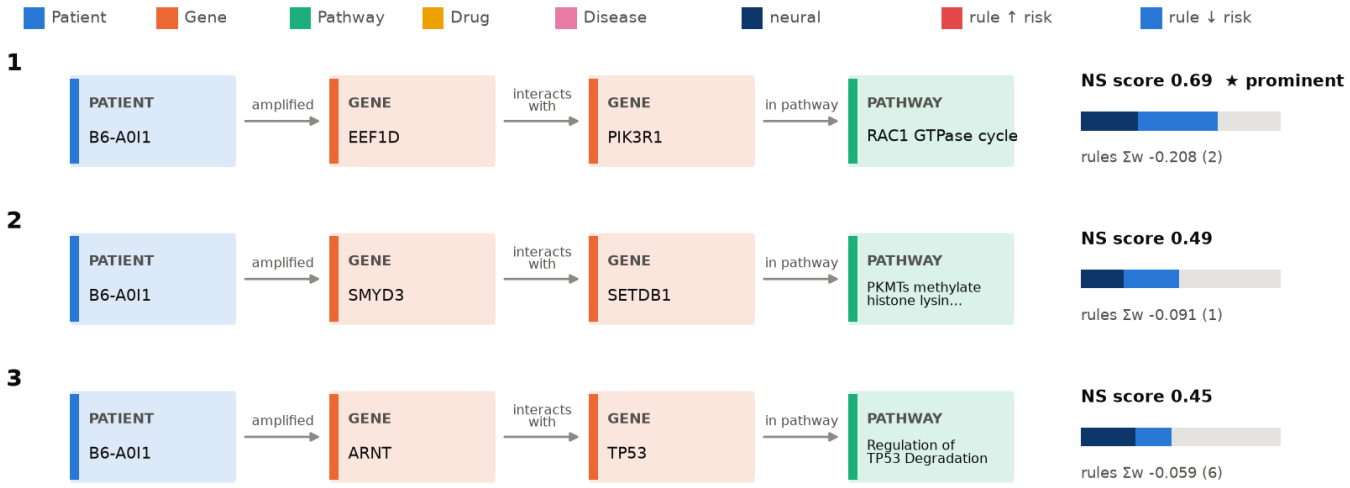
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **8th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **EEF1D** to **RAC1 GTPase cycle**; the learned formulas it instantiates point to slower progression (Σw -0.208). Progression was recorded at 36.8 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

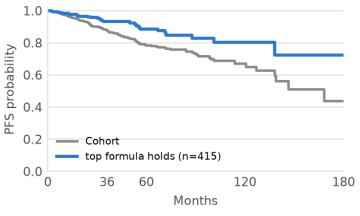


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A0I1** amplified → **EEF1D** interacts with → **PIK3R1** in pathway → **RAC1 GTPase cycle**	48	↓ **slower progression** · Σw -0.208 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 0.98, n = 419 carriers	**0.69**
2	**TCGA-B6-A0I1** amplified → **SMYD3** interacts with → **SETDB1** in pathway → **PKMTs methylate histone lysines**	36	↓ **slower progression** · Σw -0.091 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Keratan sulfate biosynthesis') ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'Major pathway of rRNA processing in the nucleolus and cytosol') ⇒ Progress(p) cohort: HR 0.98, n = 319 carriers	**0.49**
3	**TCGA-B6-A0I1** amplified → **ARNT** interacts with → **TP53** in pathway → **Regulation of TP53 Degradation**	46	↓ **slower progression** · Σw -0.059 over 6 formula(s) Altered(p, g) ∧ InPathway(g, 'Stimuli-sensing channels') ∧ Altered(p, g) ∧ InPathway(g, 'Pyroptosis') ∧ Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) cohort: HR 0.87, n = 382 carriers	**0.45**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

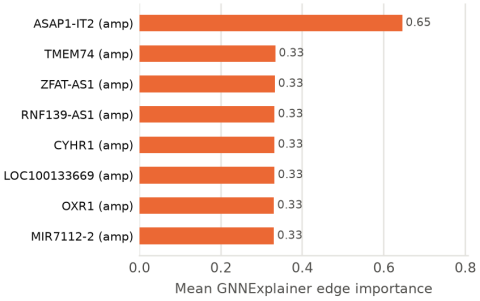
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
-0.141 ↓ risk	Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) [KG]	RL-proposed	93 pts
+0.101 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Mitochondrial translation initiation') ⇒ Progress(p) [KG]	single atom	104 pts
-0.097 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Stimuli-sensing channels') ∧ Altered(p, g) ∧ InPathway(g, 'Pyroptosis') ∧ Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	RL-proposed	70 pts
-0.091 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Keratan sulfate biosynthesis') ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'Major pathway of rRNA processing in the nucleolus and cytosol') ⇒ Progress(p) [KG]	RL-proposed	45 pts
-0.082 ↓ risk	Amplified(p, ZFHX4) ⇒ Progress(p)	single atom	54 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
MYC · driver	Mutation	p.P233T	20	0.32
TP53 · driver	Mutation	p.R248Q	57	0.31
ASAP1-IT2	High-level amplification	GISTIC ±2	0	0.65
TMEM74	High-level amplification	GISTIC ±2	0	0.33
ZFAT-AS1	High-level amplification	GISTIC ±2	0	0.33
RNF139-AS1	High-level amplification	GISTIC ±2	0	0.33
CYHR1	High-level amplification	GISTIC ±2	0	0.33
LOC100133669	High-level amplification	GISTIC ±2	0	0.33
OXR1	High-level amplification	GISTIC ±2	0	0.33
MIR7112-2	High-level amplification	GISTIC ±2	0	0.33
TATDN1	High-level amplification	GISTIC ±2	0	0.33



405 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 140 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A04T	62	II	Basal	0.92	no progression, followed 73.8 mo
TCGA-E2-A14R	62	II	Basal	0.91	progressed at 31.9 mo (group C)
TCGA-A7-A0DA	62	II	Basal	0.90	no progression, followed 35.7 mo
TCGA-D8-A1XQ	69	II	Basal	0.90	no progression, followed 16.4 mo
TCGA-AR-A0TP	43	II	Basal	0.89	no progression, followed 140.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A0C1 · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-BH-A0C1-01** · Aneuploidy score: **19** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **0.08** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

77th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

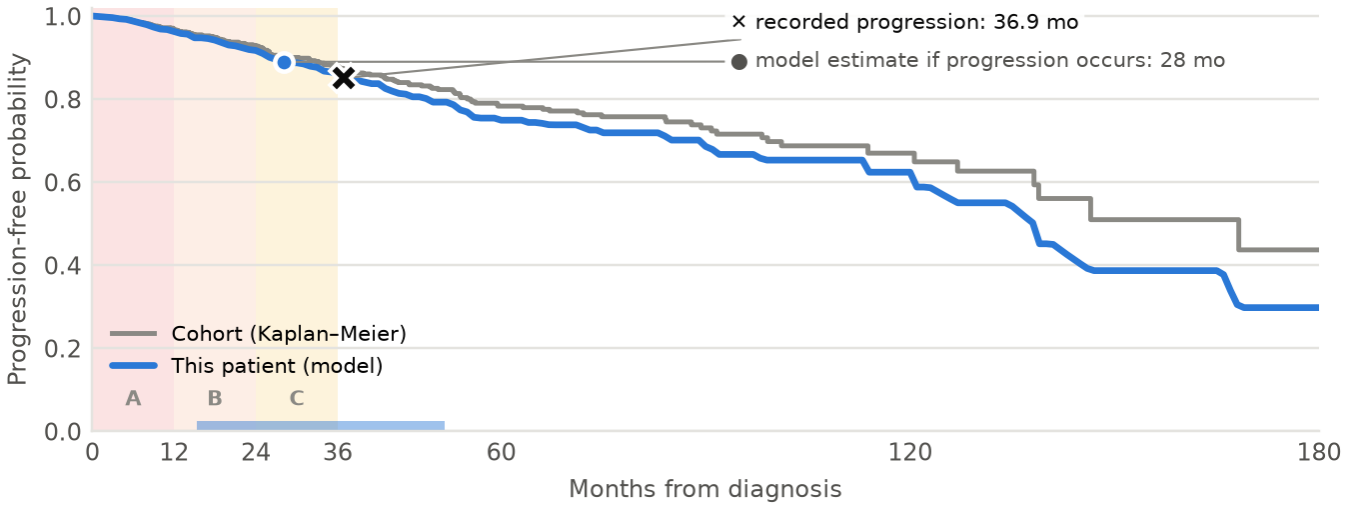
≈ 28 months

Window 24–36 months · IQR 15–52 mo
if progression occurs

Recorded progression

36.9 months

Group D · Late progression
✗ recorded window differs · |error| 9 mo



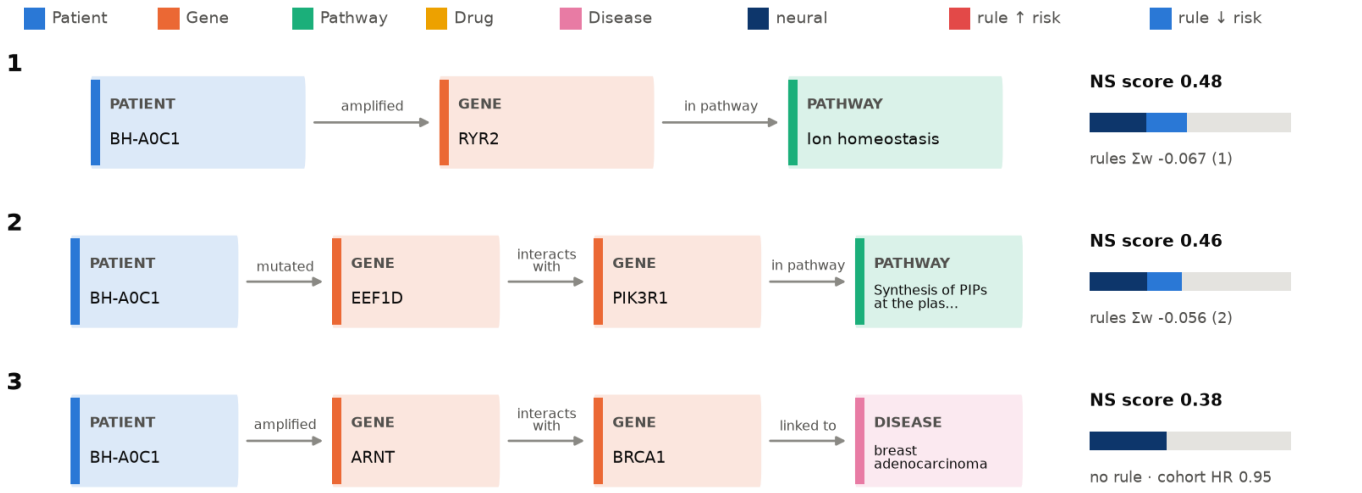
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **77th risk percentile** (high tier). The strongest learned formula that holds for them (*Stage(p, III)* ∧ *Amplified(p, FAM72C)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.081 per SD). The highest-scoring knowledge-graph path runs from **RYR2** to **Ion homeostasis**; the learned formulas it instantiates point to slower progression (Σw -0.067). Progression was recorded at 36.9 months; the model's estimate given progression was 28 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

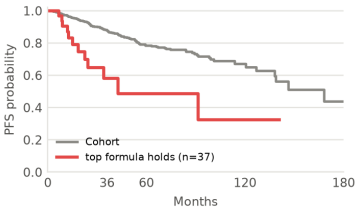


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A0C1** amplified → **RYR2** in pathway → **Ion homeostasis**	47	↓ **slower progression** · Σw -0.067 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) cohort: HR 0.96, n = 184 carriers	**0.48**
2	**TCGA-BH-A0C1** mutated → **EEF1D** interacts with → **PIK3R1** in pathway → **Synthesis of PIPs at the plasma membrane**	48	↓ **slower progression** · Σw -0.056 over 2 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.78, n = 435 carriers	**0.46**
3	**TCGA-BH-A0C1** amplified → **ARNT** interacts with → **BRCA1** linked to → **breast adenocarcinoma**	64	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 734 carriers	**0.38**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

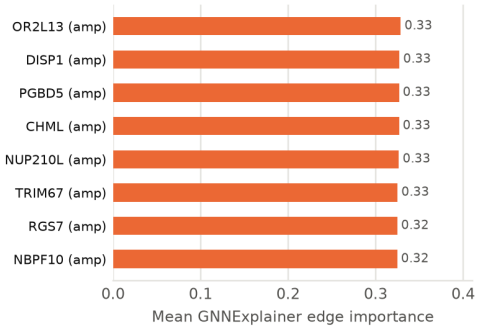
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.081 ↑ risk	Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	mined pair	37 pts
+0.073 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ⇒ Progress(p) [KG]	mined pair	53 pts
-0.067 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	143 pts
+0.053 ↑ risk	Amplified(p, FAM72C) ∧ Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ⇒ Progress(p) [KG]	mined pair	82 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
+0.027 ↑ risk	Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	RL-proposed	95 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.P159Lfs*56	17	0.32
RUNX1 · driver	Mutation	p.G135D	18	0.32
ERBB2 · driver	Mutation	p.V777L	55	0.31
OR2L13	High-level amplification	GISTIC ±2	1	0.33
DISP1	High-level amplification	GISTIC ±2	0	0.33
PGBD5	High-level amplification	GISTIC ±2	0	0.33
CHML	High-level amplification	GISTIC ±2	2	0.33
NUP210L	High-level amplification	GISTIC ±2	0	0.33
TRIM67	High-level amplification	GISTIC ±2	0	0.33
RGS7	High-level amplification	GISTIC ±2	2	0.32
NBPF10	High-level amplification	GISTIC ±2	0	0.32
FMN2	High-level amplification	GISTIC ±2	0	0.32



114 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 62 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A27I	58	III	LumA	0.88	no progression, followed 14.4 mo
TCGA-A8-A0AB	54	II	LumA	0.88	no progression, followed 17.0 mo
TCGA-OL-A66N	59	III	LumA	0.88	no progression, followed 26.0 mo
TCGA-BH-A0B5	40	III	LumA	0.85	no progression, followed 70.2 mo
TCGA-BH-A0BA	51	III	LumA	0.84	no progression, followed 37.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A8-A08J · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**52**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**Her2**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8541/3**

Sample: **TCGA-A8-A08J-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **1.2** · MSIsensor score: **0.49** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

100th

High risk tier · fused neuro-symbolic model



Probability of progression

87% by 36 mo

12 mo 41% · 24 mo 70% · 60 mo 98%

Model-estimated progression time

≈ 7 months

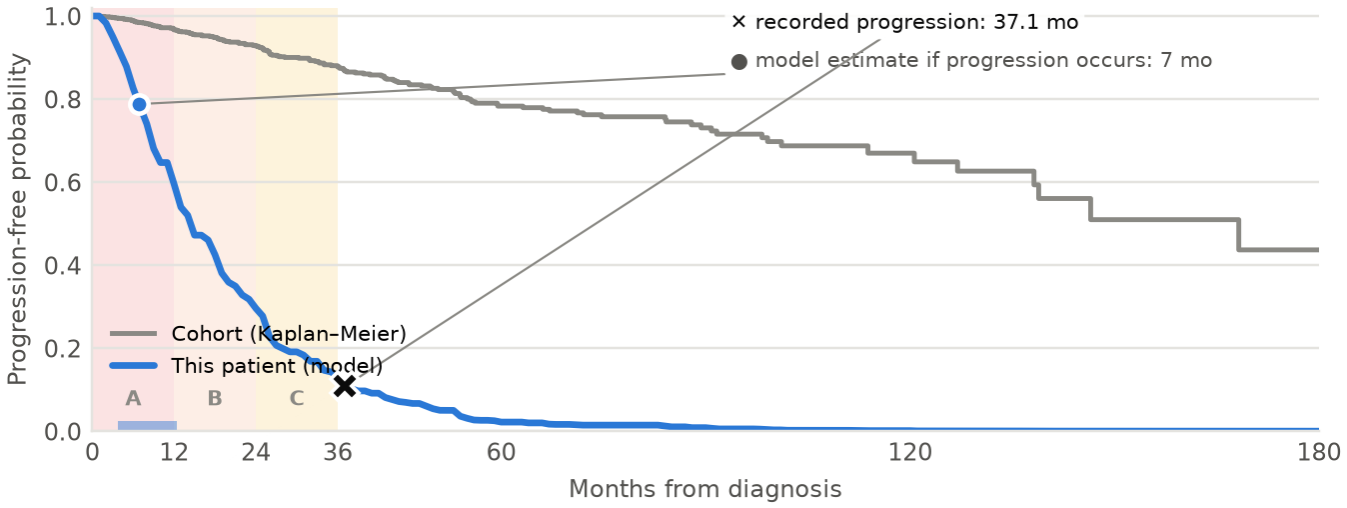
Window ≤ 12 months · IQR 4–12 mo if progression occurs

Recorded progression

37.1 months

Group D · Late progression

✗ recorded window differs · |error| 30 mo



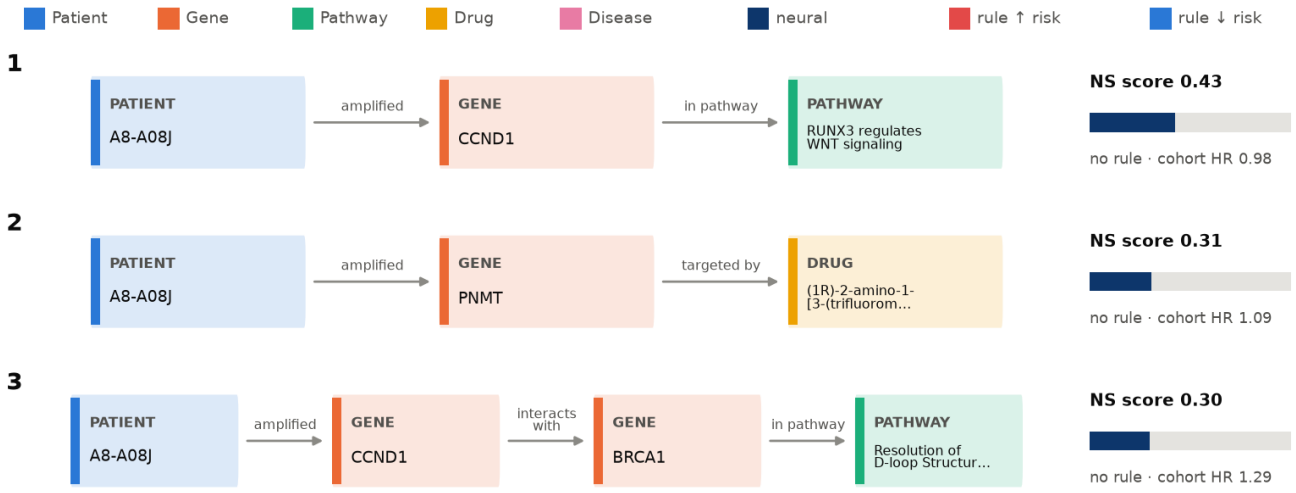
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 52-year-old patient has a Her2 tumour, AJCC stage IV (T4, N3, M1). The fused neuro-symbolic model places them in the **100th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.140 per SD). The highest-scoring knowledge-graph path runs from **CCND1** to **RUNX3 regulates WNT signaling**. Progression was recorded at 37.1 months; the model's estimate given progression was 7 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

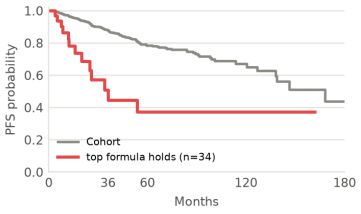


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A8-A08J** amplified → **CCND1** in pathway → **RUNX3 regulates WNT signaling**	71	No learned formula fires on this path; its score is neural only. cohort: HR 0.98, n = 246 carriers	**0.43**
2	**TCGA-A8-A08J** amplified → **PNMT** targeted by → **(1R)-2-amino-1-[3-(trifluoromethyl)phenyl]ethanol**	51	No learned formula fires on this path; its score is neural only. cohort: HR 1.09, n = 104 carriers	**0.31**
3	**TCGA-A8-A08J** amplified → **CCND1** interacts with → **BRCA1** in pathway → **Resolution of D-loop Structures through Synthesis-Dependent Strand Annealing (SDSA)**	50	No learned formula fires on this path; its score is neural only. cohort: HR 1.29, n = 179 carriers	**0.30**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

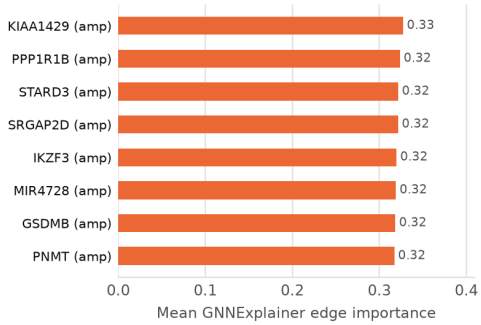
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.140 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	32 pts
+0.098 ↑ risk	NStage(p, N3) ⇒ Progress(p)	single atom	57 pts
+0.072 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ⇒ Progress(p) [KG]	mined pair	32 pts
+0.036 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Interaction between L1 and Ankyrins') ⇒ Progress(p) [KG]	mined pair	27 pts
+0.004 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) [KG]	mined pair	45 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
GATA3 · driver	Mutation	p.S426Ifs*81	13	0.31
KIAA1429	High-level amplification	GISTIC ±2	0	0.33
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
IKZF3	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
PNMT	High-level amplification	GISTIC ±2	16	0.32
ORMDL3	High-level amplification	GISTIC ±2	2	0.32



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 23 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-UU-A93S	63	IV	Her2	0.70	progressed at 3.8 mo (group A)
TCGA-B6-A0I9	62	IV	Her2	0.55	progressed at 11.9 mo (group A)
TCGA-PL-A8LX	35	IV	LumA	0.55	no progression, followed 0.2 mo
TCGA-E9-A22D	38	II	LumB	0.55	no progression, followed 41.0 mo
TCGA-A8-A08X	43	III	Her2	0.53	progressed at 12.0 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**51**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A1AH-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **2.67** · MSIsensor score: **1.35** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

71st

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 23%

Model-estimated progression time

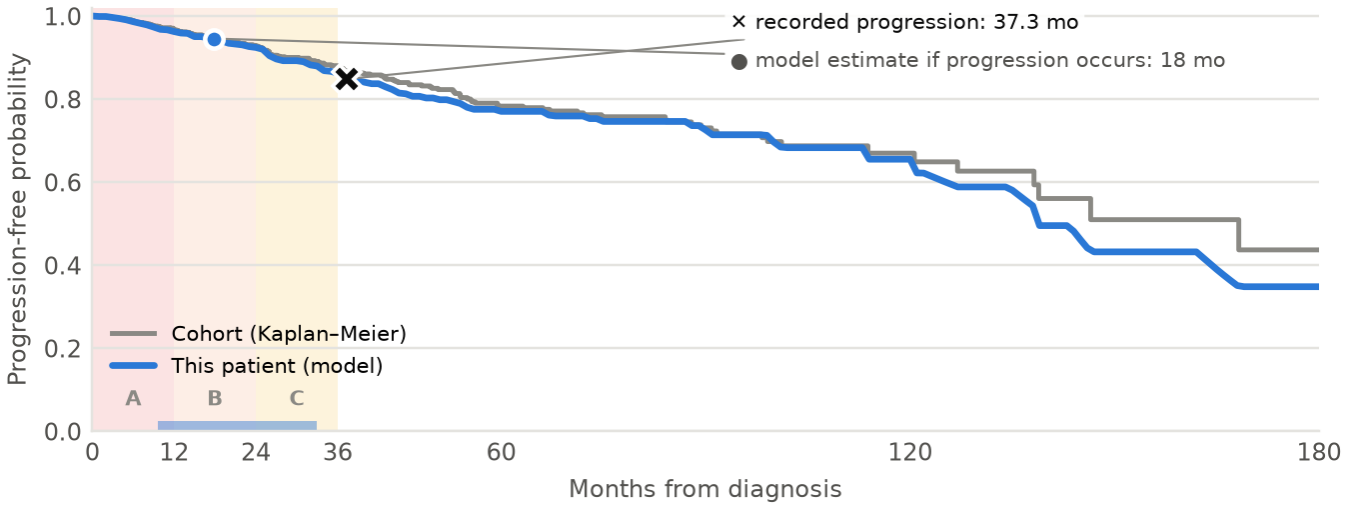
≈ 18 months

Window 12–24 months · IQR 10–33 mo
if progression occurs

Recorded progression

37.3 months

Group D · Late progression
✗ recorded window differs · |error| 20 mo



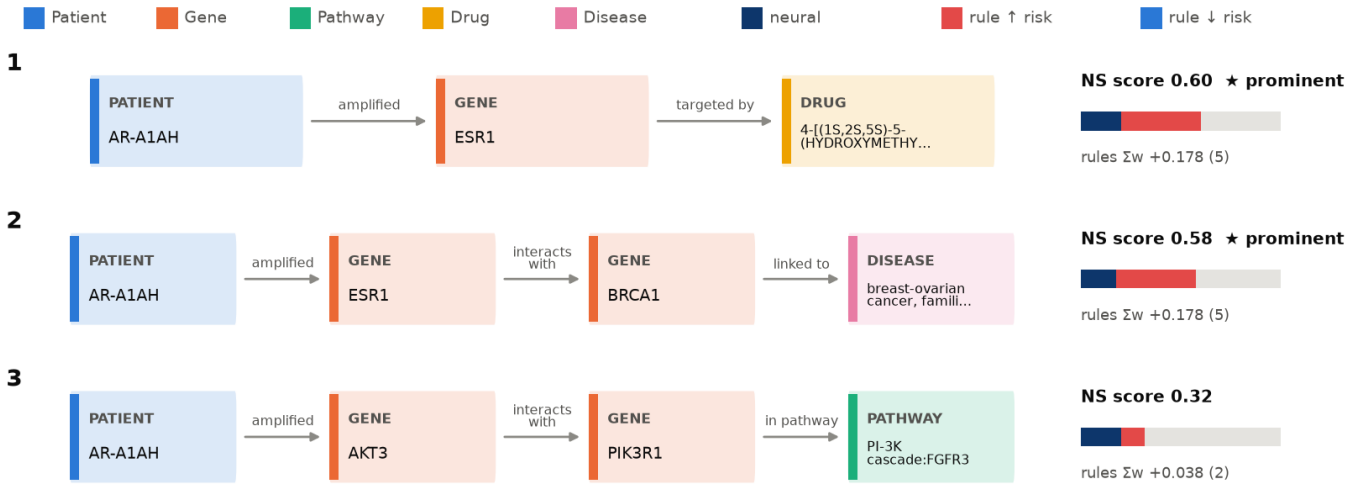
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 51-year-old patient has a Basal tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **71st risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'HATs acetylate histones')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'NrCAM interactions')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Ovarian tumor domain proteases')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.063 per SD). The highest-scoring knowledge-graph path runs from **ESR1** to **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**; the learned formulas it instantiates point to faster progression (Σw +0.178). Progression was recorded at 37.3 months; the model's estimate given progression was 18 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

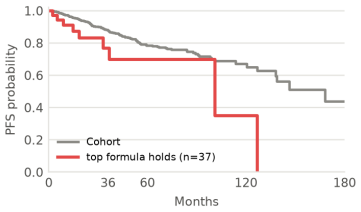


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AR-A1AH** amplified → **ESR1** targeted by → **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**	33	↑ **faster progression** · Σw +0.178 over 5 formula(s) Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'NrCAM interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ⇒ Progress(p) cohort: HR 2.93, n = 28 carriers	**0.60**
2 ★	**TCGA-AR-A1AH** amplified → **ESR1** interacts with → **BRCA1** linked to → **breast-ovarian cancer, familial, susceptibility to**	29	↑ **faster progression** · Σw +0.178 over 5 formula(s) Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'NrCAM interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ⇒ Progress(p) cohort: HR 1.18, n = 77 carriers	**0.58**
3	**TCGA-AR-A1AH** amplified → **AKT3** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR3**	34	↑ **faster progression** · Σw +0.038 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Laminin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'The NLRP3 inflammasome') ∧ Altered(p, g) ∧ LinkedTo(g, 'mammary neoplasms, animal') ⇒ Progress(p) cohort: HR 0.86, n = 387 carriers	**0.32**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

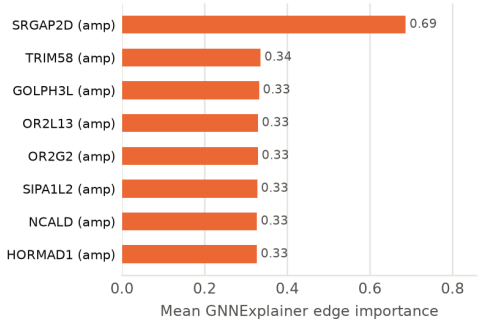
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.063 ↑ risk	Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'NrCAM interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ⇒ Progress(p) [KG]	RL-proposed	37 pts
+0.052 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Nuclear signaling by ERBB4') ⇒ Progress(p) [KG]	single atom	26 pts
+0.046 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Laminin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'The NLRP3 inflammasome') ∧ Altered(p, g) ∧ LinkedTo(g, 'mammary neoplasms, animal') ⇒ Progress(p) [KG]	RL-proposed	54 pts
+0.012 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	mined pair	45 pts
+0.012 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Formation of the beta-catenin:TCF transactivating complex') ∧ Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ⇒ Progress(p) [KG]	RL-proposed	35 pts
+0.010 ↑ risk	Amplified(p, ANK1) ∧ Altered(p, g) ∧ InPathway(g, 'Cytoprotection by HMOX1') ⇒ Progress(p) [KG]	mined pair	34 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ESR1 · driver	High-level amplification	GISTIC ±2	141	0.31
TP53 · driver	Mutation	p.G108Ffs*40	57	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.69
TRIM58	High-level amplification	GISTIC ±2	0	0.34
GOLPH3L	High-level amplification	GISTIC ±2	0	0.33
OR2L13	High-level amplification	GISTIC ±2	1	0.33
OR2G2	High-level amplification	GISTIC ±2	1	0.33
SIPA1L2	High-level amplification	GISTIC ±2	0	0.33
NCALD	High-level amplification	GISTIC ±2	1	0.33
HORMAD1	High-level amplification	GISTIC ±2	0	0.33
NBPF10	High-level amplification	GISTIC ±2	0	0.32
TPCN2	High-level amplification	GISTIC ±2	1	0.32



117 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 164 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A0CE	57	II	Basal	0.91	no progression, followed 35.3 mo
TCGA-A7-A4SE	54	II	Basal	0.88	no progression, followed 21.2 mo
TCGA-A8-A07F	65	II	LumA	0.86	no progression, followed 19.0 mo
TCGA-A8-A06Y	66	II	LumA	0.84	progressed at 26.0 mo (group C)
TCGA-B6-A0WX	40	III	Basal	0.84	progressed at 18.5 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month

is always shown beside it.

■ **Group D** · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**78**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A203-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **2.03** · MSIsensor score: **0.31** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

66th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 21%

Model-estimated progression time

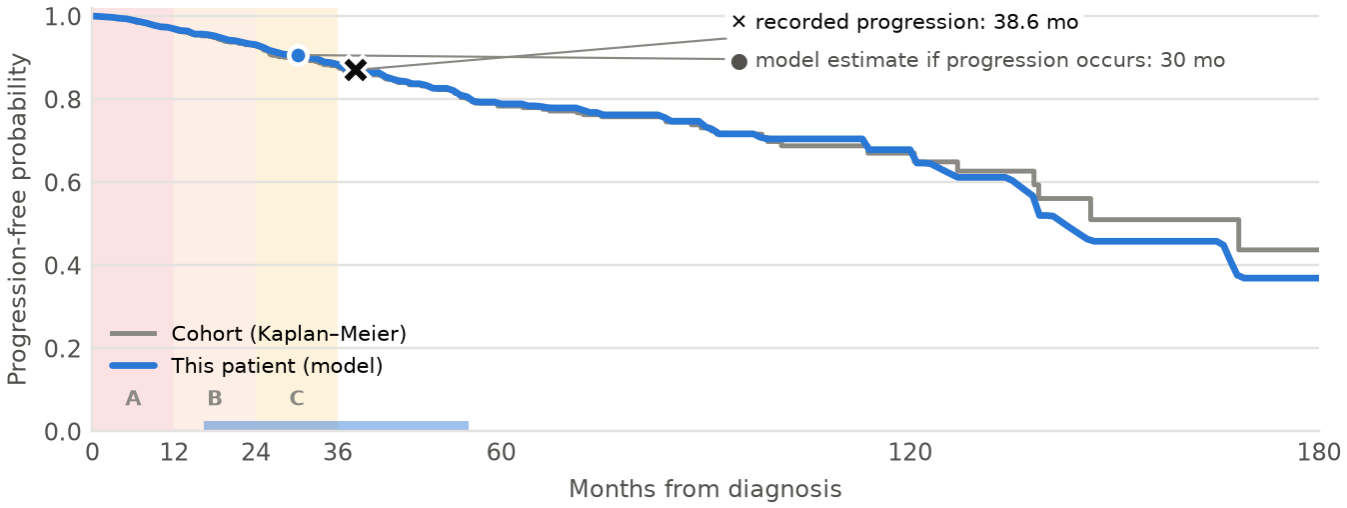
≈ 30 months

Window 24–36 months · IQR 16–55 mo
if progression occurs

Recorded progression

38.6 months

■ **Group D** · Late progression
✗ **recorded window differs** · |error| 9 mo



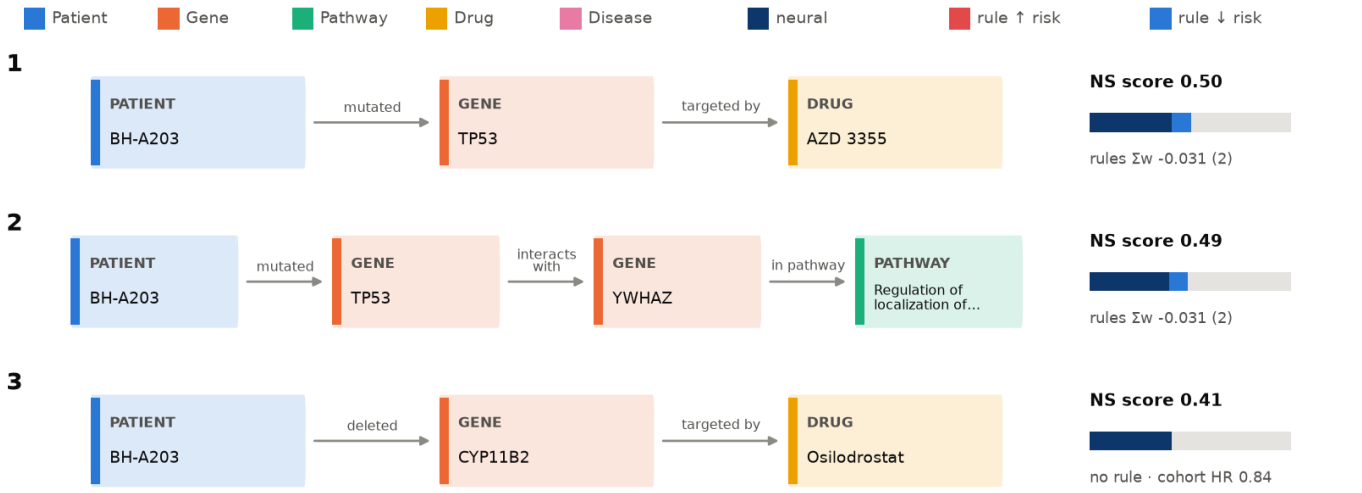
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 78-year-old patient has a Her2 tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **66th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **TP53** to **AZD 3355**; the learned formulas it instantiates point to slower progression (Σw -0.031). Progression was recorded at 38.6 months; the model's estimate given progression was 30 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

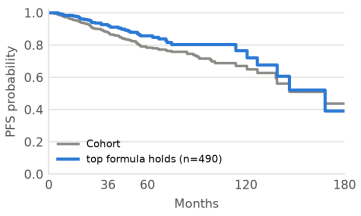


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A203** mutated → **TP53** targeted by → **AZD 3355**	68	↓ **slower progression** · Σw -0.031 over 2 formula(s) Stage(p, II) ∧ Mutated(p, TP53) ∧ Altered(p, g) ∧ InPathway(g, 'Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants') ⇒ Progress(p) cohort: HR 0.95, n = 283 carriers	**0.50**
2	**TCGA-BH-A203** mutated → **TP53** interacts with → **YWHAZ** in pathway → **Regulation of localization of FOXO transcription factors**	66	↓ **slower progression** · Σw -0.031 over 2 formula(s) Stage(p, II) ∧ Mutated(p, TP53) ∧ Altered(p, g) ∧ InPathway(g, 'Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants') ⇒ Progress(p) cohort: HR 1.00, n = 210 carriers	**0.49**
3	**TCGA-BH-A203** deleted → **CYP11B2** targeted by → **Osilodrostat**	68	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 108 carriers	**0.41**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

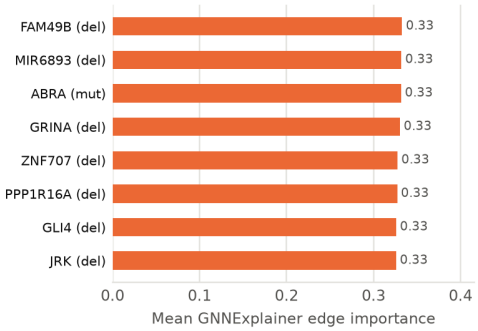
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
+0.026 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K Cascade') ⇒ Progress(p) [KG]	mined pair	31 pts
-0.024 ↓ risk	Stage(p, II) ∧ Mutated(p, TP53) ∧ Altered(p, g) ∧ InPathway(g, 'Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants') ⇒ Progress(p) [KG]	RL-proposed	71 pts
+0.023 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ⇒ Progress(p) [KG]	mined pair	55 pts
-0.007 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	306 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.32
TP53 · driver	Mutation	p.R273H	57	0.31
FAM49B	Homozygous deletion	GISTIC ±2	0	0.33
MIR6893	Homozygous deletion	GISTIC ±2	0	0.33
ABRA	Mutation	p.E5K	0	0.33
GRINA	Homozygous deletion	GISTIC ±2	0	0.33
ZNF707	Homozygous deletion	GISTIC ±2	1	0.33
PPP1R16A	Homozygous deletion	GISTIC ±2	0	0.33
GLI4	Homozygous deletion	GISTIC ±2	0	0.33
JRK	Homozygous deletion	GISTIC ±2	0	0.33
EPPK1	Homozygous deletion	GISTIC ±2	0	0.33
LYNX1	Homozygous deletion	GISTIC ±2	0	0.33



163 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347) … and 78 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A6SD	32	II	Her2	0.80	no progression, followed 33.2 mo
TCGA-A2-A0D1	76	II	Her2	0.79	no progression, followed 34.6 mo
TCGA-C8-A12K	80	II	Basal	0.78	no progression, followed 0.0 mo
TCGA-C8-A135	64	II	Her2	0.77	no progression, followed 12.9 mo
TCGA-AR-A0TS	46	II	Basal	0.76	no progression, followed 84.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**89**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**—**	Pathologic T	**T1C**
Prior diagnosis	**Yes**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

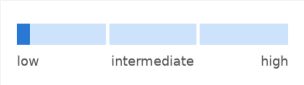
Sample: **TCGA-E2-A15O-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **0.9** · MSIsensor score: **0.19** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

5th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

≈ 16 months

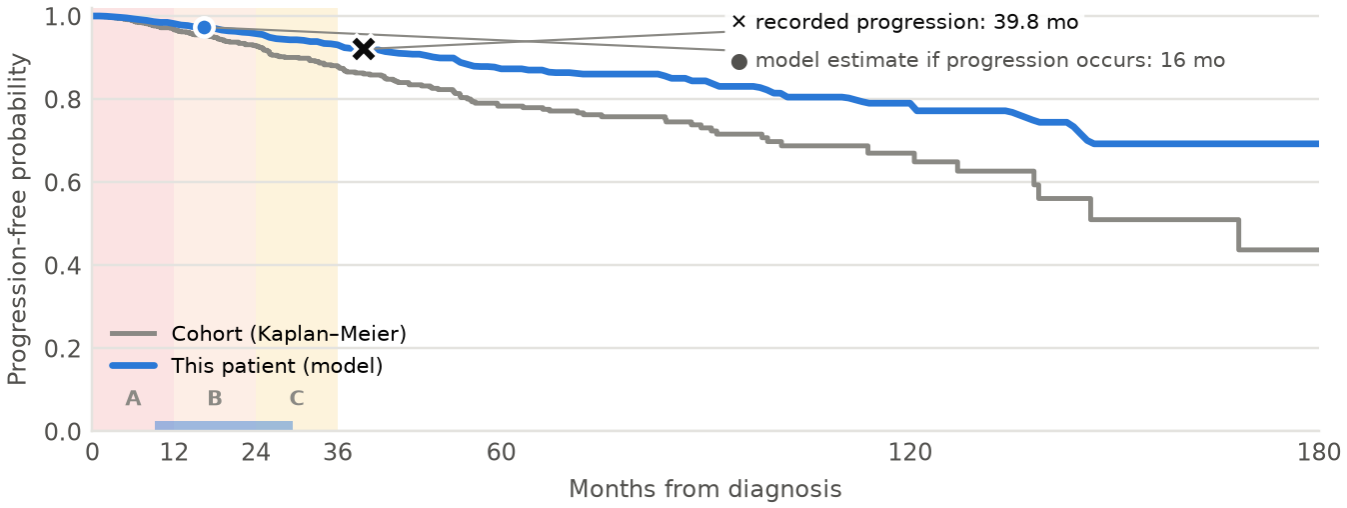
Window 12–24 months · IQR 9–29 mo if progression occurs

Recorded progression

39.8 months

Group D · Late progression

✗ **recorded window differs** · |error| 23 mo



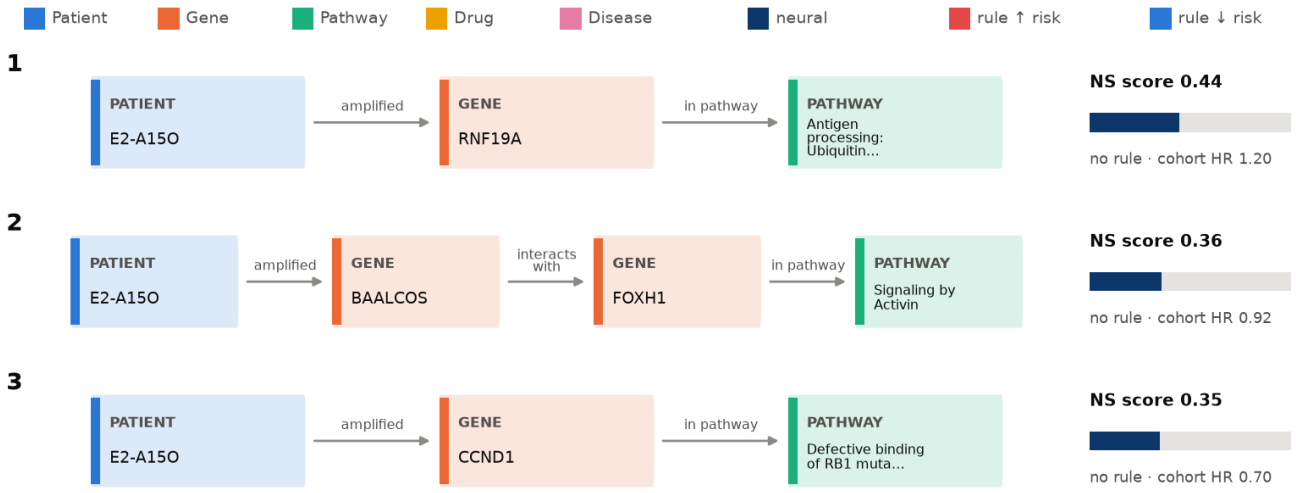
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 89-year-old patient has a LumA tumour, AJCC stage I (T1, missing, M0). The fused neuro-symbolic model places them in the **5th risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, I)* ∧ *Subtype(p, LumA)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.055 per SD). The highest-scoring knowledge-graph path runs from **RNF19A** to **Antigen processing: Ubiquitination & Proteasome degradation**. Progression was recorded at 39.8 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

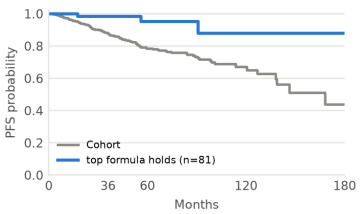


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-E2-A15O** amplified → **RNF19A** in pathway → **Antigen processing: Ubiquitination & Proteasome degradation**	74	No learned formula fires on this path; its score is neural only. cohort: HR 1.20, n = 141 carriers	**0.44**
2	**TCGA-E2-A15O** amplified → **BAALCOS** interacts with → **FOXH1** in pathway → **Signaling by Activin**	60	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 98 carriers	**0.36**
3	**TCGA-E2-A15O** amplified → **CCND1** in pathway → **Defective binding of RB1 mutants to E2F1,(E2F2, E2F3)**	58	No learned formula fires on this path; its score is neural only. cohort: HR 0.70, n = 182 carriers	**0.35**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

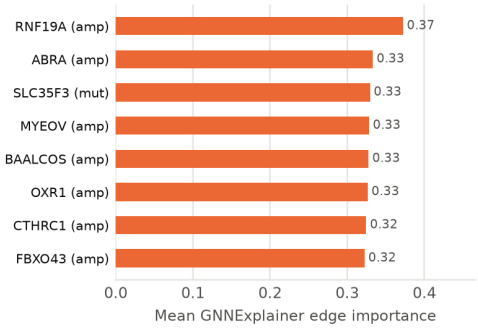
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.055 ↓ risk	Stage(p, I) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	81 pts
-0.051 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	118 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
MAP3K1 · driver	Mutation	p.P286Tfs*14	13	0.31
RNF19A	High-level amplification	GISTIC ±2	1	0.37
ABRA	High-level amplification	GISTIC ±2	0	0.33
SLC35F3	Mutation	p.D95=	0	0.33
MYEOV	High-level amplification	GISTIC ±2	0	0.33
BAALCOS	High-level amplification	GISTIC ±2	0	0.33
OXR1	High-level amplification	GISTIC ±2	0	0.33
CTHRC1	High-level amplification	GISTIC ±2	0	0.32
FBXO43	High-level amplification	GISTIC ±2	0	0.32
INTS8	High-level amplification	GISTIC ±2	1	0.32
ZNF703	High-level amplification	GISTIC ±2	1	0.32



86 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010); FGF19 → Heparin (DB01109); SLC25A32 → Folic acid (DB00158); FGF4 → Pentosan polysulfate (DB00686) … and 48 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0ER	63	I	LumA	0.71	no progression, followed 74.4 mo
TCGA-BH-A201	64	I	LumA	0.71	no progression, followed 28.1 mo
TCGA-E2-A15C	61	I	LumA	0.68	no progression, followed 22.8 mo
TCGA-A2-A0CP	60	I	LumA	0.68	no progression, followed 92.5 mo
TCGA-E2-A15R	64	II	LumA	0.67	no progression, followed 56.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**48**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

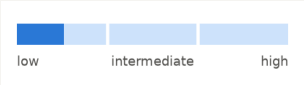
Sample: **TCGA-A2-A0SX-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **0.77** · MSIsensor score: **0.31** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

17th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 15%

Model-estimated progression time

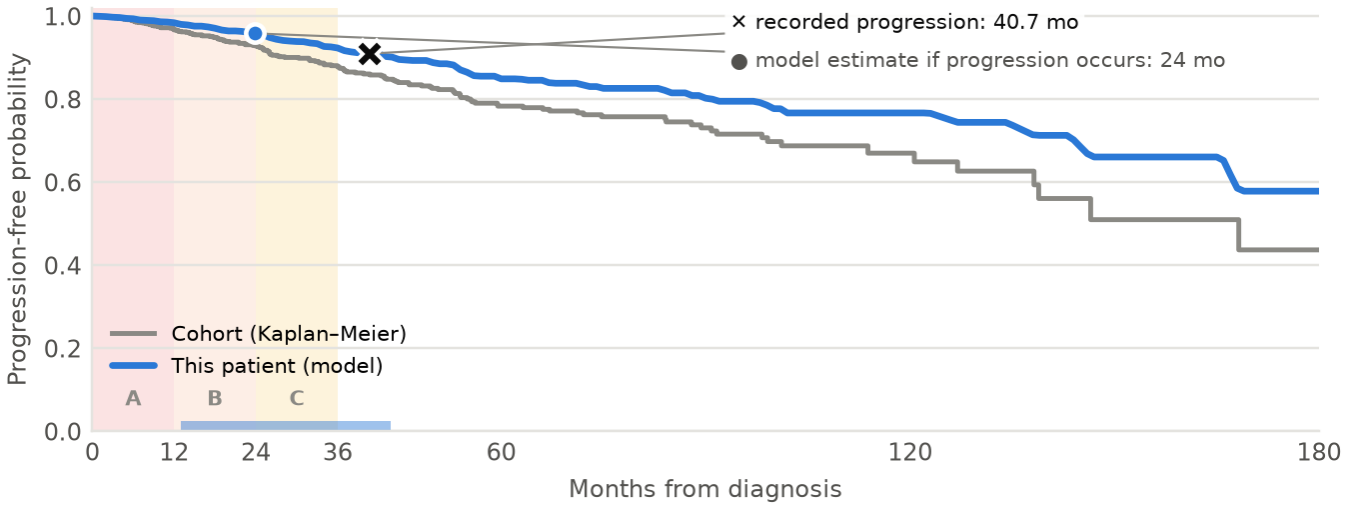
≈ 24 months

Window 12–24 months · IQR 13–44 mo
if progression occurs

Recorded progression

40.7 months

Group D · Late progression
✗ recorded window differs · |error| 17 mo



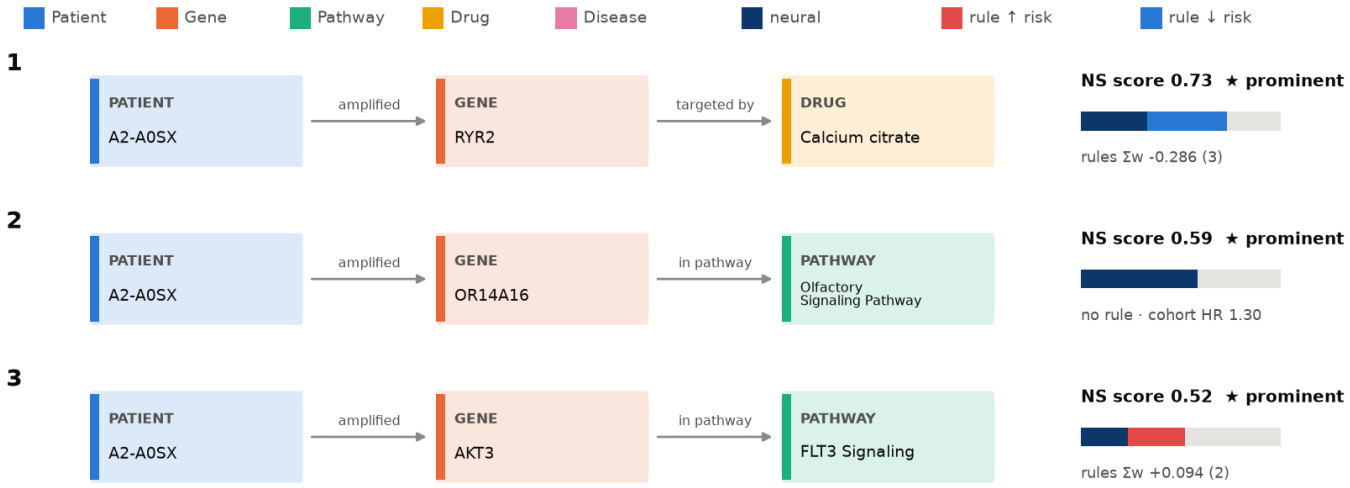
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 48-year-old patient has a Basal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **17th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **RYR2** to **Calcium citrate**; the learned formulas it instantiates point to slower progression (Σw -0.286). Progression was recorded at 40.7 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

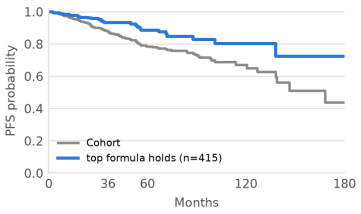


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A0SX** amplified → **RYR2** targeted by → **Calcium citrate**	55	↓ **slower progression** · Σw -0.286 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 0.77, n = 196 carriers	**0.73**
2 ★	**TCGA-A2-A0SX** amplified → **OR14A16** in pathway → **Olfactory Signaling Pathway**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.30, n = 162 carriers	**0.59**
3 ★	**TCGA-A2-A0SX** amplified → **AKT3** in pathway → **FLT3 Signaling**	40	↑ **faster progression** · Σw +0.094 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 0.98, n = 401 carriers	**0.52**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

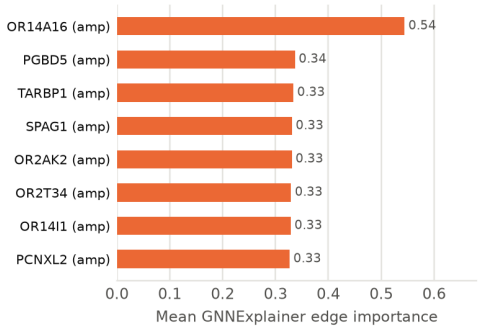
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
-0.141 ↓ risk	Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) [KG]	RL-proposed	93 pts
-0.097 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Stimuli-sensing channels') ∧ Altered(p, g) ∧ InPathway(g, 'Pyroptosis') ∧ Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	RL-proposed	70 pts
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) [KG]	RL-proposed	71 pts
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ⇒ Progress(p) [KG]	single atom	97 pts
-0.047 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	147 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.32
PIK3R1 · driver	Homozygous deletion	GISTIC ±2	79	0.32
TP53 · driver	Mutation	p.G199V	57	0.31
MAP3K1 · driver	Homozygous deletion	GISTIC ±2	13	0.31
OR14A16	High-level amplification	GISTIC ±2	1	0.54
PGBD5	High-level amplification	GISTIC ±2	0	0.34
TARBP1	High-level amplification	GISTIC ±2	0	0.33
SPAG1	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
OR2T34	High-level amplification	GISTIC ±2	1	0.33
OR14I1	High-level amplification	GISTIC ±2	1	0.33
PCNXL2	High-level amplification	GISTIC ±2	0	0.33



86 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486) … and 56 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A229	37	I	LumA	0.79	no progression, followed 37.7 mo
TCGA-E9-A22G	47	II	Basal	0.78	no progression, followed 40.7 mo
TCGA-AO-A0J4	41	I	Basal	0.78	no progression, followed 52.2 mo
TCGA-B6-A409	44	III	Basal	0.76	progressed at 15.0 mo (group B)
TCGA-A2-A0D0	60	II	Basal	0.76	no progression, followed 67.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

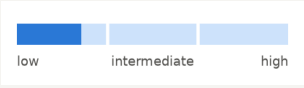
Sample: **TCGA-A2-A25B-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **3.27** · MSIsensor score: **0.61** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

24th

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

≈ 37 months

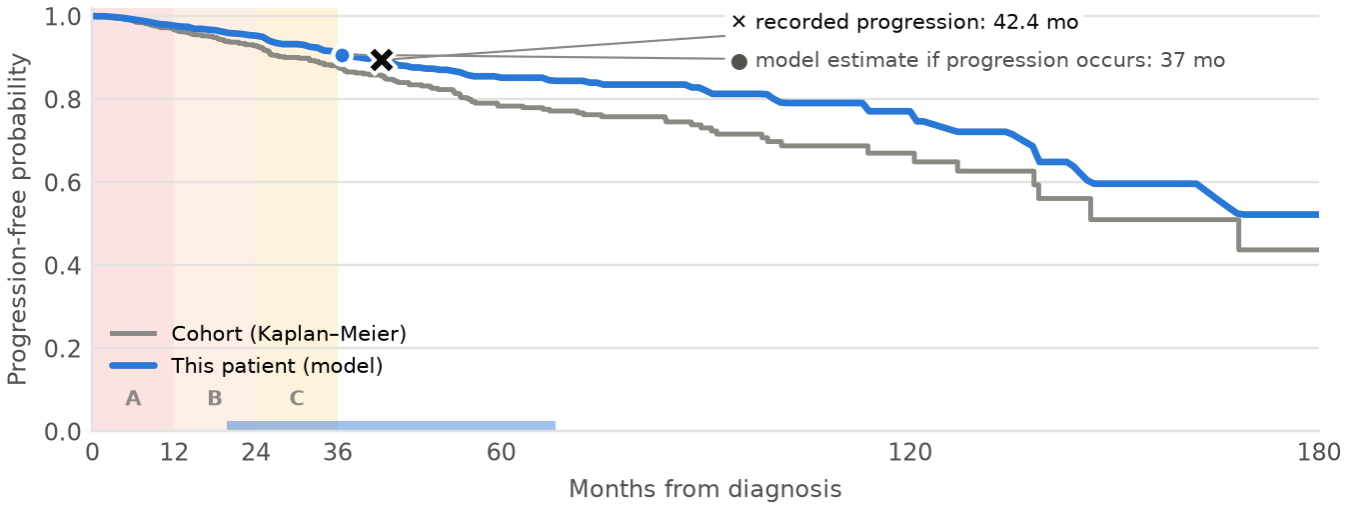
Window > 36 months · IQR 20–68 mo if progression occurs

Recorded progression

42.4 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 6 mo



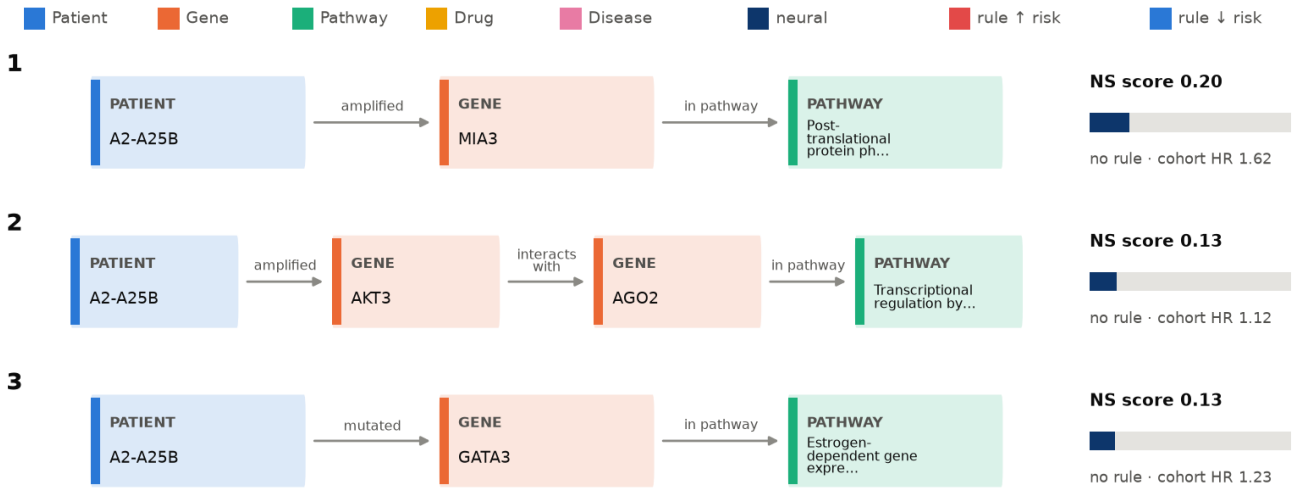
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **24th risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II)* ∧ *Age(p)* < *40* ⇒ *Progress(p)*) raises the progression hazard (weight +0.033 per SD). The highest-scoring knowledge-graph path runs from **MIA3** to **Post-translational protein phosphorylation**. Progression was recorded at 42.4 months; the model's estimate given progression was 37 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

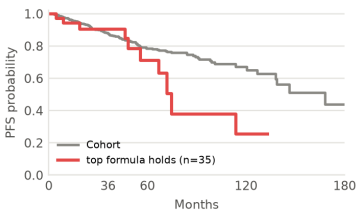


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A25B** amplified → **MIA3** in pathway → **Post-translational protein phosphorylation**	33	No learned formula fires on this path; its score is neural only. cohort: HR 1.62, n = 116 carriers	**0.20**
2	**TCGA-A2-A25B** amplified → **AKT3** interacts with → **AGO2** in pathway → **Transcriptional regulation by small RNAs**	22	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 137 carriers	**0.13**
3	**TCGA-A2-A25B** mutated → **GATA3** in pathway → **Estrogen-dependent gene expression**	21	No learned formula fires on this path; its score is neural only. cohort: HR 1.23, n = 442 carriers	**0.13**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

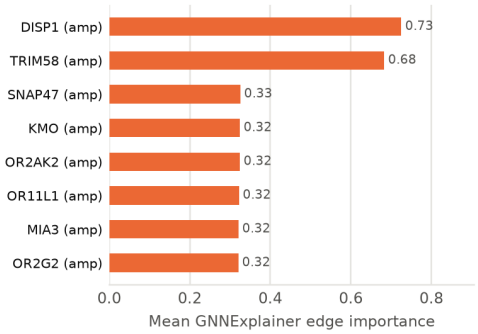
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.033 ↑ risk	Stage(p, II) ∧ Age(p) < 40 ⇒ Progress(p)	mined pair	35 pts
-0.001 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	147 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.M400Vfs*106	13	0.32
BRCA1 · driver	Mutation	p.E720*	29	0.31
FOXA1 · driver	High-level amplification	GISTIC ±2	9	0.31
DISP1	High-level amplification	GISTIC ±2	0	0.73
TRIM58	High-level amplification	GISTIC ±2	0	0.68
SNAP47	High-level amplification	GISTIC ±2	0	0.33
KMO	High-level amplification	GISTIC ±2	1	0.32
OR2AK2	High-level amplification	GISTIC ±2	1	0.32
OR11L1	High-level amplification	GISTIC ±2	1	0.32
MIA3	High-level amplification	GISTIC ±2	3	0.32
OR2G2	High-level amplification	GISTIC ±2	1	0.32
OBSCN	High-level amplification	GISTIC ±2	8	0.32



60 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RYR2 → Flecainide (DB01195); RYR2 → Tetracaine (DB09085); RYR2 → Calcium citrate (DB11093); RYR2 → Calcium Phosphate (DB11348); RYR2 → Calcium phosphate dihydrate (DB14481); TARS2 → Threonine (DB00156); PARP1 → Theophylline (DB00277); PARP1 → Zinc (DB01593); PARP1 → Carba-nicotinamide-adenine-dinucleotide (DB02498); PARP1 → NU1025 (DB02690) … and 31 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A03N	59	II	LumB	0.80	progressed at 54.1 mo (group D)
TCGA-D8-A1JB	54	II	LumA	0.78	no progression, followed 55.5 mo
TCGA-BH-A28Q	46	II	LumA	0.77	no progression, followed 36.8 mo
TCGA-A7-A6VW	48	II	Basal	0.75	no progression, followed 9.4 mo
TCGA-E9-A1NI	51	II	LumB	0.73	no progression, followed 9.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A1LL · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**—**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

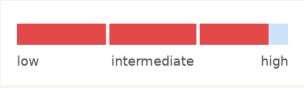
Sample: **TCGA-E2-A1LL-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **1.43** · MSIsensor score: **2.8** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

93rd

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 6% · 24 mo 13% · 60 mo 35%

Model-estimated progression time

≈ 16 months

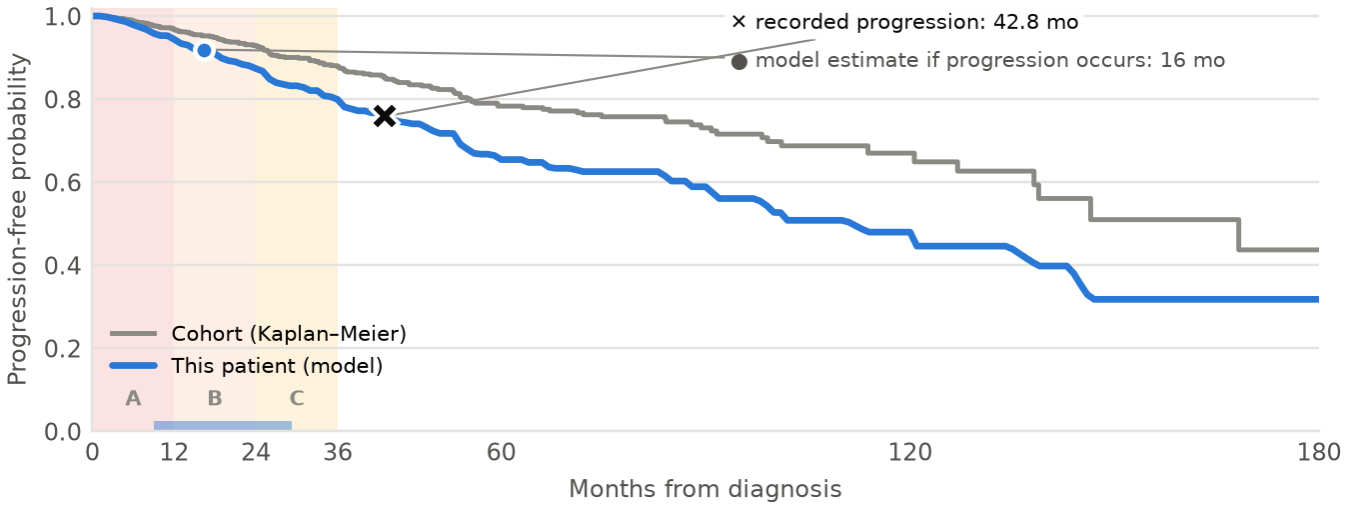
Window 12–24 months · IQR 9–29 mo if progression occurs

Recorded progression

42.8 months

Group D · Late progression

✗ **recorded window differs** · |error| 27 mo



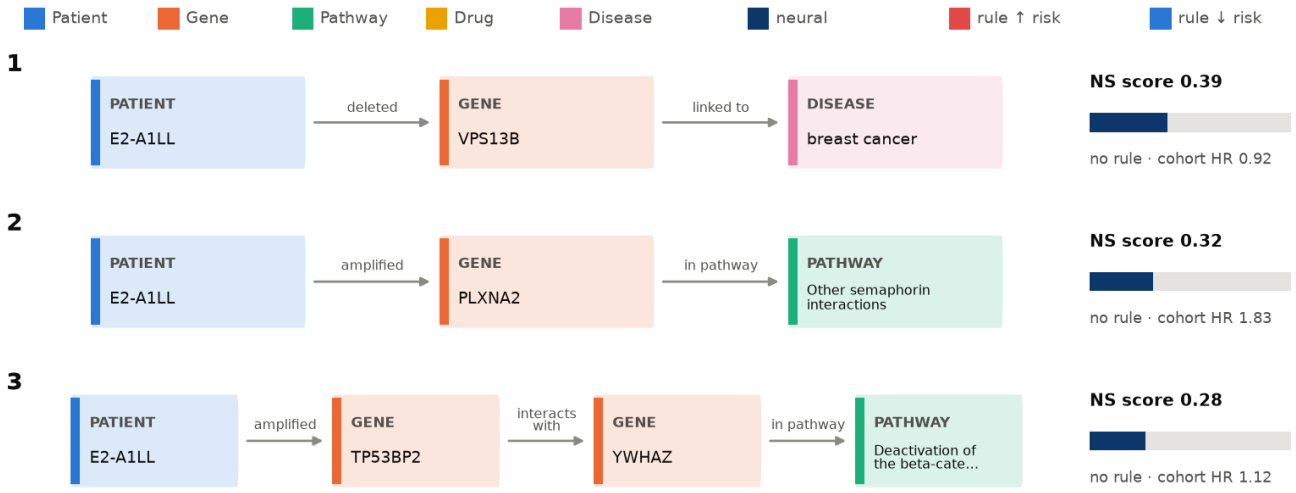
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a Basal tumour, AJCC stage III (T3, N2, MX). The fused neuro-symbolic model places them in the **93rd risk percentile** (high tier). The strongest learned formula that holds for them (*Stage(p, III)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Neurofascin interactions')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.087 per SD). The highest-scoring knowledge-graph path runs from **VPS13B** to **breast cancer**. Progression was recorded at 42.8 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

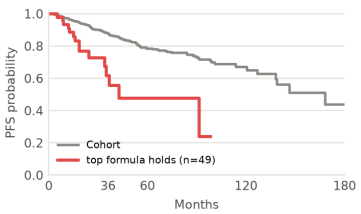


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-E2-A1LL** deleted → **VPS13B** linked to → **breast cancer**	64	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 793 carriers	**0.39**
2	**TCGA-E2-A1LL** amplified → **PLXNA2** in pathway → **Other semaphorin interactions**	53	No learned formula fires on this path; its score is neural only. cohort: HR 1.83, n = 92 carriers	**0.32**
3	**TCGA-E2-A1LL** amplified → **TP53BP2** interacts with → **YWHAZ** in pathway → **Deactivation of the beta-catenin transactivating complex**	46	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 283 carriers	**0.28**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

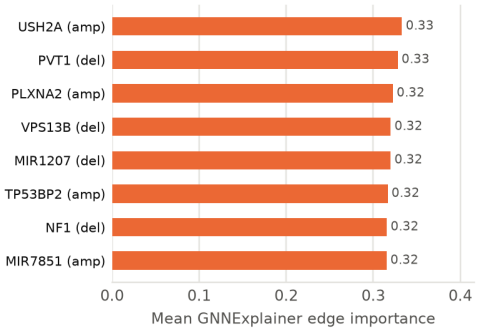
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.087 ↑ risk	Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ⇒ Progress(p) [KG]	mined pair	49 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
NF1 · driver	Homozygous deletion	GISTIC ±2	6	0.32
RUNX1 · driver	Homozygous deletion	GISTIC ±2	18	0.31
USH2A	High-level amplification	GISTIC ±2	0	0.33
PVT1	Homozygous deletion	GISTIC ±2	0	0.33
PLXNA2	High-level amplification	GISTIC ±2	4	0.32
VPS13B	Homozygous deletion	GISTIC ±2	4	0.32
MIR1207	Homozygous deletion	GISTIC ±2	0	0.32
TP53BP2	High-level amplification	GISTIC ±2	12	0.32
MIR7851	High-level amplification	GISTIC ±2	0	0.32
CR1	High-level amplification	GISTIC ±2	3	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.31
STK3	Homozygous deletion	GISTIC ±2	2	0.31



23 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

STK3 → Fostamatinib (DB12010). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1X6	80	III	LumB	0.62	no progression, followed 17.8 mo
TCGA-AC-A2BK	78	III	Basal	0.51	no progression, followed 73.1 mo
TCGA-AR-A24R	45	III	LumB	0.51	no progression, followed 112.8 mo
TCGA-A7-A26E	71	III	LumA	0.50	no progression, followed 31.4 mo
TCGA-OL-A5RV	43	III	LumA	0.47	no progression, followed 34.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE III**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-AR-A2LK-01** · Aneuploidy score: **26** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **1.05** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

82nd

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 28%

Model-estimated progression time

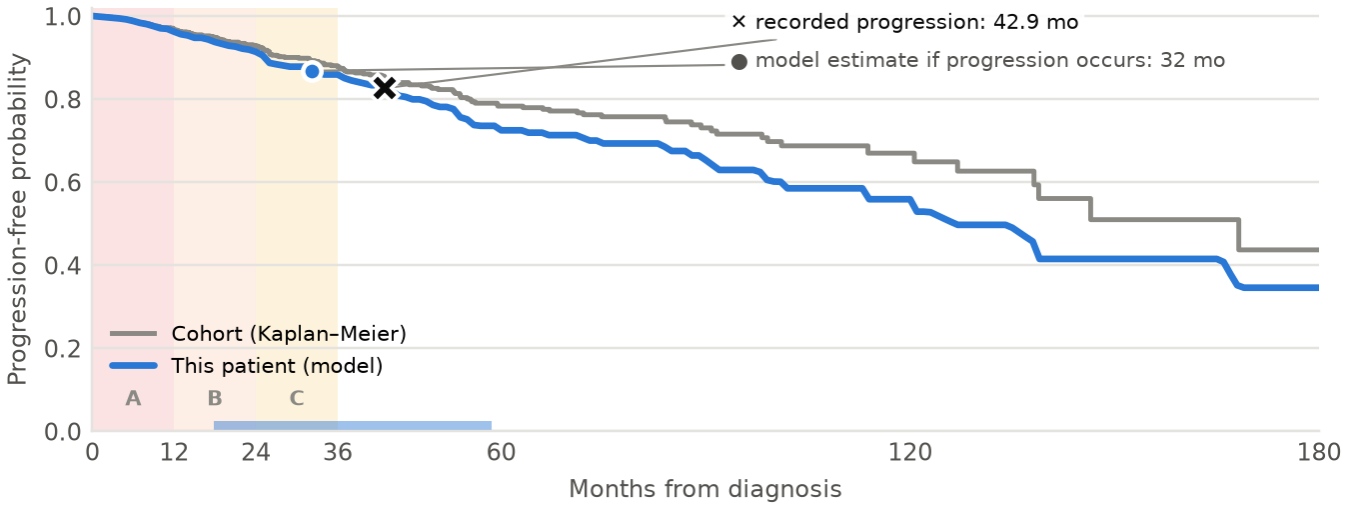
≈ 32 months

Window 24–36 months · IQR 18–59 mo
if progression occurs

Recorded progression

42.9 months

Group D · Late progression
✗ **recorded window differs** · |error| 11 mo



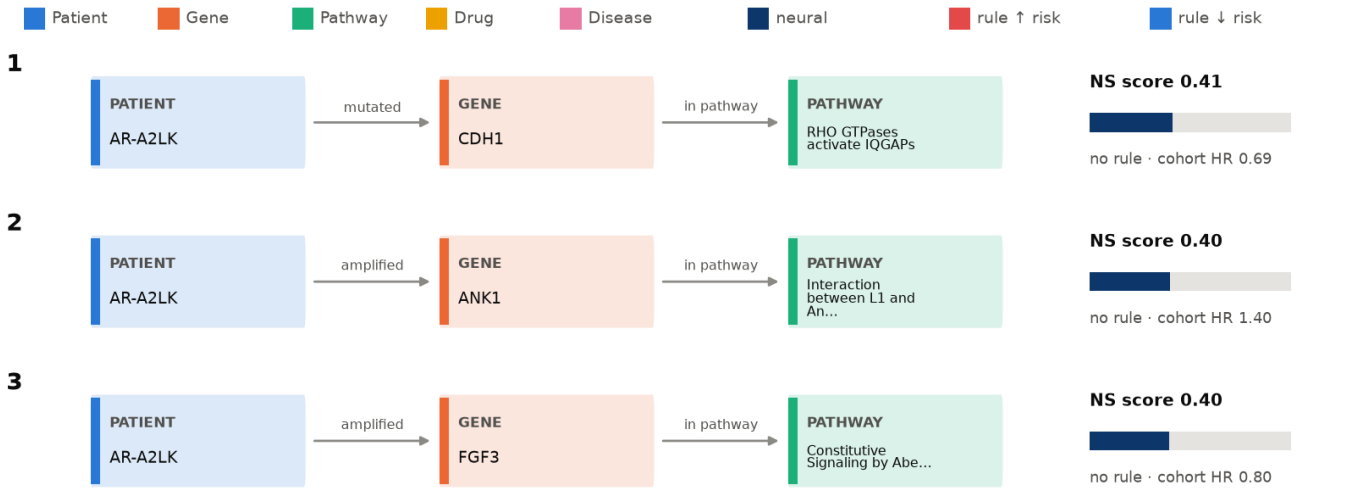
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumB tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **82nd risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **CDH1** to **RHO GTPases activate IQGAPs**. Progression was recorded at 42.9 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

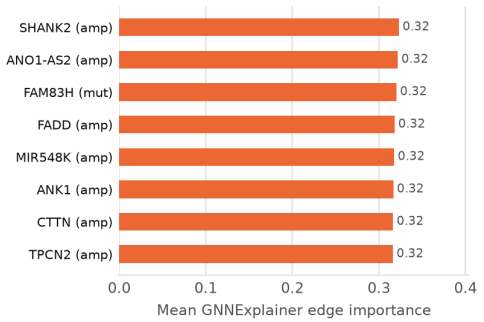
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A2LK** mutated → **CDH1** in pathway → **RHO GTPases activate IQGAPs**	69	No learned formula fires on this path; its score is neural only. cohort: HR 0.69, n = 113 carriers	**0.41**
2	**TCGA-AR-A2LK** amplified → **ANK1** in pathway → **Interaction between L1 and Ankyrins**	67	No learned formula fires on this path; its score is neural only. cohort: HR 1.40, n = 296 carriers	**0.40**
3	**TCGA-AR-A2LK** amplified → **FGF3** in pathway → **Constitutive Signaling by Aberrant PI3K in Cancer**	66	No learned formula fires on this path; its score is neural only. cohort: HR 0.80, n = 532 carriers	**0.40**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.A788Efs*14	17	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
SHANK2	High-level amplification	GISTIC ±2	1	0.32
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.32
FAM83H	Mutation	p.A238T	0	0.32
FADD	High-level amplification	GISTIC ±2	14	0.32
MIR548K	High-level amplification	GISTIC ±2	0	0.32
ANK1	High-level amplification	GISTIC ±2	9	0.32
CTTN	High-level amplification	GISTIC ±2	2	0.32
TPCN2	High-level amplification	GISTIC ±2	1	0.32
WISP1	Mutation	p.R171Q	0	0.32



25 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718) … and 6 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CK	60	III	LumA	0.82	progressed at 44.8 mo (group D)
TCGA-AN-A03Y	66	II	LumB	0.74	no progression, followed 0.3 mo
TCGA-A2-A1FX	61	III	LumB	0.74	no progression, followed 60.7 mo
TCGA-PE-A5DC	72	III	LumA	0.72	no progression, followed 47.0 mo
TCGA-A8-A0A6	64	III	LumA	0.69	no progression, followed 21.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AR-A0TZ · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**43**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A0TZ-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.87** · MSIsensor score: **0.45** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

45% by 36 mo

12 mo 11% · 24 mo 27% · 60 mo 70%

Model-estimated progression time

≈ 40 months

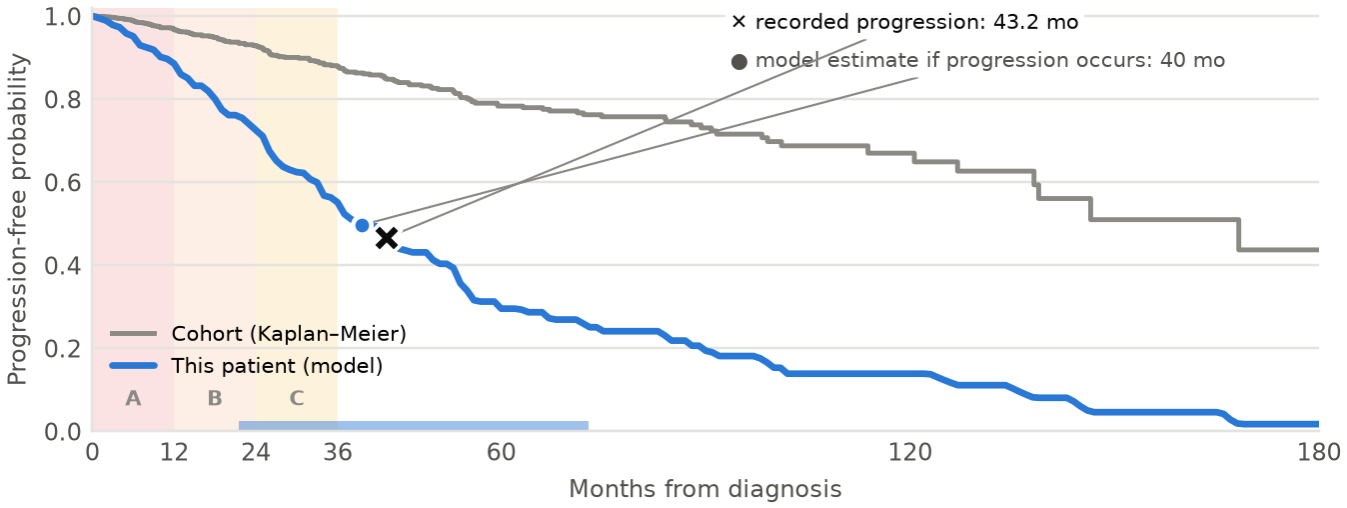
Window > 36 months · IQR 22–73 mo if progression occurs

Recorded progression

43.2 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 4 mo



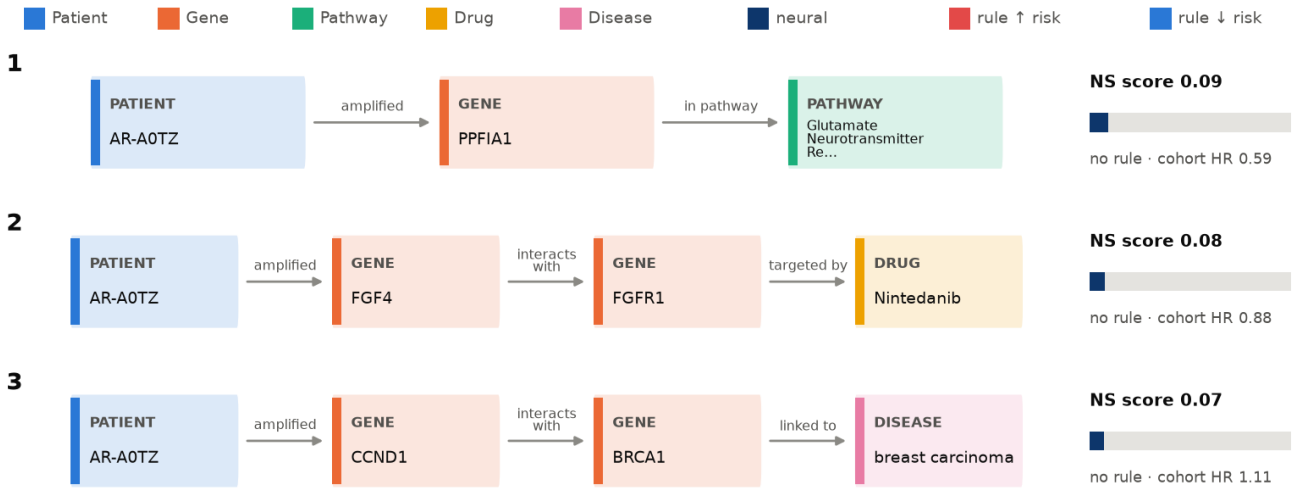
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 43-year-old patient has a LumA tumour, AJCC stage III (T2, N2, M1). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The strongest learned formula that holds for them (*Subtype(p, LumA) ⇒ Progress(p)*) lowers the progression hazard (weight -0.016 per SD). The highest-scoring knowledge-graph path runs from **PPFIA1** to **Glutamate Neurotransmitter Release Cycle**. Progression was recorded at 43.2 months; the model's estimate given progression was 40 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

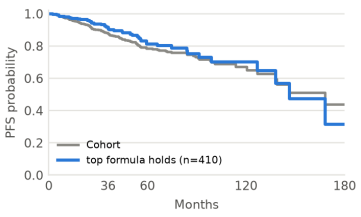


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A0TZ** amplified → **PPFIA1** in pathway → **Glutamate Neurotransmitter Release Cycle**	15	No learned formula fires on this path; its score is neural only. cohort: HR 0.59, n = 129 carriers	**0.09**
2	**TCGA-AR-A0TZ** amplified → **FGF4** interacts with → **FGFR1** targeted by → **Nintedanib**	13	No learned formula fires on this path; its score is neural only. cohort: HR 0.88, n = 114 carriers	**0.08**
3	**TCGA-AR-A0TZ** amplified → **CCND1** interacts with → **BRCA1** linked to → **breast carcinoma**	12	No learned formula fires on this path; its score is neural only. cohort: HR 1.11, n = 782 carriers	**0.07**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

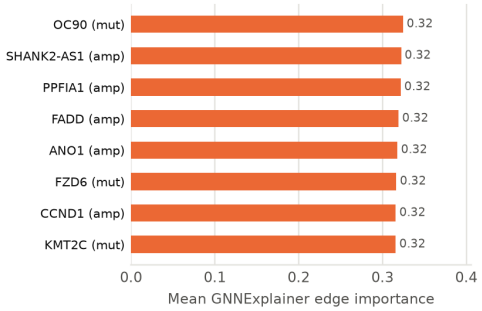
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
KMT2C · driver	Mutation	p.K4499R	4	0.32
PIK3CA · driver	Mutation	p.H1047R	70	0.31
OC90	Mutation	p.E285*	0	0.32
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
PPFIA1	High-level amplification	GISTIC ±2	6	0.32
FADD	High-level amplification	GISTIC ±2	14	0.32
ANO1	High-level amplification	GISTIC ±2	2	0.32
FZD6	Mutation	3'UTR	5	0.32
RNU6-83P	High-level amplification	GISTIC ±2	0	0.32
TRAPPC9	Mutation	p.Q801*	2	0.31
FGF19	High-level amplification	GISTIC ±2	13	0.31



23 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 7 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A0JG	49	III	LumA	0.76	no progression, followed 26.2 mo
TCGA-AN-A0FW	67	III	LumA	0.74	no progression, followed 0.4 mo
TCGA-BH-A0BC	60	III	LumA	0.69	no progression, followed 32.0 mo
TCGA-AO-A126	39	II	LumA	0.69	progressed at 8.8 mo (group A)
TCGA-C8-A12Y	44	II	LumA	0.69	no progression, followed 48.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0X5 · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

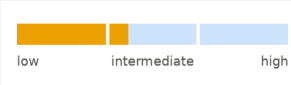
Sample: **TCGA-B6-A0X5-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **1.2** · MSIsensor score: **0.01** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

41st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 17%

Model-estimated progression time

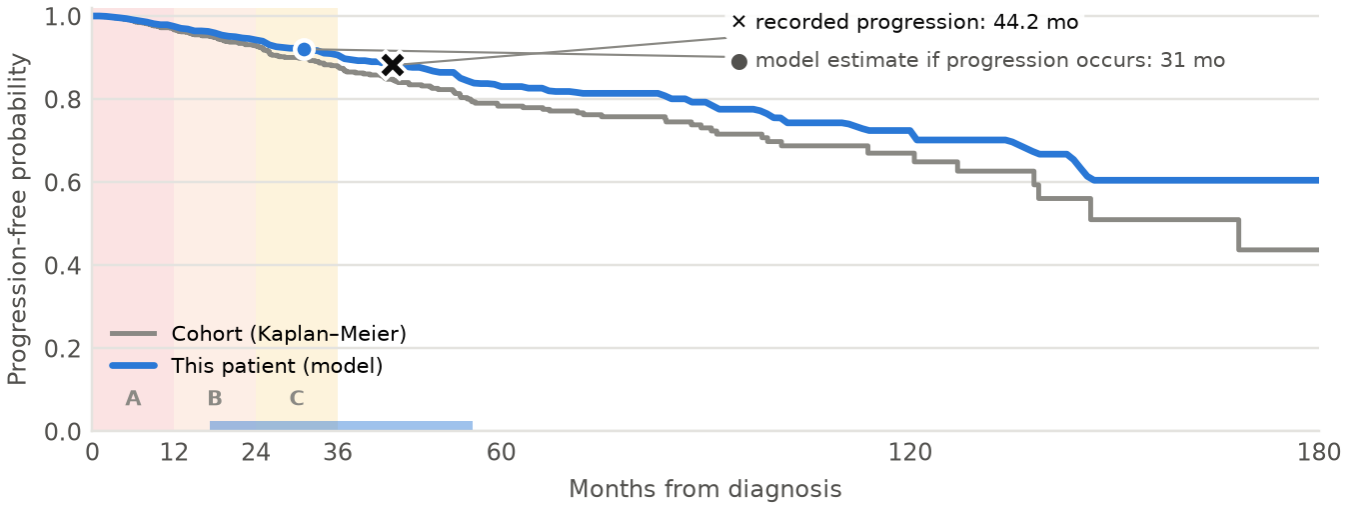
≈ 31 months

Window 24–36 months · IQR 17–56 mo
if progression occurs

Recorded progression

44.2 months

Group D · Late progression
✗ **recorded window differs** · |error| 13 mo



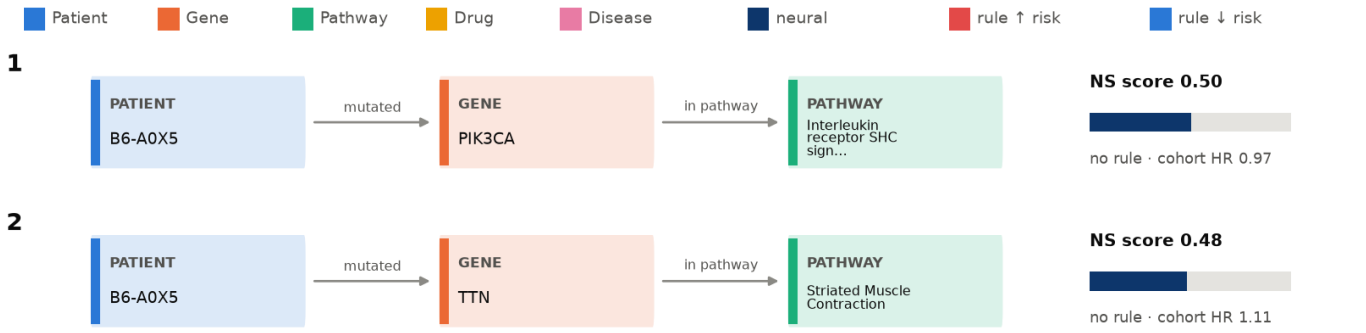
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **41st risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.006 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Interleukin receptor SHC signaling**. Progression was recorded at 44.2 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

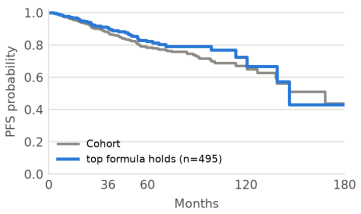


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0X5** mutated → **PIK3CA** in pathway → **Interleukin receptor SHC signaling**	84	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 324 carriers	**0.50**
2	**TCGA-B6-A0X5** mutated → **TTN** in pathway → **Striated Muscle Contraction**	81	No learned formula fires on this path; its score is neural only. cohort: HR 1.11, n = 334 carriers	**0.48**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

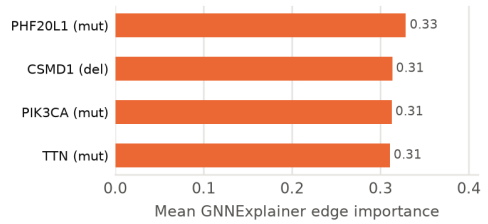
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.31
PHF20L1	Mutation	p.A525Gfs*23	0	0.33
CSMD1	Homozygous deletion	GISTIC ±2	0	0.31
TTN	Mutation	p.N26237=	2	0.31



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A25C	50	II	LumB	0.97	no progression, followed 17.2 mo
TCGA-BH-A0BF	56	II	LumB	0.83	no progression, followed 43.5 mo
TCGA-A2-A0YC	59	II	LumA	0.83	progressed at 24.9 mo (group C)
TCGA-E2-A10B	67	II	LumA	0.83	progressed at 6.4 mo (group A)
TCGA-D8-A1JE	62	II	LumB	0.82	no progression, followed 18.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**53**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T1B**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-A2-A0EW-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.5** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

76th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 24%

Model-estimated progression time

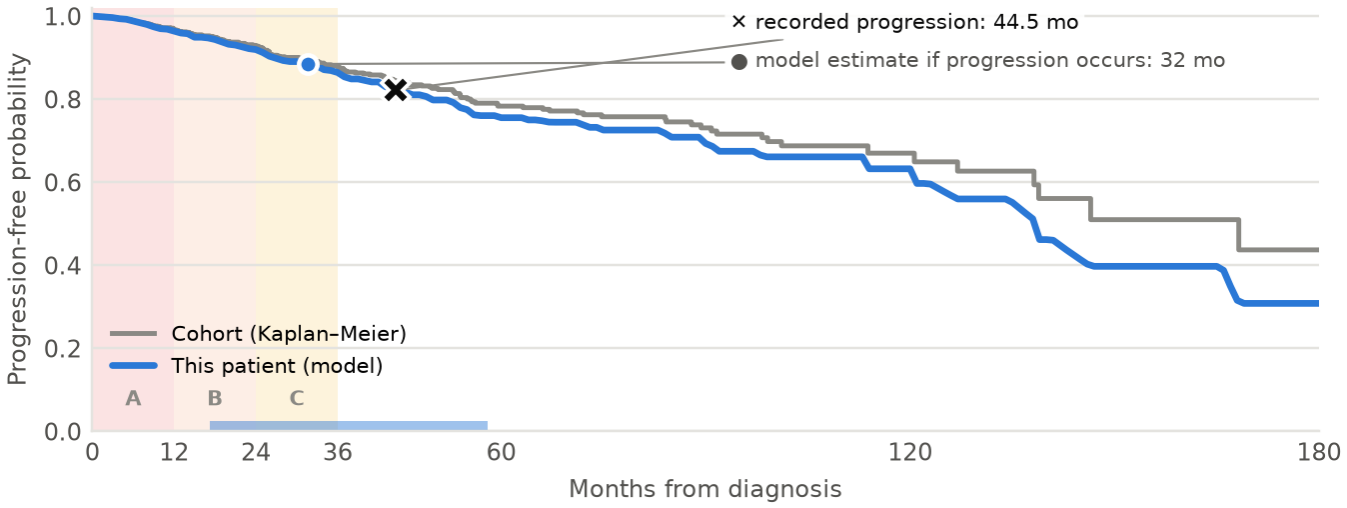
≈ 32 months

Window 24–36 months · IQR 17–58 mo
if progression occurs

Recorded progression

44.5 months

Group D · Late progression
✗ recorded window differs · |error| 13 mo



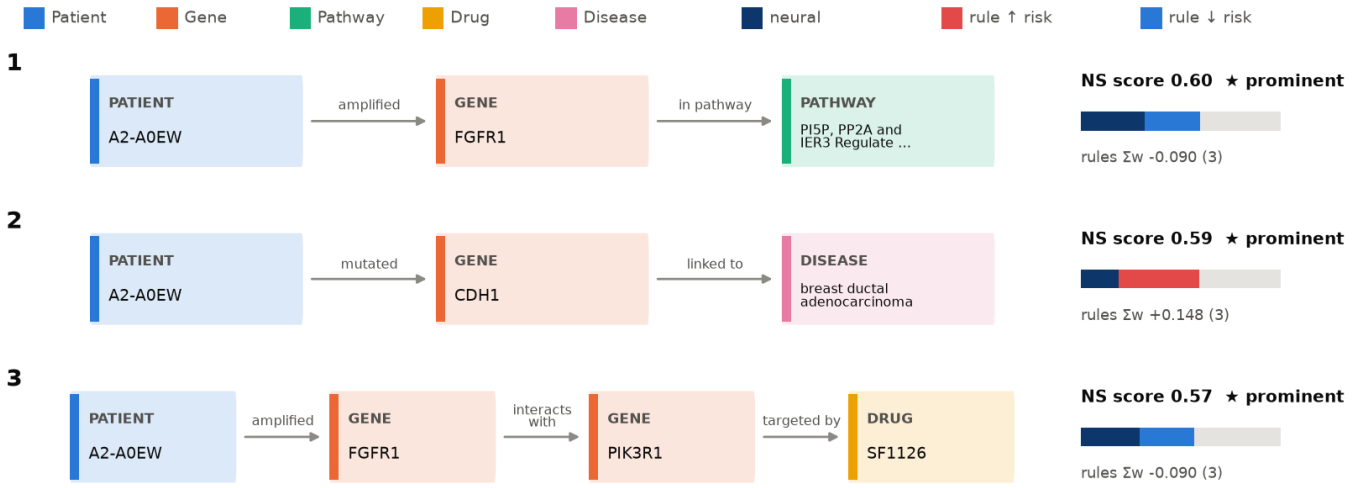
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 53-year-old patient has a LumA tumour, AJCC stage III (T1, N3, M0). The fused neuro-symbolic model places them in the **76th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T1)* ∧ *Subtype(p, LumA)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.125 per SD). The highest-scoring knowledge-graph path runs from **FGFR1** to **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**; the learned formulas it instantiates point to slower progression (Σw -0.090). Progression was recorded at 44.5 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

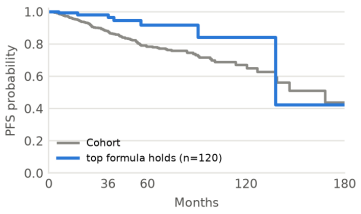


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A0EW** amplified → **FGFR1** in pathway → **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**	54	↓ **slower progression** · Σw -0.090 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.73, n = 536 carriers	**0.60**
2 ★	**TCGA-A2-A0EW** mutated → **CDH1** linked to → **breast ductal adenocarcinoma**	32	↑ **faster progression** · Σw +0.148 over 3 formula(s) NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 1.29, n = 521 carriers	**0.59**
3 ★	**TCGA-A2-A0EW** amplified → **FGFR1** interacts with → **PIK3R1** targeted by → **SF1126**	49	↓ **slower progression** · Σw -0.090 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.95, n = 37 carriers	**0.57**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

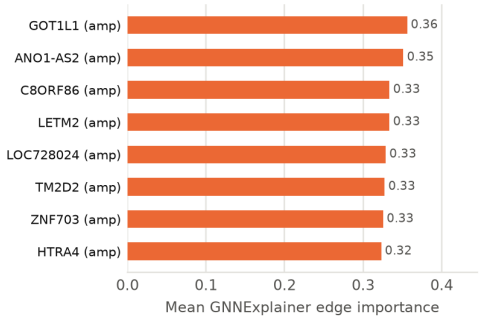
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.125 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	120 pts
+0.097 ↑ risk	NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	50 pts
+0.073 ↑ risk	LowTmb(p) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	73 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.034 ↓ risk	LowAneuploidy(p) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Deactivation of the beta-catenin transactivating complex') ⇒ Progress(p) [KG]	RL-proposed	41 pts
-0.022 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ LinkedTo(g, 'invasive ductal breast carcinoma') ⇒ Progress(p) [KG]	RL-proposed	38 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
CDH1 · driver	Mutation	p.X230_splice	17	0.32
PIK3CA · driver	Mutation	p.Q546R	70	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.36
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.35
C8ORF86	High-level amplification	GISTIC ±2	0	0.33
LETM2	High-level amplification	GISTIC ±2	0	0.33
LOC728024	High-level amplification	GISTIC ±2	0	0.33
TM2D2	High-level amplification	GISTIC ±2	0	0.33
ZNF703	High-level amplification	GISTIC ±2	1	0.33
HTRA4	High-level amplification	GISTIC ±2	0	0.32



47 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 71 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-GM-A2DO	54	I	LumB	0.74	no progression, followed 85.3 mo
TCGA-A2-A0YK	61	II	Normal	0.72	no progression, followed 19.3 mo
TCGA-A8-A0A6	64	III	LumA	0.72	no progression, followed 21.0 mo
TCGA-B6-A0IP	74	I	LumA	0.71	no progression, followed 129.1 mo
TCGA-BH-A201	64	I	LumA	0.69	no progression, followed 28.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

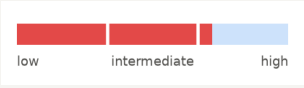
Sample: **TCGA-A2-A0CK-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.43** · MSIsensor score: **0** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

72nd

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 24%

Model-estimated progression time

≈ 42 months

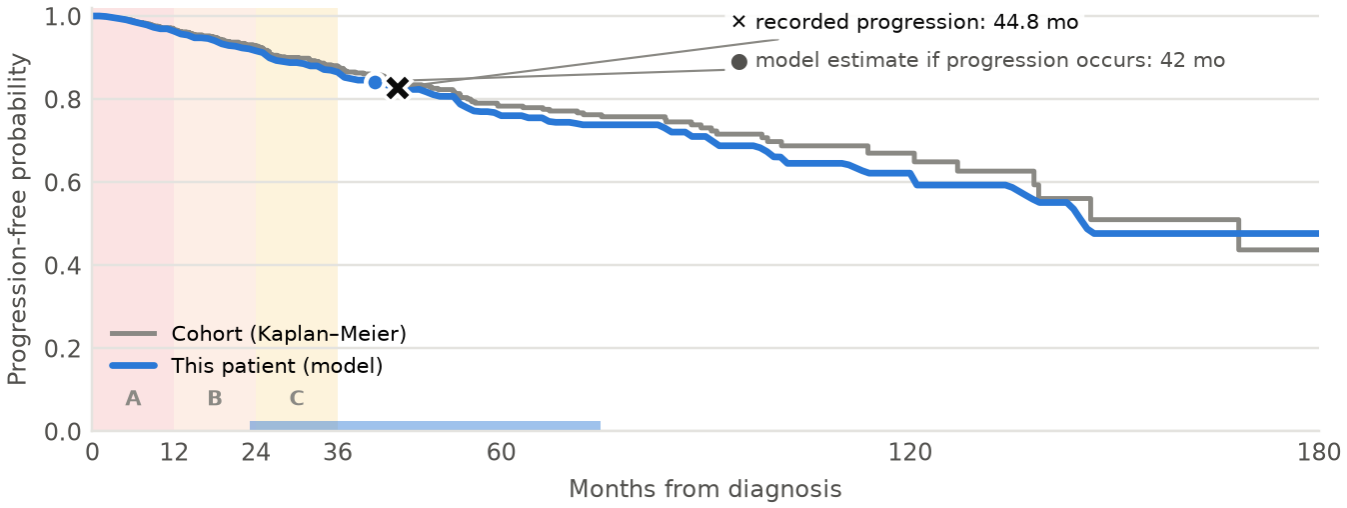
Window > 36 months · IQR 23–75 mo if progression occurs

Recorded progression

44.8 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 3 mo



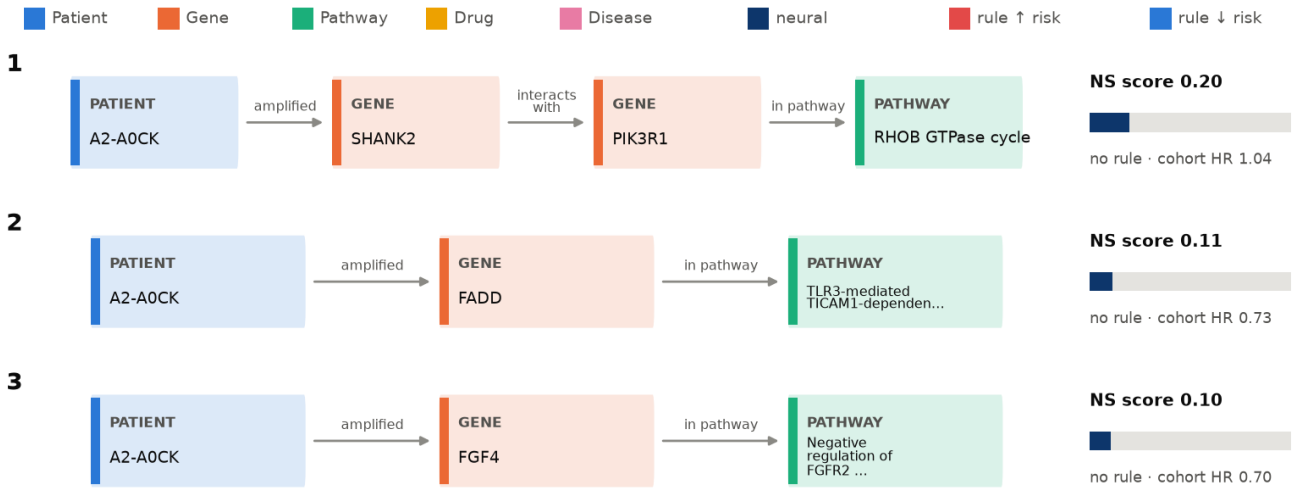
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a LumA tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **72nd risk percentile** (high tier). The strongest learned formula that holds for them (*Subtype(p, LumA) ⇒ Progress(p)*) lowers the progression hazard (weight -0.010 per SD). The highest-scoring knowledge-graph path runs from **SHANK2** to **RHOB GTPase cycle**. Progression was recorded at 44.8 months; the model's estimate given progression was 42 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

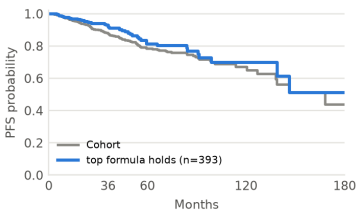


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A0CK** amplified → **SHANK2** interacts with → **PIK3R1** in pathway → **RHOB GTPase cycle**	33	No learned formula fires on this path; its score is neural only. cohort: HR 1.04, n = 132 carriers	**0.20**
2	**TCGA-A2-A0CK** amplified → **FADD** in pathway → **TLR3-mediated TICAM1-dependent programmed cell death**	19	No learned formula fires on this path; its score is neural only. cohort: HR 0.73, n = 121 carriers	**0.11**
3	**TCGA-A2-A0CK** amplified → **FGF4** in pathway → **Negative regulation of FGFR2 signaling**	17	No learned formula fires on this path; its score is neural only. cohort: HR 0.70, n = 128 carriers	**0.10**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

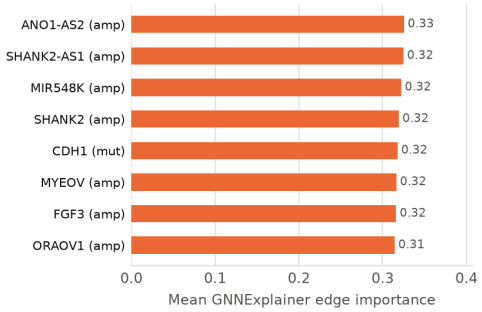
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.G278*	17	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.33
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
MIR548K	High-level amplification	GISTIC ±2	0	0.32
SHANK2	High-level amplification	GISTIC ±2	1	0.32
MYEOV	High-level amplification	GISTIC ±2	0	0.32
FGF3	High-level amplification	GISTIC ±2	28	0.32
ORAOV1	High-level amplification	GISTIC ±2	0	0.31
FGF4	High-level amplification	GISTIC ±2	48	0.31
FADD	High-level amplification	GISTIC ±2	14	0.31
PPFIA1	High-level amplification	GISTIC ±2	6	0.31



15 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); FGF4 → Pentosan polysulfate (DB00686); FGF4 → Heparin (DB01109); FGF4 → TP-508 (DB05309); FGF19 → Heparin (DB01109); ANO1 → Crofelemer (DB04941). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-PE-A5DC	72	III	LumA	0.87	no progression, followed 47.0 mo
TCGA-AR-A2LK	62	III	LumB	0.82	progressed at 42.9 mo (group D)
TCGA-AC-A4ZE	63	II	LumA	0.80	no progression, followed 29.3 mo
TCGA-A7-A4SB	56	III	missing	0.75	no progression, followed 13.7 mo
TCGA-A2-A1FX	61	III	LumB	0.74	no progression, followed 60.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8523/3**

Sample: **TCGA-B6-A0IN-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **0.97** · MSIsensor score: **0.12** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

19% by 36 mo

12 mo 5% · 24 mo 12% · 60 mo 33%

Model-estimated progression time

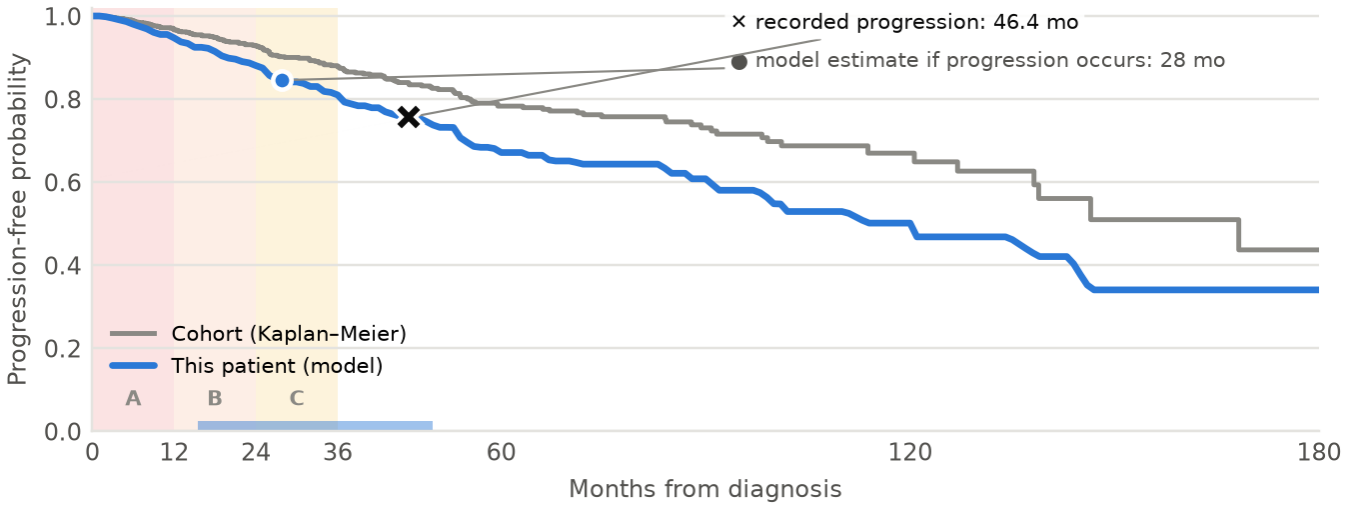
≈ 28 months

Window 24–36 months · IQR 15–50 mo
if progression occurs

Recorded progression

46.4 months

Group D · Late progression
✗ **recorded window differs** · |error| 19 mo



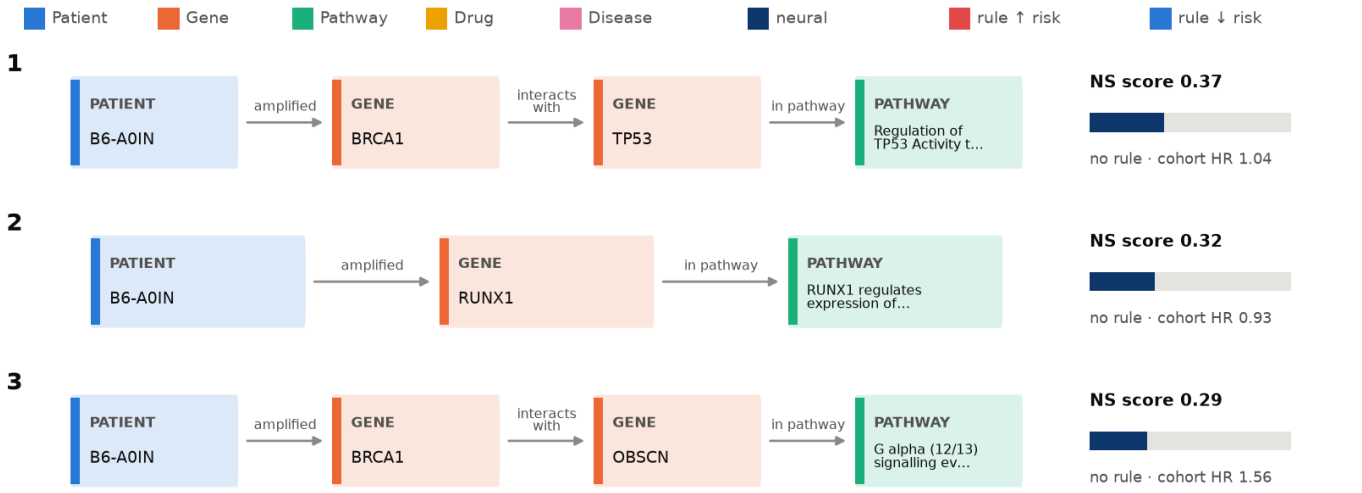
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a LumB tumour, AJCC stage X (T1, missing, MX). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **BRCA1** to **Regulation of TP53 Activity through Association with Co-factors**. Progression was recorded at 46.4 months; the model's estimate given progression was 28 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

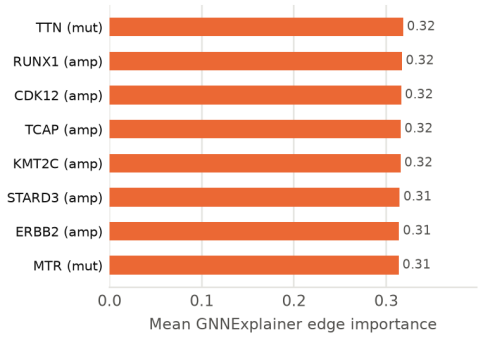
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0IN** amplified → **BRCA1** interacts with → **TP53** in pathway → **Regulation of TP53 Activity through Association with Co-factors**	62	No learned formula fires on this path; its score is neural only. cohort: HR 1.04, n = 368 carriers	**0.37**
2	**TCGA-B6-A0IN** amplified → **RUNX1** in pathway → **RUNX1 regulates expression of components of tight junctions**	54	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 84 carriers	**0.32**
3	**TCGA-B6-A0IN** amplified → **BRCA1** interacts with → **OBSCN** in pathway → **G alpha (12/13) signalling events**	48	No learned formula fires on this path; its score is neural only. cohort: HR 1.56, n = 119 carriers	**0.29**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.32
KMT2C · driver	High-level amplification	GISTIC ±2	4	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
BRCA1 · driver	High-level amplification	GISTIC ±2	29	0.31
GATA3 · driver	Mutation	p.S437Qfs*70	13	0.31
TTN	Mutation	p.D30372Y, p.P26895H, p.R19963*	2	0.32
CDK12	High-level amplification	GISTIC ±2	2	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.31
MTR	Mutation	p.A4T	18	0.31
PPP1R1B	High-level amplification	GISTIC ±2	1	0.31
PNMT	High-level amplification	GISTIC ±2	16	0.31



18 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 26 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0WS	58	X	LumA	0.54	no progression, followed 97.5 mo
TCGA-A1-A0SN	50	II	LumB	0.53	no progression, followed 39.3 mo
TCGA-EW-A1OX	43	II	LumB	0.52	no progression, followed 30.0 mo
TCGA-EW-A1IY	38	I	LumB	0.50	no progression, followed 8.5 mo
TCGA-AC-A3QQ	54	I	LumA	0.50	no progression, followed 24.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**26**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A24L-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **1.33** · MSIsensor score: **0.09** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

49th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 19%

Model-estimated progression time

≈ 52 months

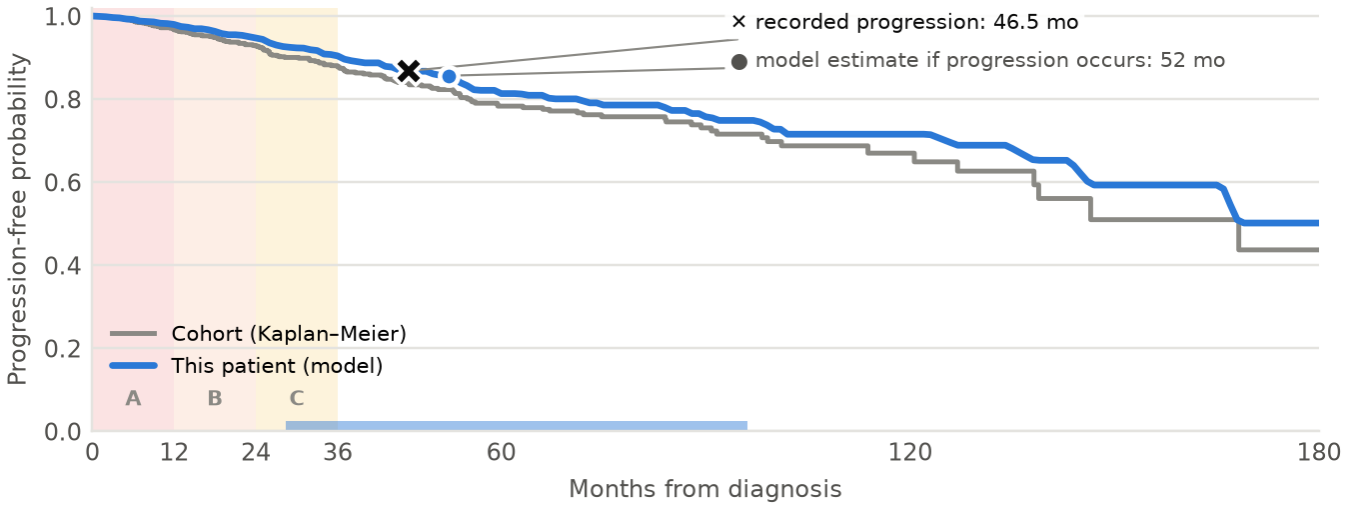
Window > 36 months · IQR 28–96 mo if progression occurs

Recorded progression

46.5 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 6 mo



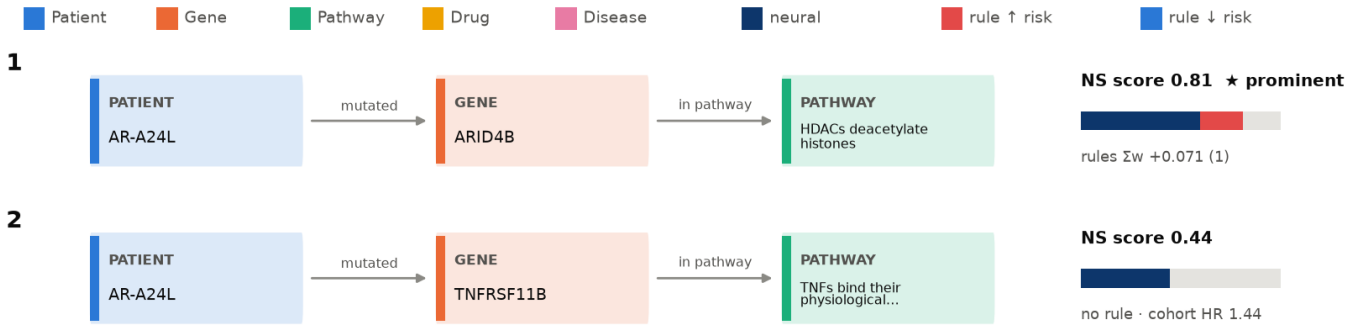
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 26-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **49th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Age(p) < 40 ⇒ Progress(p)*) raises the progression hazard (weight +0.084 per SD). The highest-scoring knowledge-graph path runs from **ARID4B** to **HDACs deacetylate histones**; the learned formulas it instantiates point to faster progression (Σw +0.071). Progression was recorded at 46.5 months; the model's estimate given progression was 52 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

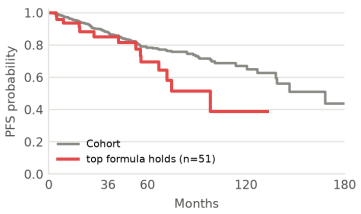


Only 1 of the 4 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AR-A24L** mutated → **ARID4B** in pathway → **HDACs deacetylate histones**	100	↑ **faster progression** · Σw +0.071 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ⇒ Progress(p) cohort: HR 1.84, n = 97 carriers	**0.81**
2	**TCGA-AR-A24L** mutated → **TNFRSF11B** in pathway → **TNFs bind their physiological receptors**	74	No learned formula fires on this path; its score is neural only. cohort: HR 1.44, n = 111 carriers	**0.44**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

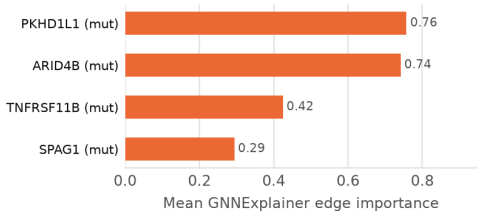
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.084 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	51 pts
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ⇒ Progress(p) [KG]	single atom	97 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PKHD1L1	Mutation	p.A2130=	0	0.76
ARID4B	Mutation	p.E734Dfs*2	3	0.74
TNFRSF11B	Mutation	p.A318G	1	0.42
SPAG1	Mutation	p.*366*	0	0.29



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1EW	38	II	LumA	0.96	progressed at 55.7 mo (group D)
TCGA-E9-A245	47	II	LumA	0.93	no progression, followed 0.9 mo
TCGA-BH-A0DQ	42	II	LumA	0.93	no progression, followed 3.2 mo
TCGA-BH-A0E1	52	II	LumA	0.90	no progression, followed 15.7 mo
TCGA-C8-A273	29	II	LumA	0.87	no progression, followed 16.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

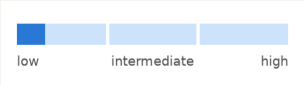
Sample: **TCGA-A2-A04V-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **0.9** · MSIsensor score: **0.04** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

10th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 15%

Model-estimated progression time

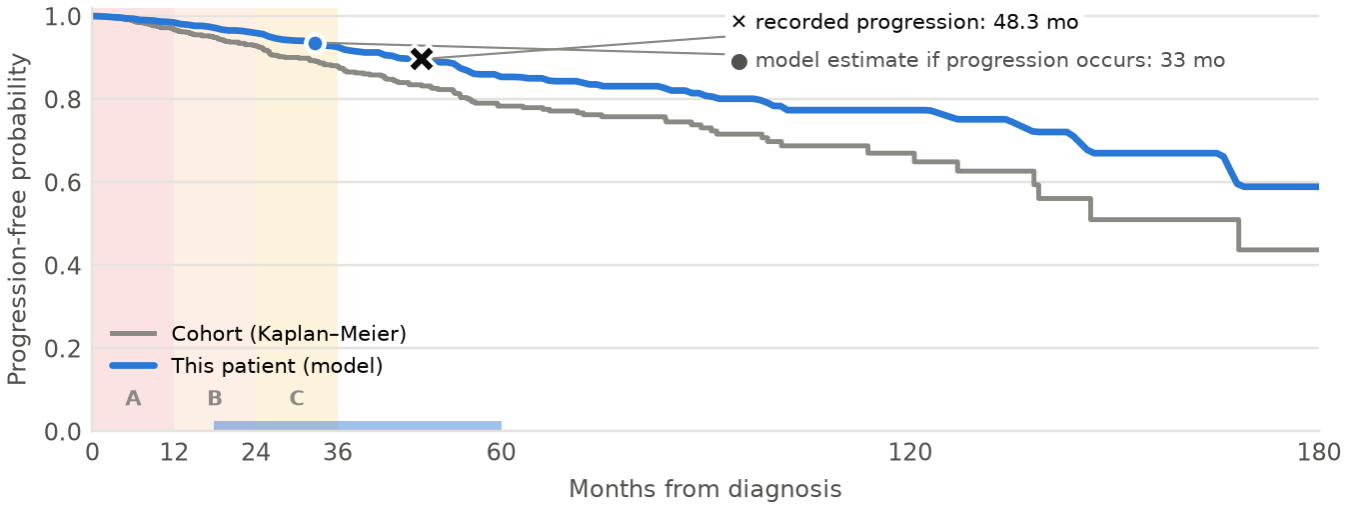
≈ 33 months

Window 24–36 months · IQR 18–60 mo
if progression occurs

Recorded progression

48.3 months

Group D · Late progression
✗ **recorded window differs** · |error| 16 mo



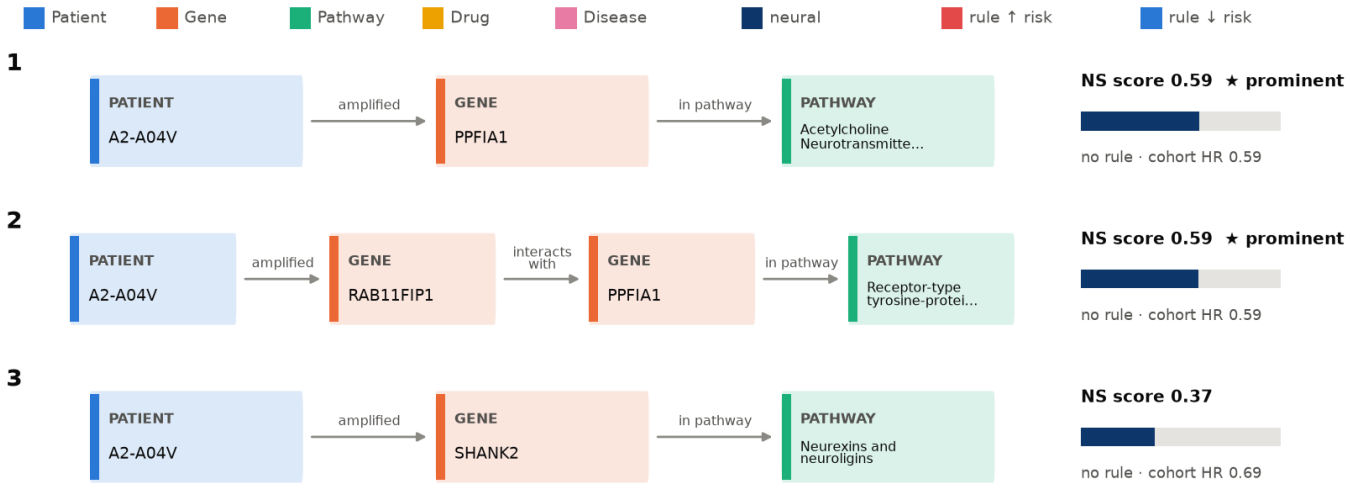
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **10th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **PPFIA1** to **Acetylcholine Neurotransmitter Release Cycle**. Progression was recorded at 48.3 months; the model's estimate given progression was 33 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

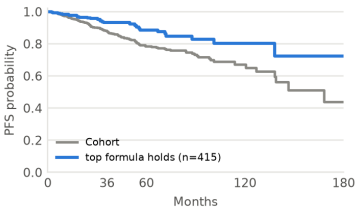


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A04V** amplified → **PPFIA1** in pathway → **Acetylcholine Neurotransmitter Release Cycle**	99	No learned formula fires on this path; its score is neural only. cohort: HR 0.59, n = 129 carriers	**0.59**
2 ★	**TCGA-A2-A04V** amplified → **RAB11FIP1** interacts with → **PPFIA1** in pathway → **Receptor-type tyrosine-protein phosphatases**	98	No learned formula fires on this path; its score is neural only. cohort: HR 0.59, n = 129 carriers	**0.59**
3	**TCGA-A2-A04V** amplified → **SHANK2** in pathway → **Neurexins and neuroligins**	61	No learned formula fires on this path; its score is neural only. cohort: HR 0.69, n = 210 carriers	**0.37**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

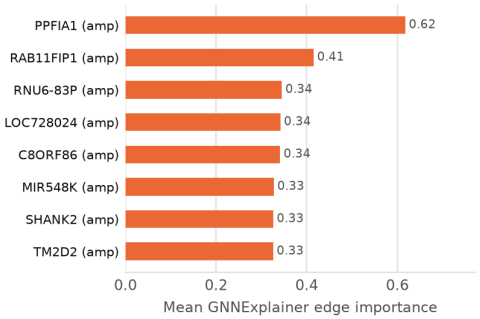
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
+0.084 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	51 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
-0.032 ↓ risk	NStage(p, N0) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	194 pts
+0.023 ↑ risk	Radiation(p, no) ⇒ Progress(p)	single atom	348 pts
-0.023 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX1 Expression and Activity') ⇒ Progress(p) [KG]	mined pair	162 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
PIK3CA · driver	Mutation	p.G118D	70	0.31
PPFIA1	High-level amplification	GISTIC ±2	6	0.62
RAB11FIP1	High-level amplification	GISTIC ±2	0	0.41
RNU6-83P	High-level amplification	GISTIC ±2	0	0.34
LOC728024	High-level amplification	GISTIC ±2	0	0.34
C8ORF86	High-level amplification	GISTIC ±2	0	0.34
MIR548K	High-level amplification	GISTIC ±2	0	0.33
SHANK2	High-level amplification	GISTIC ±2	1	0.33
TM2D2	High-level amplification	GISTIC ±2	0	0.33
LETM2	High-level amplification	GISTIC ±2	0	0.32



56 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 64 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-GM-A3NW	63	II	LumA	0.87	no progression, followed 110.5 mo
TCGA-C8-A12O	50	II	LumA	0.84	no progression, followed 12.7 mo
TCGA-BH-A201	64	I	LumA	0.83	no progression, followed 28.1 mo
TCGA-E2-A15R	64	II	LumA	0.82	no progression, followed 56.9 mo
TCGA-D8-A1XF	45	II	LumB	0.82	no progression, followed 15.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**67**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EAS**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

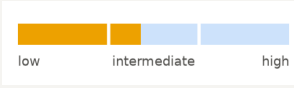
Sample: **TCGA-BH-A1EX-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **0.8** · MSIsensor score: **0.23** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

45th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

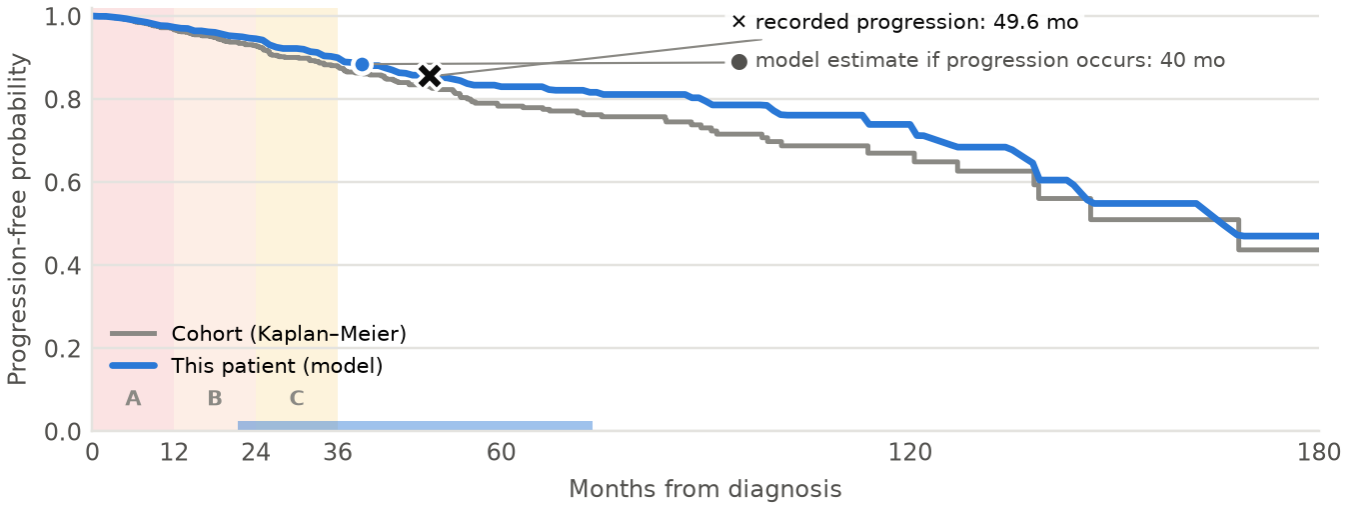
≈ 40 months

Window > 36 months · IQR 21–73 mo if progression occurs

Recorded progression

49.6 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 10 mo



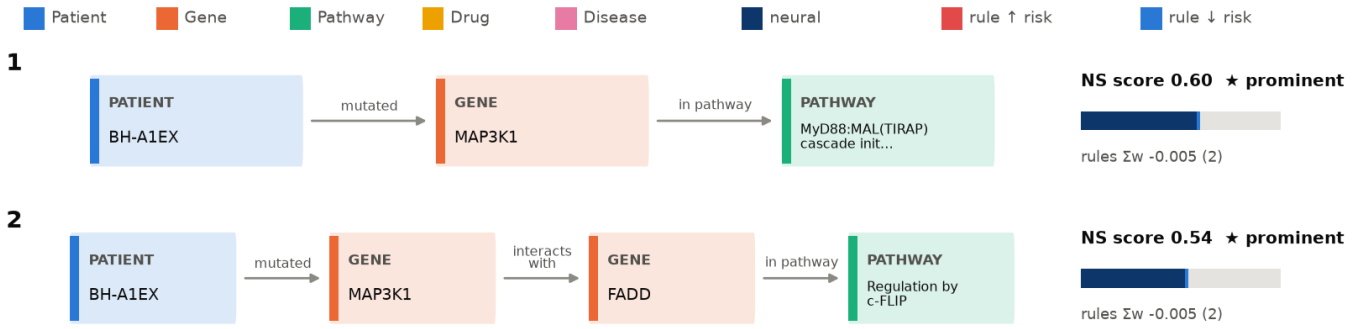
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 67-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **45th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Subtype(p, LumA)* ∧ *Mutated(p, MAP3K1)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.003 per SD). The highest-scoring knowledge-graph path runs from **MAP3K1** to **MyD88:MAL(TIRAP) cascade initiated on plasma membrane**; the learned formulas it instantiates point to slower progression (Σw -0.005). Progression was recorded at 49.6 months; the model's estimate given progression was 40 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

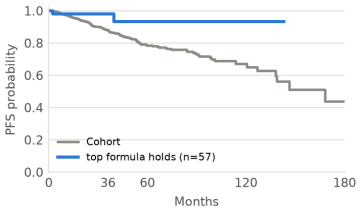


2 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1EX** mutated → **MAP3K1** in pathway → **MyD88:MAL(TIRAP) cascade initiated on plasma membrane**	97	↓ **slower progression** · Σw -0.005 over 2 formula(s) Subtype(p, LumA) ∧ Mutated(p, MAP3K1) ⇒ Progress(p) cohort: HR 0.47, n = 85 carriers	**0.60**
2 ★	**TCGA-BH-A1EX** mutated → **MAP3K1** interacts with → **FADD** in pathway → **Regulation by c-FLIP**	87	↓ **slower progression** · Σw -0.005 over 2 formula(s) Subtype(p, LumA) ∧ Mutated(p, MAP3K1) ⇒ Progress(p) cohort: HR 0.91, n = 112 carriers	**0.54**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

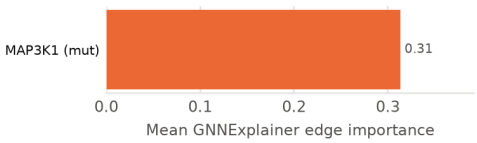
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.003 ↓ risk	Subtype(p, LumA) ∧ Mutated(p, MAP3K1) ⇒ Progress(p)	mined pair	56 pts
-0.002 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'MyD88:MAL(TIRAP) cascade initiated on plasma membrane') ⇒ Progress(p) [KG]	mined pair	57 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP3K1 · driver	Mutation	p.W201R	13	0.31



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1P6	64	II	LumA	0.96	no progression, followed 18.5 mo
TCGA-BH-A0E1	52	II	LumA	0.94	no progression, followed 15.7 mo
TCGA-E9-A245	47	II	LumA	0.92	no progression, followed 0.9 mo
TCGA-BH-A0DH	63	II	LumA	0.90	no progression, followed 38.0 mo
TCGA-D8-A1X9	66	II	LumA	0.89	no progression, followed 23.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**AFR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

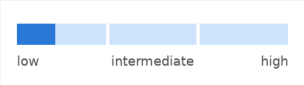
Sample: **TCGA-B6-A402-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.48** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

14th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 14%

Model-estimated progression time

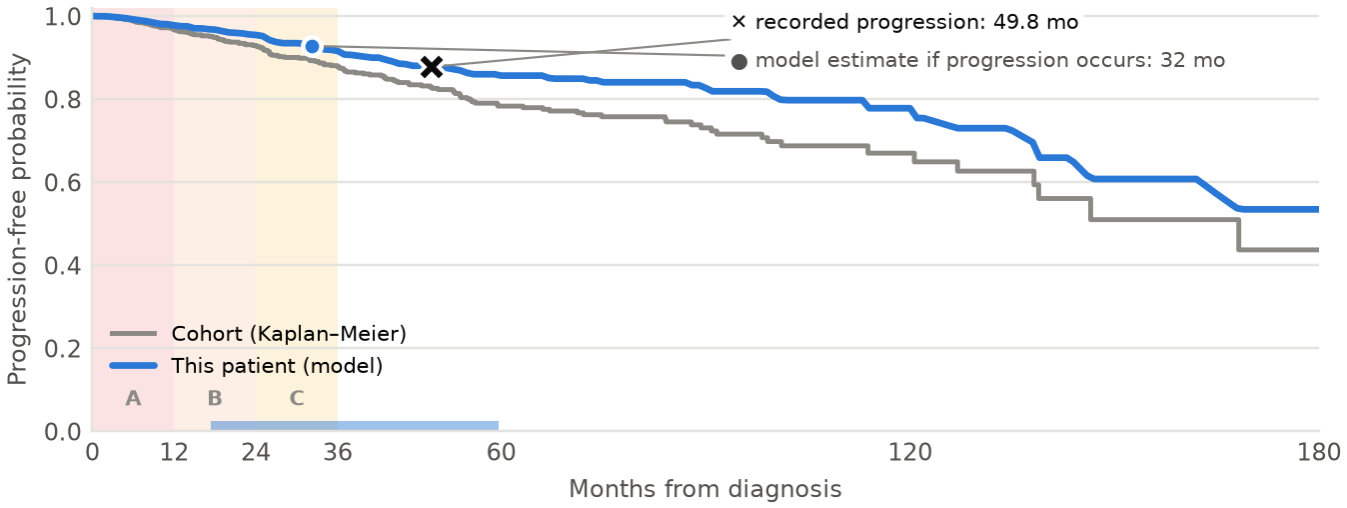
≈ 32 months

Window 24–36 months · IQR 17–60 mo
if progression occurs

Recorded progression

49.8 months

Group D · Late progression
✗ recorded window differs · |error| 18 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a subtype-unassigned tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **14th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.199 per SD). Progression was recorded at 49.8 months; the model's estimate given progression was 32 months.

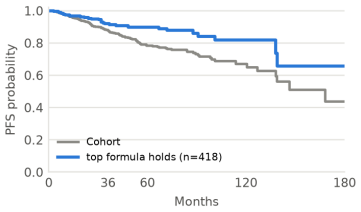
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

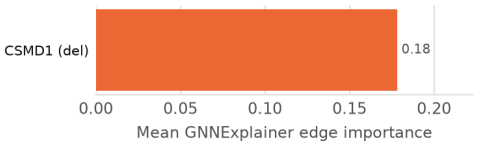
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.021 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	139 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Stage(p, I) ⇒ Progress(p)	mined pair	138 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CSMD1	Homozygous deletion	GISTIC ±2	0	0.18



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A14S	65	I	missing	0.94	no progression, followed 33.2 mo
TCGA-BH-A0B8	64	I	missing	0.91	no progression, followed 51.6 mo
TCGA-AR-A252	50	I	LumA	0.82	no progression, followed 93.3 mo
TCGA-A2-A04R	36	I	LumB	0.79	no progression, followed 121.9 mo
TCGA-A2-A0CQ	62	I	LumA	0.78	no progression, followed 88.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

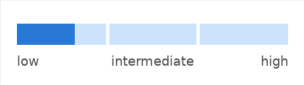
Sample: **TCGA-HN-A2OB-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

21st

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

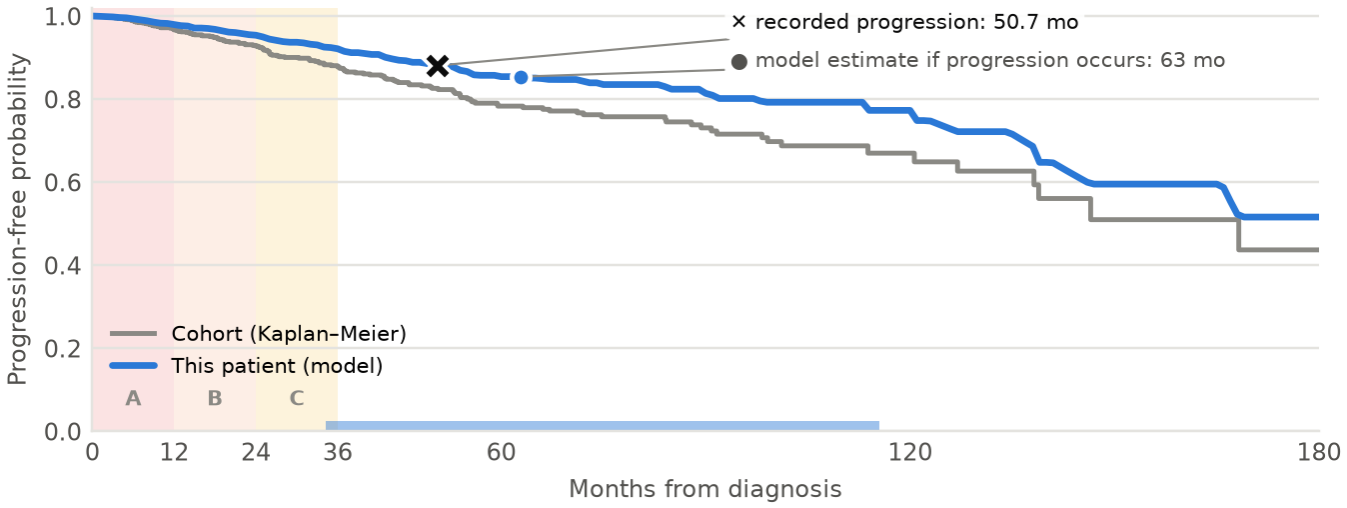
≈ 63 months

Window > 36 months · IQR 34–115 mo
if progression occurs

Recorded progression

50.7 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 12 mo



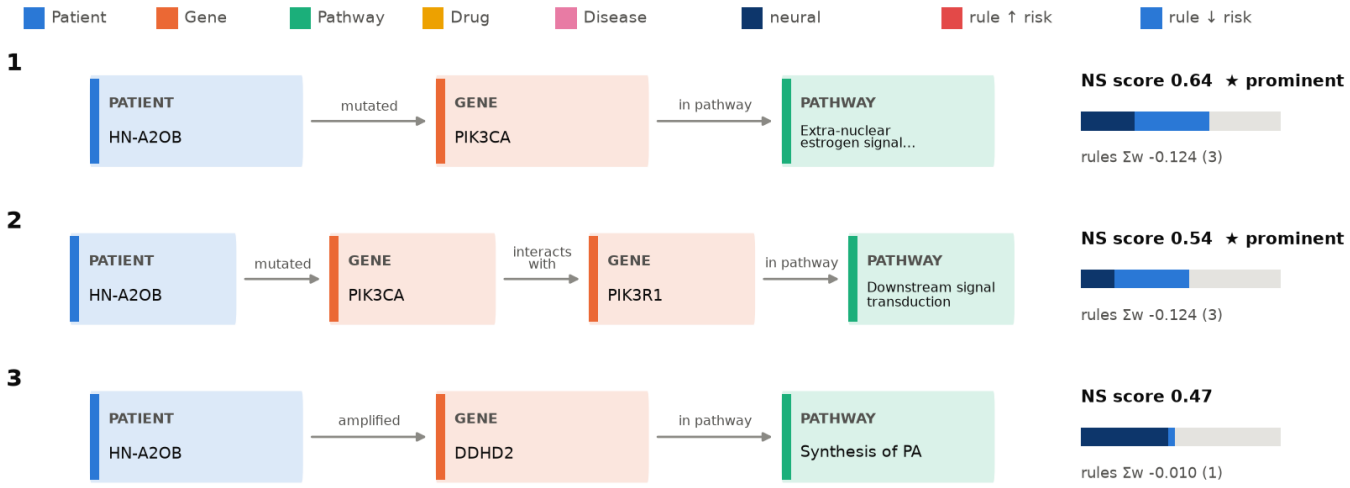
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **21st risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Extra-nuclear estrogen signaling**; the learned formulas it instantiates point to slower progression (Σw -0.124). Progression was recorded at 50.7 months; the model's estimate given progression was 63 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

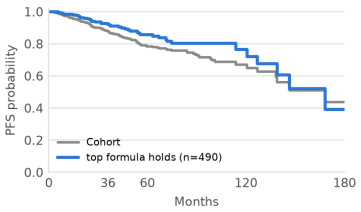


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-HN-A2OB** mutated → **PIK3CA** in pathway → **Extra-nuclear estrogen signaling**	45	↓ **slower progression** · Σw -0.124 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.76, n = 435 carriers	**0.64**
2 ★	**TCGA-HN-A2OB** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Downstream signal transduction**	28	↓ **slower progression** · Σw -0.124 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.98, n = 373 carriers	**0.54**
3	**TCGA-HN-A2OB** amplified → **DDHD2** in pathway → **Synthesis of PA**	73	↓ **slower progression** · Σw -0.010 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) cohort: HR 0.80, n = 189 carriers	**0.47**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

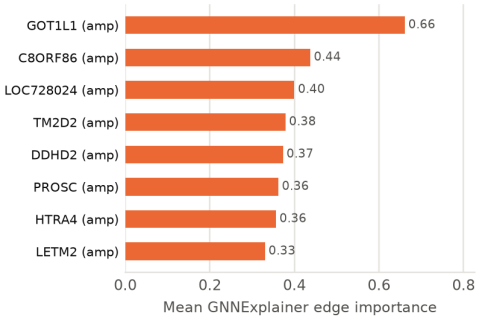
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.034 ↓ risk	LowAneuploidy(p) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Deactivation of the beta-catenin transactivating complex') ⇒ Progress(p) [KG]	RL-proposed	41 pts
-0.017 ↓ risk	Histology(p, lobular) ⇒ Progress(p)	single atom	154 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BAP1 · driver	Mutation	p.I214M	9	0.32
PIK3CA · driver	Mutation	p.E542K	70	0.31
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.66
C8ORF86	High-level amplification	GISTIC ±2	0	0.44
LOC728024	High-level amplification	GISTIC ±2	0	0.40
TM2D2	High-level amplification	GISTIC ±2	0	0.38
DDHD2	High-level amplification	GISTIC ±2	1	0.37
PROSC	High-level amplification	GISTIC ±2	0	0.36
HTRA4	High-level amplification	GISTIC ±2	0	0.36
LETM2	High-level amplification	GISTIC ±2	0	0.33
RNF5P1	High-level amplification	GISTIC ±2	0	0.33



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109) … and 52 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0ES	52	II	LumA	0.81	no progression, followed 72.0 mo
TCGA-BH-A1ES	35	II	missing	0.80	progressed at 113.8 mo (group D)
TCGA-E2-A1IE	61	II	LumA	0.77	progressed at 59.5 mo (group D)
TCGA-WT-AB41	55	II	LumA	0.77	no progression, followed 53.0 mo
TCGA-BH-A0EI	51	II	LumA	0.74	no progression, followed 63.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AR-A0U2 · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A0U2-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **1.43** · MSIsensor score: **1.19** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

79th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 27%

Model-estimated progression time

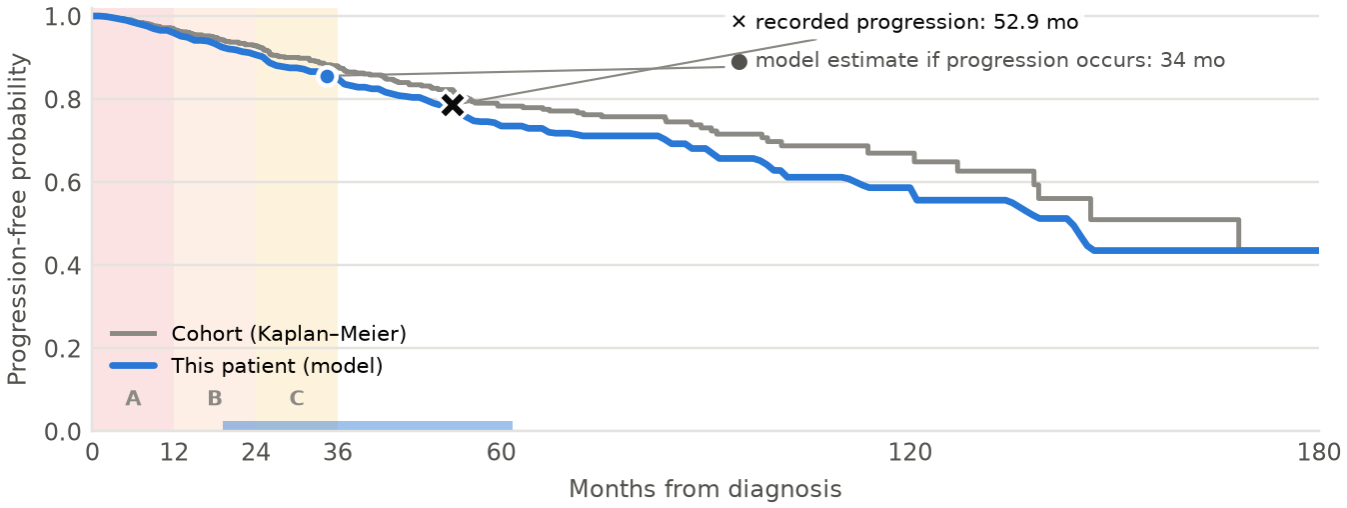
≈ 34 months

Window 24–36 months · IQR 19–62 mo
if progression occurs

Recorded progression

52.9 months

Group D · Late progression
✗ **recorded window differs** · |error| 19 mo



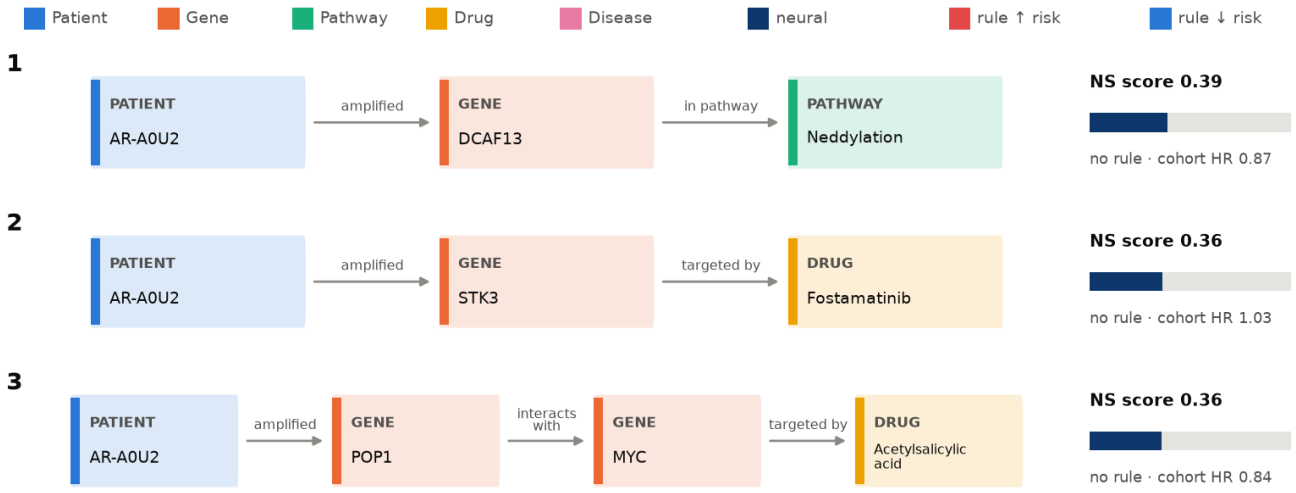
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a LumB tumour, AJCC stage III (T2, N2, M0). The fused neuro-symbolic model places them in the **79th risk percentile** (high tier). The strongest learned formula that holds for them (*HighAneuploidy(p)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.005 per SD). The highest-scoring knowledge-graph path runs from **DCAF13** to **Neddylation**. Progression was recorded at 52.9 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

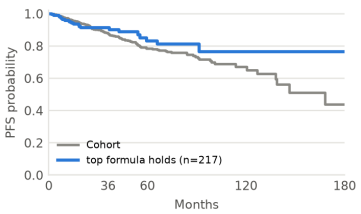


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A0U2** amplified → **DCAF13** in pathway → **Neddylation**	64	No learned formula fires on this path; its score is neural only. cohort: HR 0.87, n = 151 carriers	**0.39**
2	**TCGA-AR-A0U2** amplified → **STK3** targeted by → **Fostamatinib**	60	No learned formula fires on this path; its score is neural only. cohort: HR 1.03, n = 390 carriers	**0.36**
3	**TCGA-AR-A0U2** amplified → **POP1** interacts with → **MYC** targeted by → **Acetylsalicylic acid**	60	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 435 carriers	**0.36**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

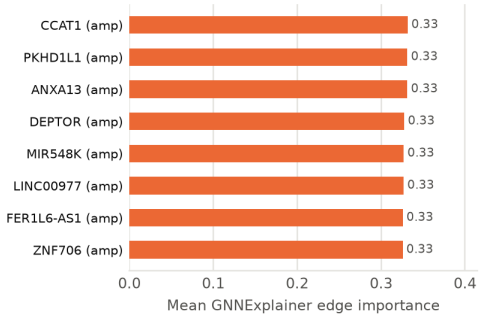
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
GATA3 · driver	Mutation	p.P408Afs*99, p.R364S	13	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TP53 · driver	Mutation	p.C176F	57	0.31
CCAT1	High-level amplification	GISTIC ±2	0	0.33
PKHD1L1	High-level amplification	GISTIC ±2	0	0.33
ANXA13	High-level amplification	GISTIC ±2	0	0.33
DEPTOR	High-level amplification	GISTIC ±2	0	0.33
MIR548K	High-level amplification	GISTIC ±2	0	0.33
LINC00977	High-level amplification	GISTIC ±2	0	0.33
FER1L6-AS1	High-level amplification	GISTIC ±2	0	0.33
ZNF706	High-level amplification	GISTIC ±2	1	0.33



197 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404) … and 33 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A130	52	III	LumB	0.83	no progression, followed 12.2 mo
TCGA-AN-A0AS	70	III	LumA	0.81	no progression, followed 0.3 mo
TCGA-A2-A0YH	53	III	LumB	0.80	no progression, followed 21.7 mo
TCGA-BH-A0HX	54	II	LumA	0.80	no progression, followed 27.3 mo
TCGA-A8-A08P	70	III	LumB	0.79	no progression, followed 31.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**31**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8502/3**

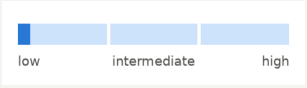
Sample: **TCGA-AO-A03U-01** · Aneuploidy score: **1** · TMB (nonsyn., /Mb): **0.17** · MSIsensor score: **0.01** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

4th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 12%

Model-estimated progression time

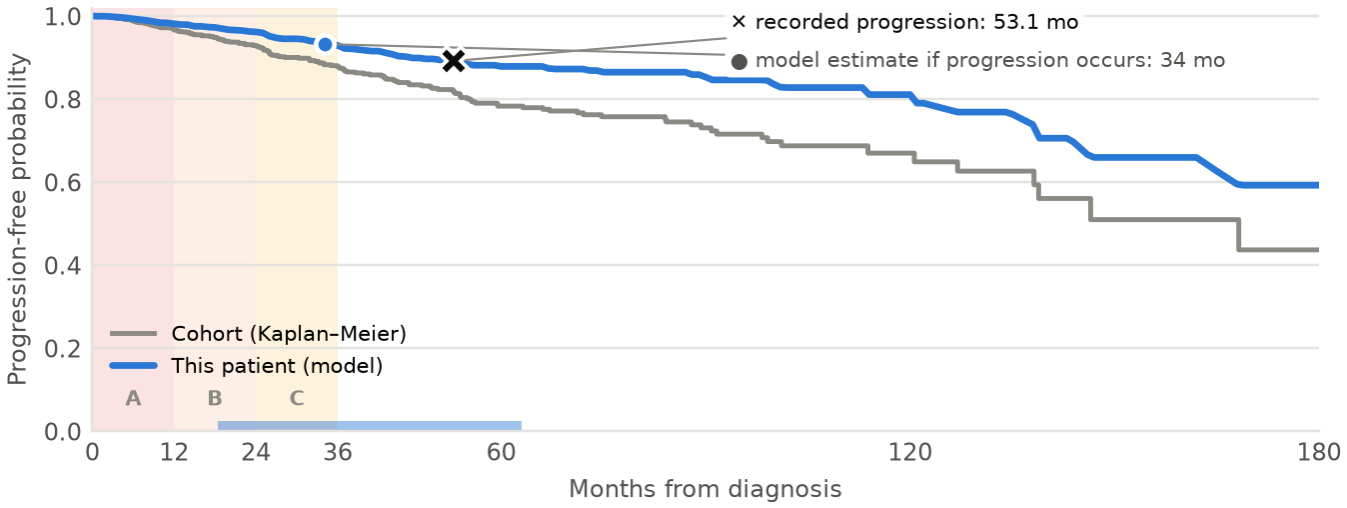
≈ 34 months

Window 24–36 months · IQR 18–63 mo
if progression occurs

Recorded progression

53.1 months

Group D · Late progression
✗ recorded window differs · |error| 19 mo



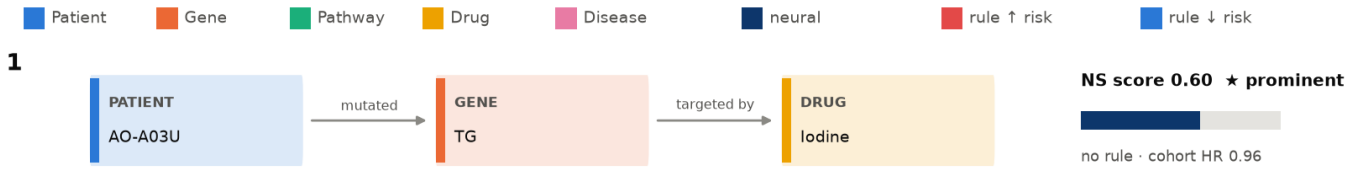
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 31-year-old patient has a Normal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **4th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **TG** to **Iodine**. Progression was recorded at 53.1 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

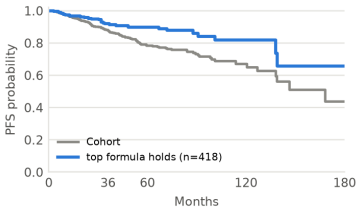


1 of 1 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AO-A03U** mutated → **TG** targeted by → **Iodine**	100	No learned formula fires on this path; its score is neural only. cohort: HR 0.96, n = 123 carriers	**0.60**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

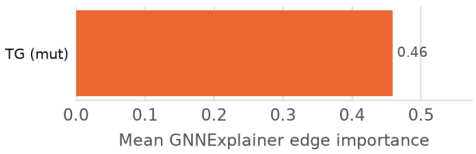
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.021 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	139 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Stage(p, I) ⇒ Progress(p)	mined pair	138 pts
+0.008 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TG	Mutation	p.R492=	1	0.46



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TG → Iodine (DB05382). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A1-A0SB	70	I	Normal	0.84	no progression, followed 8.5 mo
TCGA-D8-A27E	66	I	LumA	0.72	no progression, followed 17.4 mo
TCGA-BH-A8FZ	58	I	Normal	0.71	no progression, followed 18.9 mo
TCGA-A2-A04R	36	I	LumB	0.68	no progression, followed 121.9 mo
TCGA-A7-A3IY	71	I	missing	0.68	no progression, followed 11.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E2-A1L7-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **2.07** · MSIsensor score: **0.93** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

77th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

≈ 38 months

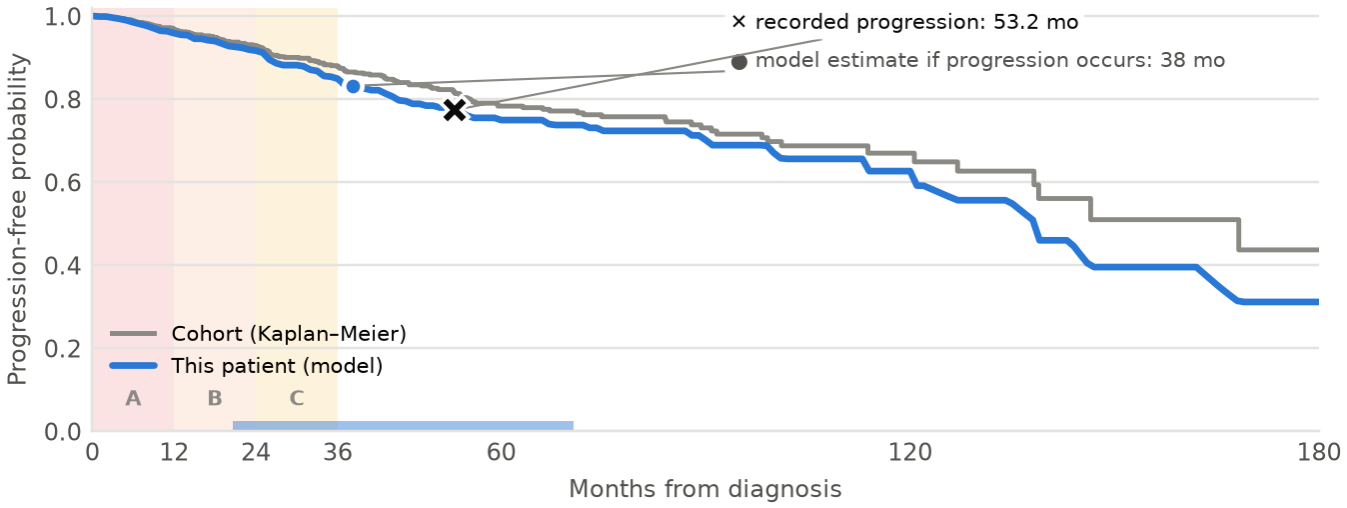
Window > 36 months · IQR 21–71 mo if progression occurs

Recorded progression

53.2 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 15 mo



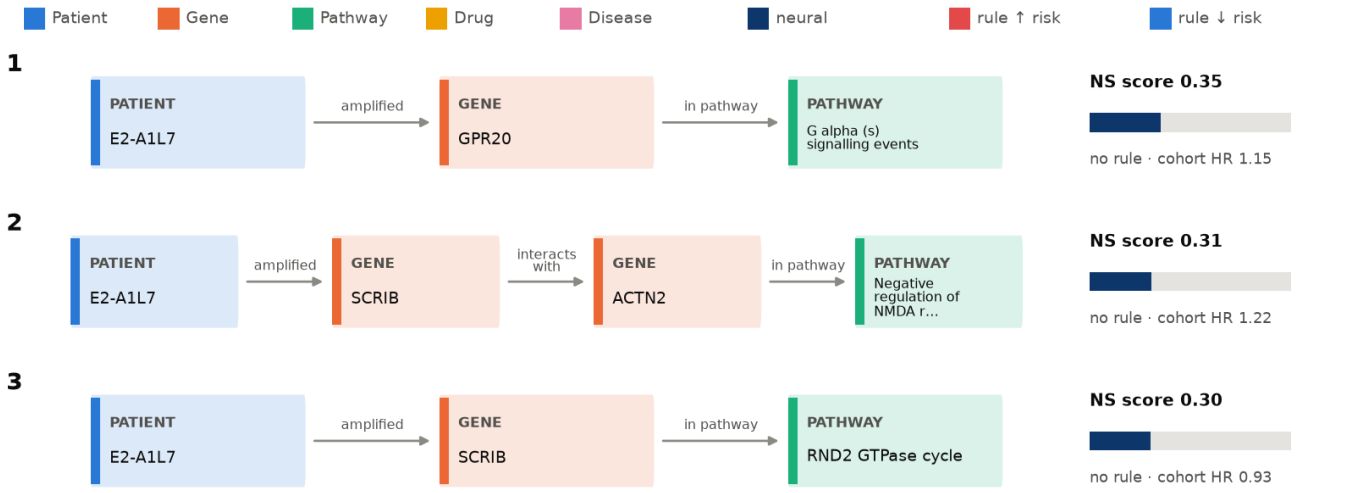
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a LumB tumour, AJCC stage III (T2, N2, M0). The fused neuro-symbolic model places them in the **77th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **GPR20** to **G alpha (s) signalling events**. Progression was recorded at 53.2 months; the model's estimate given progression was 38 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

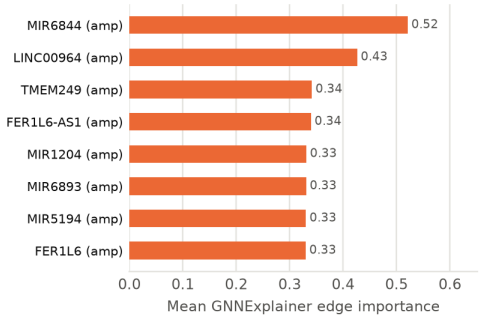
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-E2-A1L7** amplified → **GPR20** in pathway → **G alpha (s) signalling events**	59	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 206 carriers	**0.35**
2	**TCGA-E2-A1L7** amplified → **SCRIB** interacts with → **ACTN2** in pathway → **Negative regulation of NMDA receptor-mediated neuronal transmission**	51	No learned formula fires on this path; its score is neural only. cohort: HR 1.22, n = 95 carriers	**0.31**
3	**TCGA-E2-A1L7** amplified → **SCRIB** in pathway → **RND2 GTPase cycle**	50	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 121 carriers	**0.30**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BRCA1 · driver	Homozygous deletion	GISTIC ±2	29	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TP53 · driver	Mutation	p.I255F	57	0.31
MIR6844	High-level amplification	GISTIC ±2	0	0.52
LINC00964	High-level amplification	GISTIC ±2	0	0.43
TMEM249	High-level amplification	GISTIC ±2	0	0.34
FER1L6-AS1	High-level amplification	GISTIC ±2	0	0.34
MIR1204	High-level amplification	GISTIC ±2	0	0.33
MIR6893	High-level amplification	GISTIC ±2	0	0.33
MIR5194	High-level amplification	GISTIC ±2	0	0.33
FER1L6	High-level amplification	GISTIC ±2	0	0.33
MIR7112-2	High-level amplification	GISTIC ±2	0	0.33



208 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 78 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A09N	57	III	LumB	0.85	no progression, followed 1.0 mo
TCGA-AR-A251	51	III	Basal	0.81	no progression, followed 99.6 mo
TCGA-E9-A22E	56	III	LumB	0.79	no progression, followed 41.7 mo
TCGA-BH-A18U	72	III	LumB	0.79	no progression, followed 51.4 mo
TCGA-BH-A0DS	71	III	LumA	0.78	no progression, followed 2.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A256-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **5.67** · MSIsensor score: **4.53** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

61st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 20%

Model-estimated progression time

≈ 16 months

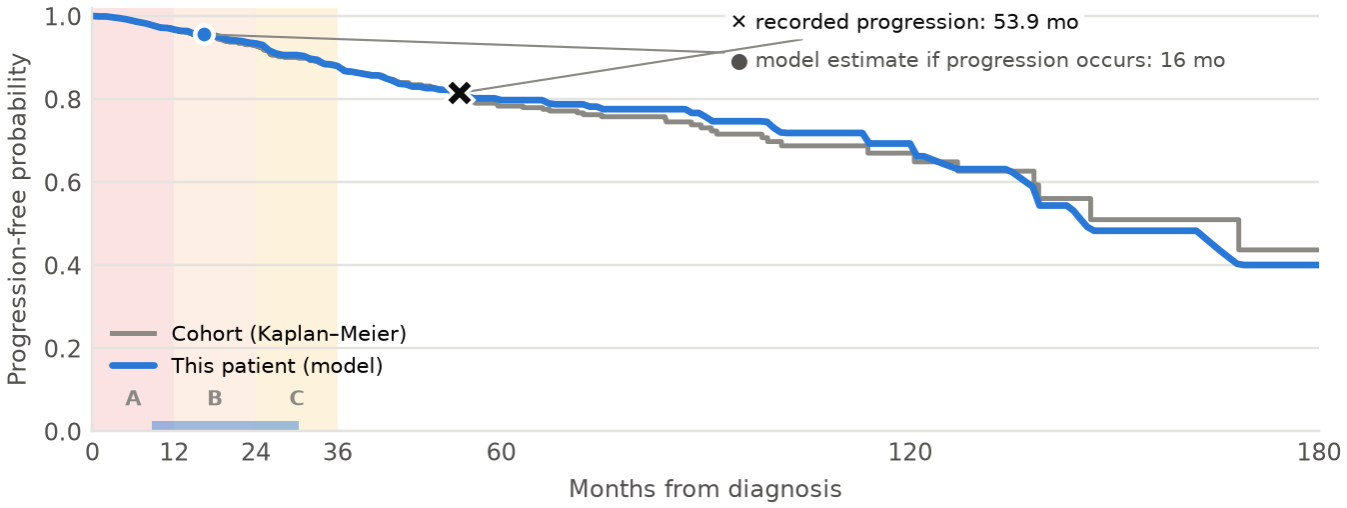
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

53.9 months

Group D · Late progression

✗ recorded window differs · |error| 38 mo



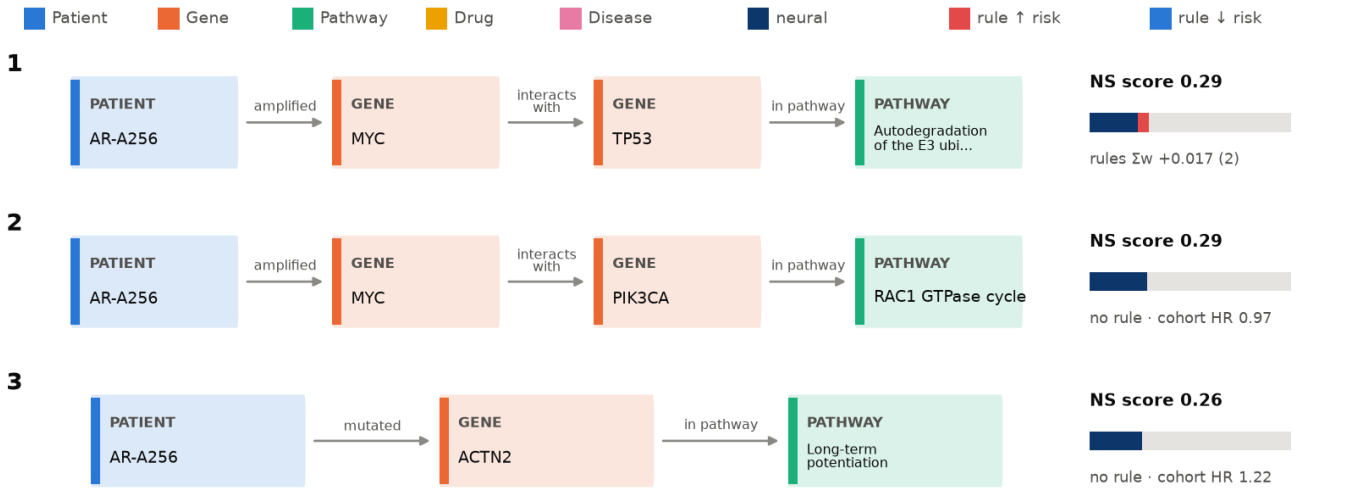
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **61st risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **MYC** to **Autodegradation of the E3 ubiquitin ligase COP1**; the learned formulas it instantiates point to faster progression (Σw +0.017). Progression was recorded at 53.9 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

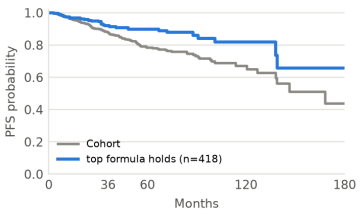


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A256** amplified → **MYC** interacts with → **TP53** in pathway → **Autodegradation of the E3 ubiquitin ligase COP1**	40	↑ **faster progression** · Σw +0.017 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Formation of the beta-catenin:TCF transactivating complex') ∧ Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ⇒ Progress(p) cohort: HR 1.01, n = 308 carriers	**0.29**
2	**TCGA-AR-A256** amplified → **MYC** interacts with → **PIK3CA** in pathway → **RAC1 GTPase cycle**	48	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 428 carriers	**0.29**
3	**TCGA-AR-A256** mutated → **ACTN2** in pathway → **Long-term potentiation**	43	No learned formula fires on this path; its score is neural only. cohort: HR 1.22, n = 95 carriers	**0.26**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

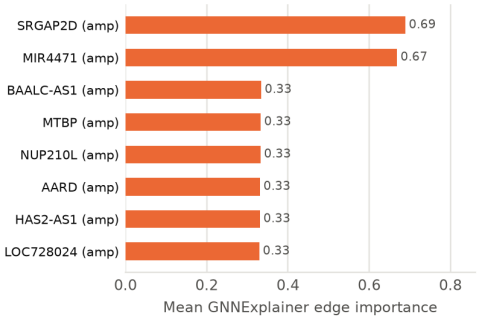
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
+0.012 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Formation of the beta-catenin:TCF transactivating complex') ∧ Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ⇒ Progress(p) [KG]	RL-proposed	35 pts
+0.006 ↑ risk	Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Ovarian tumor domain proteases') ∧ Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	RL-proposed	37 pts
-0.004 ↓ risk	Altered(p, g) ∧ InPathway(g, 'G alpha (s) signalling events') ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 regulates genes involved in megakaryocyte differentiation and platelet function') ∧ Altered(p, g) ∧ InPathway(g, 'FOXO-mediated transcription of oxidative stress, metabolic and neuronal genes') ⇒ Progress(p) [KG]	RL-proposed	37 pts
-0.000 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Pregnenolone biosynthesis') ⇒ Progress(p) [KG]	mined pair	69 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
TP53 · driver	Mutation	p.R110Sfs*14	57	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.69
MIR4471	High-level amplification	GISTIC ±2	0	0.67
BAALC-AS1	High-level amplification	GISTIC ±2	0	0.33
MTBP	High-level amplification	GISTIC ±2	0	0.33
NUP210L	High-level amplification	GISTIC ±2	0	0.33
AARD	High-level amplification	GISTIC ±2	0	0.33
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.33
LOC728024	High-level amplification	GISTIC ±2	0	0.33
OC90	High-level amplification	GISTIC ±2	0	0.33



204 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 89 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A3QA	33	II	Basal	0.86	no progression, followed 30.2 mo
TCGA-A7-A13D	46	II	Basal	0.86	no progression, followed 31.7 mo
TCGA-E2-A14R	62	II	Basal	0.85	progressed at 31.9 mo (group C)
TCGA-AN-A0FX	52	II	Basal	0.84	no progression, followed 0.3 mo
TCGA-AC-A6IW	73	II	Basal	0.84	no progression, followed 13.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8523/3**

Sample: **TCGA-BH-A1FR-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **0.47** · MSIsensor score: **0.11** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

74th

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 3% · 24 mo 8% · 60 mo 24%

Model-estimated progression time

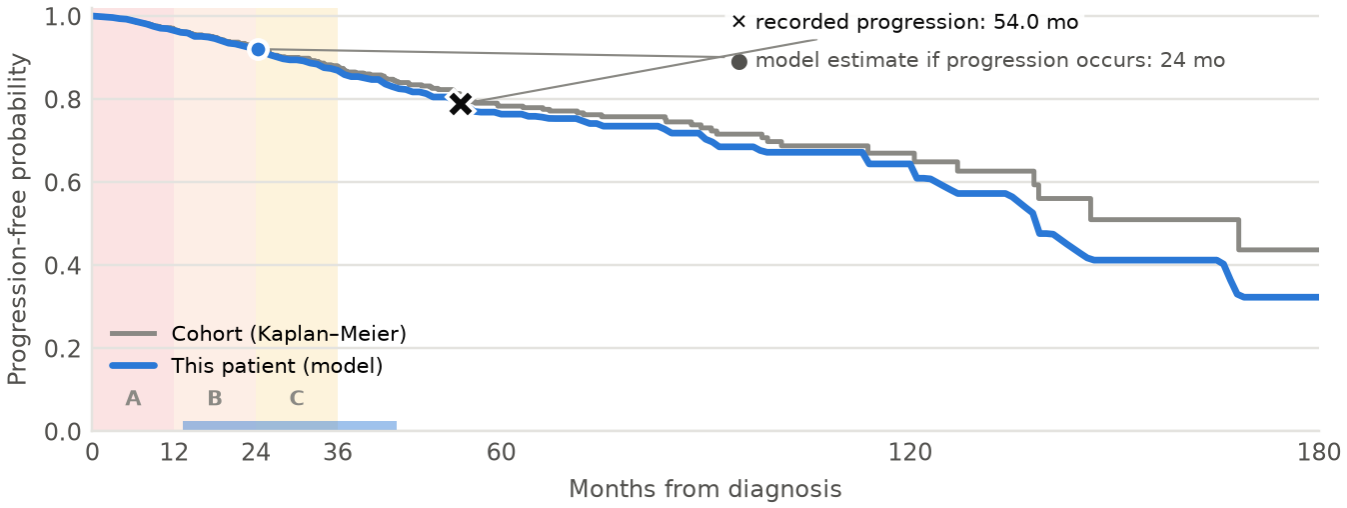
≈ 24 months

Window 24–36 months · IQR 13–45 mo
if progression occurs

Recorded progression

54.0 months

Group D · Late progression
✗ recorded window differs · |error| 30 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a LumA tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **74th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.113 per SD). Progression was recorded at 54.0 months; the model's estimate given progression was 24 months.

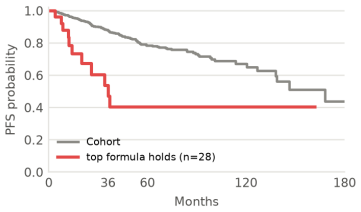
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.113 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	26 pts
+0.019 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts
+0.001 ↑ risk	Histology(p, other) ∧ LowTmb(p) ⇒ Progress(p)	mined pair	27 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XC	85	III	LumA	0.94	progressed at 12.4 mo (group B)
TCGA-A2-A3KD	47	III	LumA	0.76	no progression, followed 39.6 mo
TCGA-B6-A0RI	44	III	LumA	0.70	progressed at 36.2 mo (group D)
TCGA-B6-A0RO	71	III	LumA	0.66	no progression, followed 162.0 mo
TCGA-D8-A73W	79	III	LumA	0.66	progressed at 0.7 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**59**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

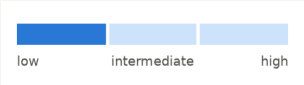
Sample: **TCGA-AO-A03N-01** · Aneuploidy score: **23** · TMB (nonsyn., /Mb): **3** · MSIsensor score: **0.23** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

33rd

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

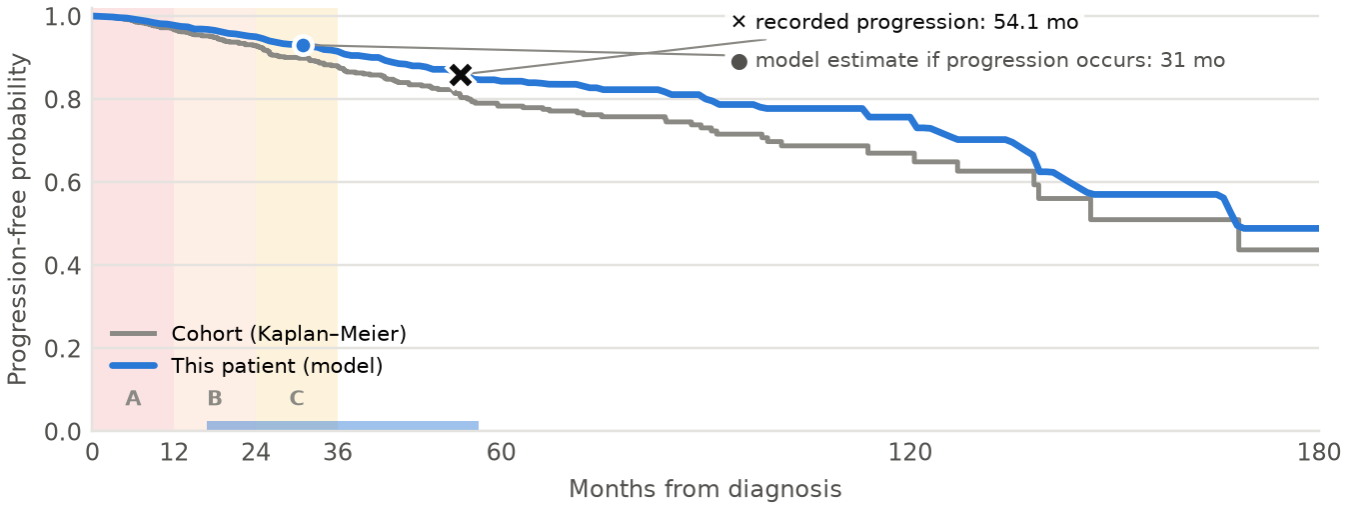
≈ 31 months

Window 24–36 months · IQR 17–57 mo
if progression occurs

Recorded progression

54.1 months

Group D · Late progression
✗ recorded window differs · |error| 23 mo



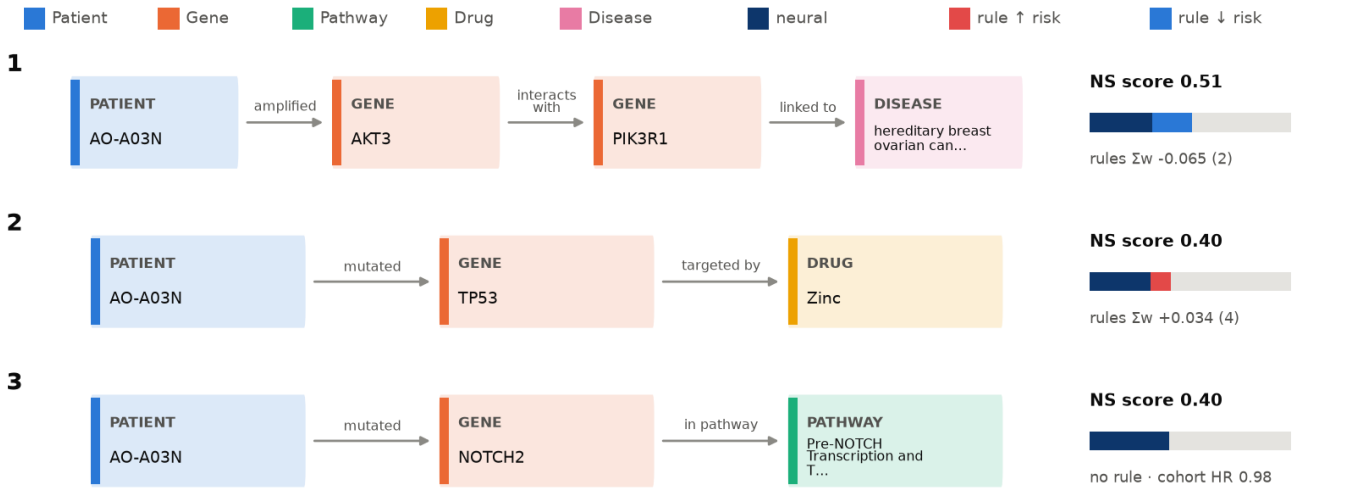
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 59-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **33rd risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **AKT3** to **hereditary breast ovarian cancer syndrome**; the learned formulas it instantiates point to slower progression (Σw -0.065). Progression was recorded at 54.1 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

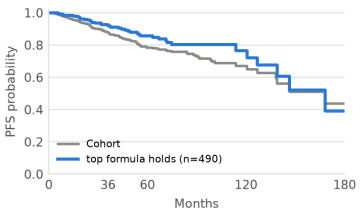


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AO-A03N** amplified → **AKT3** interacts with → **PIK3R1** linked to → **hereditary breast ovarian cancer syndrome**	52	↓ **slower progression** · Σw -0.065 over 2 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.99, n = 798 carriers	**0.51**
2	**TCGA-AO-A03N** mutated → **TP53** targeted by → **Zinc**	50	↑ **faster progression** · Σw +0.034 over 4 formula(s) Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) cohort: HR 1.06, n = 379 carriers	**0.40**
3	**TCGA-AO-A03N** mutated → **NOTCH2** in pathway → **Pre-NOTCH Transcription and Translation**	66	No learned formula fires on this path; its score is neural only. cohort: HR 0.98, n = 490 carriers	**0.40**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

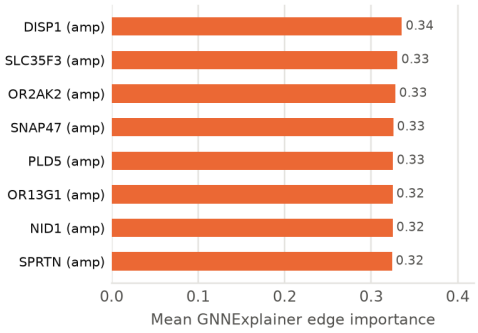
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
+0.091 ↑ risk	Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) [KG]	RL-proposed	84 pts
-0.067 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	143 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.027 ↓ risk	NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'Pyroptosis') ⇒ Progress(p) [KG]	RL-proposed	46 pts
-0.024 ↓ risk	Stage(p, II) ∧ Mutated(p, TP53) ∧ Altered(p, g) ∧ InPathway(g, 'Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants') ⇒ Progress(p) [KG]	RL-proposed	71 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
TP53 · driver	Mutation	p.A161T	57	0.31
PIK3CA · driver	Mutation	p.E542K	70	0.31
FOXA1 · driver	High-level amplification	GISTIC ±2	9	0.31
DISP1	High-level amplification	GISTIC ±2	0	0.34
SLC35F3	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
SNAP47	High-level amplification	GISTIC ±2	0	0.33
PLD5	High-level amplification	GISTIC ±2	0	0.33
OR13G1	High-level amplification	GISTIC ±2	1	0.32
NID1	High-level amplification	GISTIC ±2	3	0.32
SPRTN	High-level amplification	GISTIC ±2	1	0.32



83 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PIK3CA → ATP (DB00171) … and 43 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1JB	54	II	LumA	0.85	no progression, followed 55.5 mo
TCGA-AR-A0U1	36	II	missing	0.84	no progression, followed 133.2 mo
TCGA-E9-A22G	47	II	Basal	0.84	no progression, followed 40.7 mo
TCGA-A2-A25B	39	II	LumB	0.80	progressed at 42.4 mo (group D)
TCGA-A2-A0D0	60	II	Basal	0.80	no progression, followed 67.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**69**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1FL-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.6** · MSIsensor score: **0.94** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

46th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 18%

Model-estimated progression time

≈ 49 months

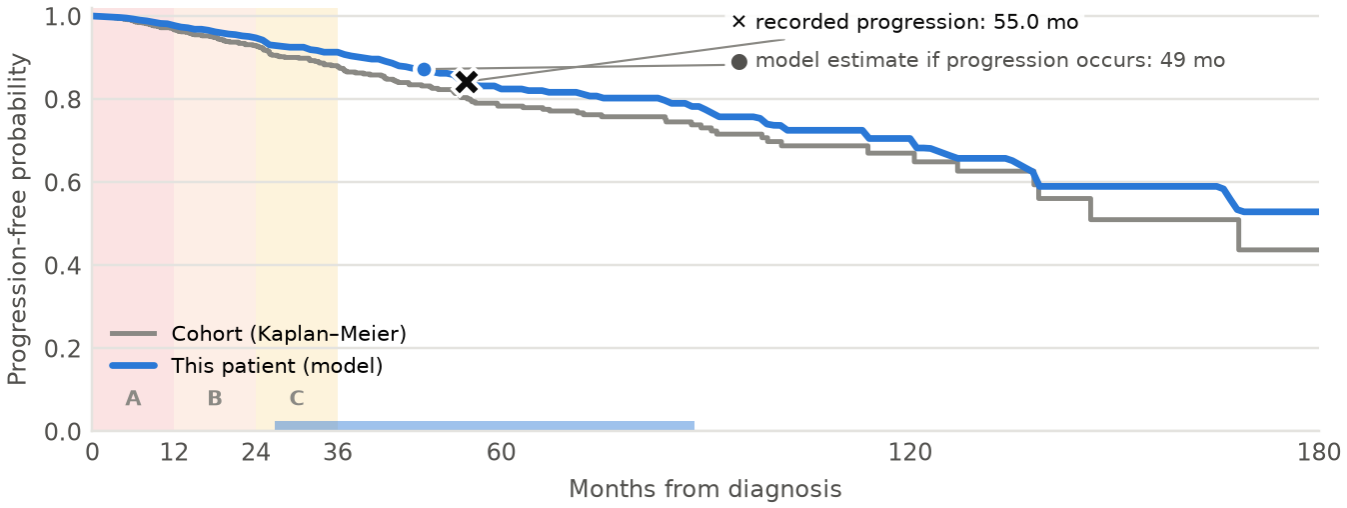
Window > 36 months · IQR 27–88 mo if progression occurs

Recorded progression

55.0 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 6 mo



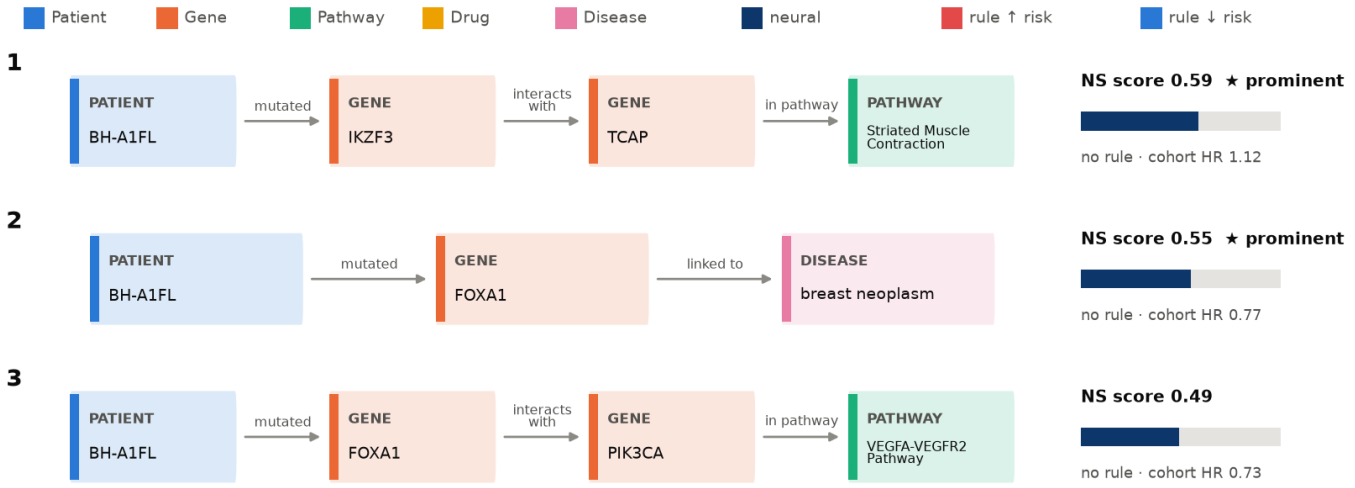
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 69-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **46th risk percentile** (intermediate tier). The highest-scoring knowledge-graph path runs from **IKZF3** to **Striated Muscle Contraction**. Progression was recorded at 55.0 months; the model's estimate given progression was 49 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

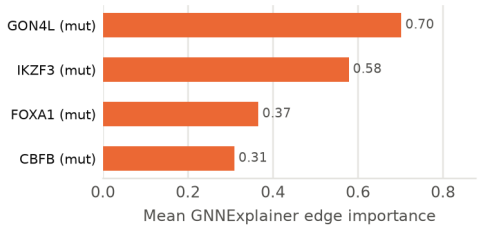
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1FL** mutated → **IKZF3** interacts with → **TCAP** in pathway → **Striated Muscle Contraction**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 328 carriers	**0.59**
2 ★	**TCGA-BH-A1FL** mutated → **FOXA1** linked to → **breast neoplasm**	92	No learned formula fires on this path; its score is neural only. cohort: HR 0.77, n = 794 carriers	**0.55**
3	**TCGA-BH-A1FL** mutated → **FOXA1** interacts with → **PIK3CA** in pathway → **VEGFA-VEGFR2 Pathway**	82	No learned formula fires on this path; its score is neural only. cohort: HR 0.73, n = 398 carriers	**0.49**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.Y259S	9	0.37
CBFB · driver	Mutation	p.I102M	29	0.31
GON4L	Mutation	p.S1477L	0	0.70
IKZF3	Mutation	p.A20S	0	0.58



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1NH	71	II	LumA	0.85	no progression, followed 18.9 mo
TCGA-BH-A0BM	54	II	LumA	0.83	no progression, followed 61.7 mo
TCGA-E2-A153	51	II	LumA	0.81	no progression, followed 23.2 mo
TCGA-BH-A0E1	52	II	LumA	0.80	no progression, followed 15.7 mo
TCGA-BH-A1EX	67	II	LumA	0.80	progressed at 49.6 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the

weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**44**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

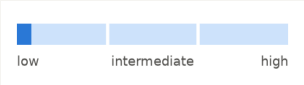
Sample: **TCGA-BH-A1FU-01** · Aneuploidy score: **26** · TMB (nonsyn., /Mb): **2.73** · MSIsensor score: **0** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

5th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

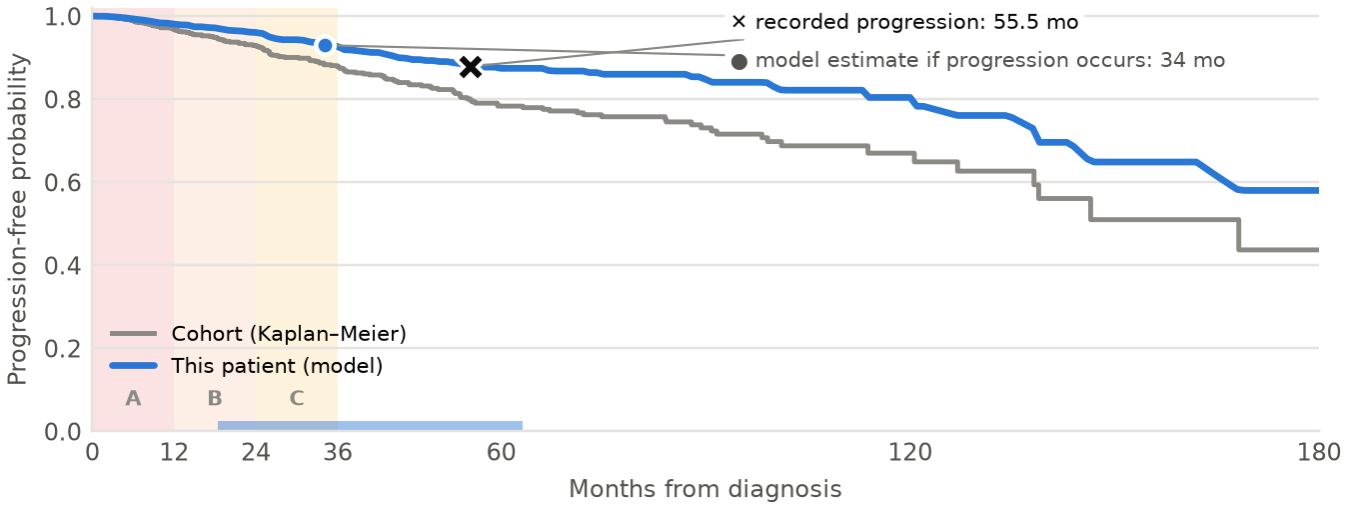
≈ 34 months

Window 24–36 months · IQR 18–63 mo
if progression occurs

Recorded progression

55.5 months

Group D · Late progression
✗ recorded window differs · |error| 21 mo



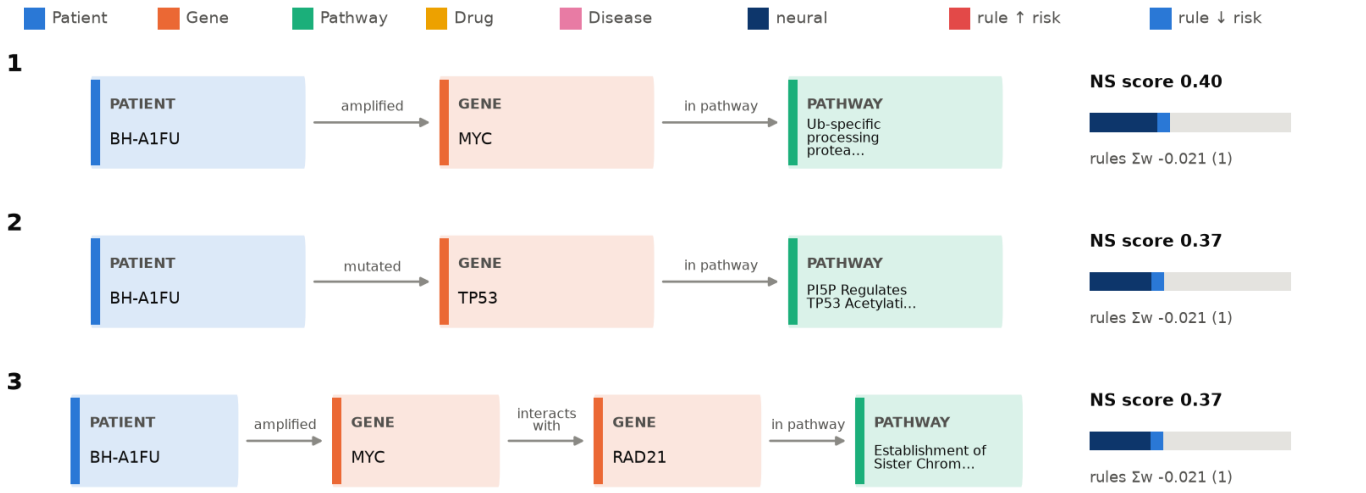
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 44-year-old patient has a Normal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **5th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **MYC** to **Ub-specific processing proteases**; the learned formulas it instantiates point to slower progression (Σw -0.021). Progression was recorded at 55.5 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

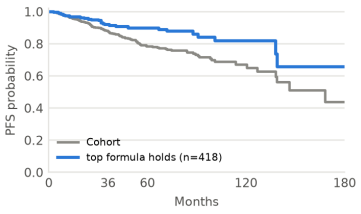


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A1FU** amplified → **MYC** in pathway → **Ub-specific processing proteases**	56	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 1.00, n = 457 carriers	**0.40**
2	**TCGA-BH-A1FU** mutated → **TP53** in pathway → **PI5P Regulates TP53 Acetylation**	51	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 1.05, n = 284 carriers	**0.37**
3	**TCGA-BH-A1FU** amplified → **MYC** interacts with → **RAD21** in pathway → **Establishment of Sister Chromatid Cohesion**	51	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 1.12, n = 117 carriers	**0.37**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

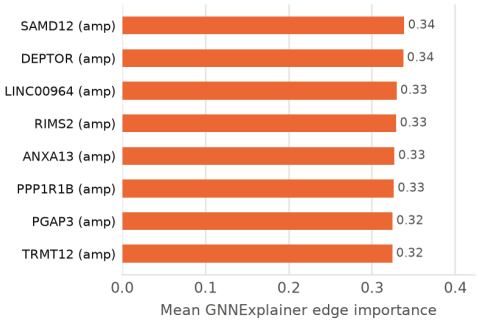
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.021 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) **[KG]**	RL-proposed	73 pts
-0.021 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	139 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Stage(p, I) ⇒ Progress(p)	mined pair	138 pts
-0.000 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Pregnenolone biosynthesis') ⇒ Progress(p) **[KG]**	mined pair	69 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R175H	57	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
ERBB2 · driver	Mutation	p.R340G	55	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
SAMD12	High-level amplification	GISTIC ±2	0	0.34
DEPTOR	High-level amplification	GISTIC ±2	0	0.34
LINC00964	High-level amplification	GISTIC ±2	0	0.33
RIMS2	High-level amplification	GISTIC ±2	0	0.33
ANXA13	High-level amplification	GISTIC ±2	0	0.33
PPP1R1B	High-level amplification	GISTIC ±2	1	0.33
PGAP3	High-level amplification	GISTIC ±2	0	0.32
TRMT12	High-level amplification	GISTIC ±2	1	0.32



118 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 51 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CX	52	II	Her2	0.75	no progression, followed 56.8 mo
TCGA-BH-A0BG	73	I	Basal	0.72	no progression, followed 61.5 mo
TCGA-D8-A1J9	48	I	LumB	0.70	no progression, followed 17.5 mo
TCGA-AO-A0J4	41	I	Basal	0.68	no progression, followed 52.2 mo
TCGA-D8-A27M	59	I	Basal	0.68	no progression, followed 13.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1EW · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**38**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

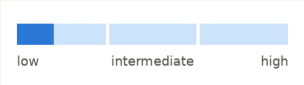
Sample: **TCGA-BH-A1EW-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **0.47** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

14th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

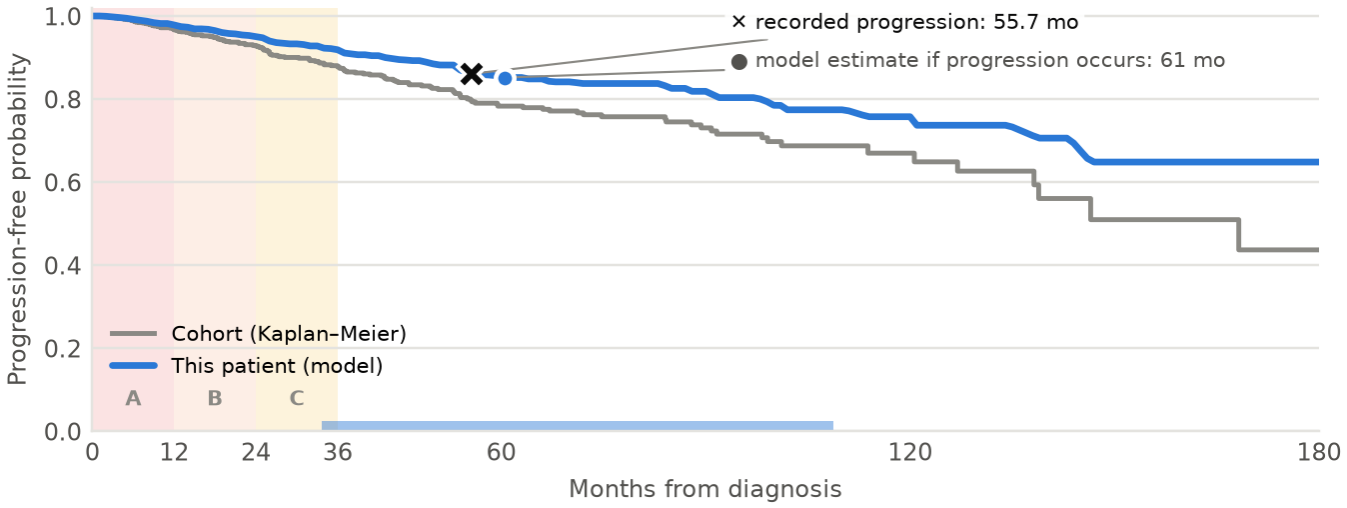
≈ 61 months

Window > 36 months · IQR 34–109 mo
if progression occurs

Recorded progression

55.7 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 5 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 38-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **14th risk percentile** (low tier). The strongest learned formula that holds for them (*Age(p) < 40 ⇒ Progress(p)*) raises the progression hazard (weight +0.025 per SD). Progression was recorded at 55.7 months; the model's estimate given progression was 61 months.

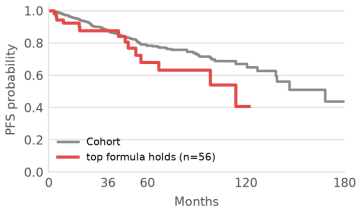
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.025 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	56 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A245	47	II	LumA	0.99	no progression, followed 0.9 mo
TCGA-BH-A0DQ	42	II	LumA	0.98	no progression, followed 3.2 mo
TCGA-BH-A0E1	52	II	LumA	0.98	no progression, followed 15.7 mo
TCGA-AR-A24L	26	II	LumA	0.96	progressed at 46.5 mo (group D)
TCGA-C8-A273	29	II	LumA	0.89	no progression, followed 16.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

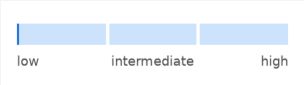
Sample: **TCGA-BH-A0BQ-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.5** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

1st

Low risk tier · fused neuro-symbolic model



Probability of progression

6% by 36 mo

12 mo 2% · 24 mo 3% · 60 mo 11%

Model-estimated progression time

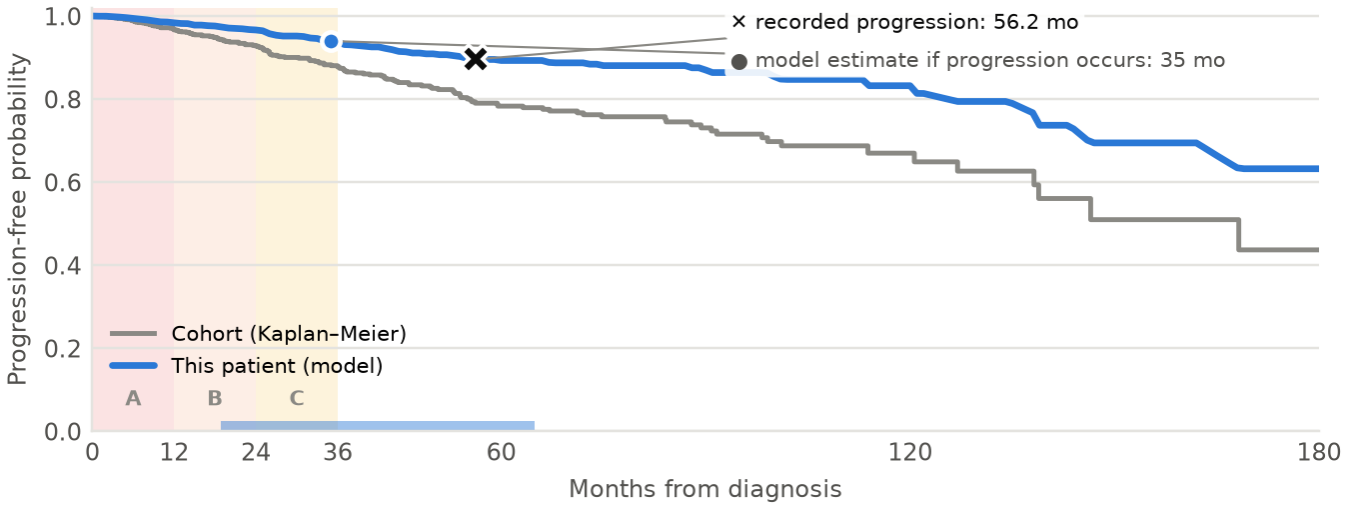
≈ 35 months

Window 24–36 months · IQR 19–65 mo
if progression occurs

Recorded progression

56.2 months

Group D · Late progression
✗ recorded window differs · |error| 21 mo



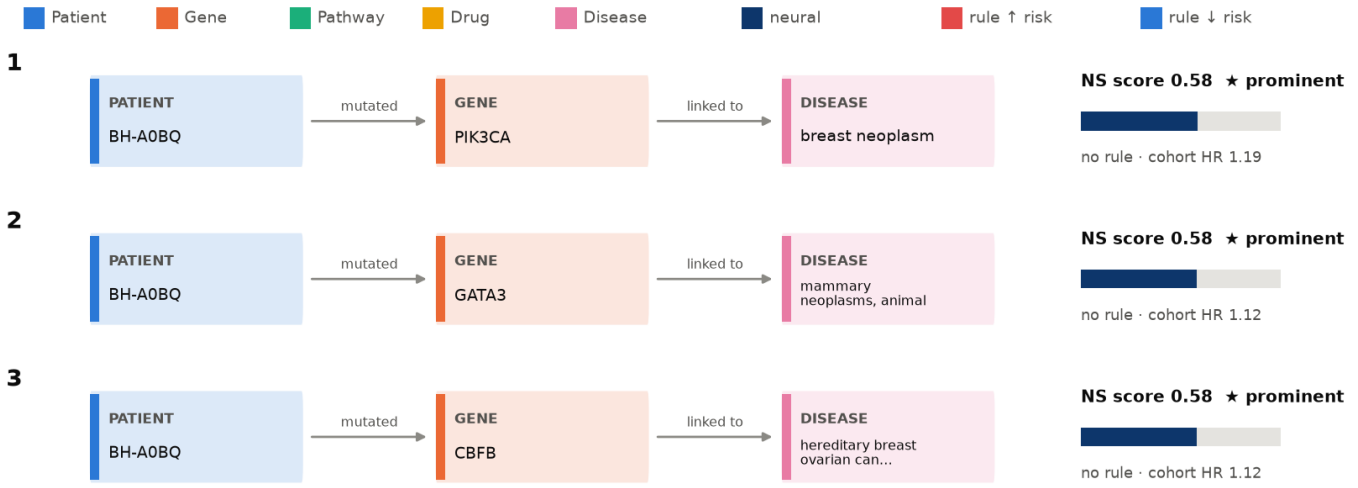
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **1st risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **breast neoplasm**. Progression was recorded at 56.2 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

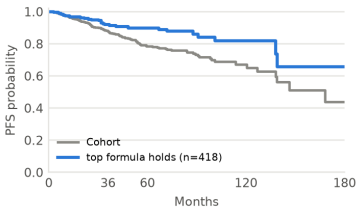


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A0BQ** mutated → **PIK3CA** linked to → **breast neoplasm**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.19, n = 797 carriers	**0.58**
2 ★	**TCGA-BH-A0BQ** mutated → **GATA3** linked to → **mammary neoplasms, animal**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 532 carriers	**0.58**
3 ★	**TCGA-BH-A0BQ** mutated → **CBFB** linked to → **hereditary breast ovarian cancer syndrome**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 800 carriers	**0.58**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

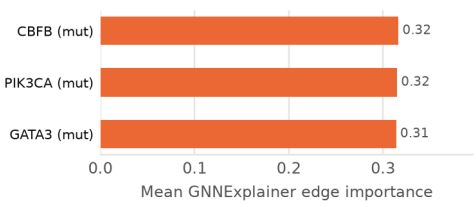
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.028 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	133 pts
-0.021 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	139 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Stage(p, I) ⇒ Progress(p)	mined pair	138 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CBFB · driver	Mutation	p.X94_splice	29	0.32
PIK3CA · driver	Mutation	p.E545K	70	0.32
GATA3 · driver	Mutation	p.K357*, p.M356I	13	0.31



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0H3	46	I	LumA	0.95	no progression, followed 63.4 mo
TCGA-B6-A0RN	60	I	LumA	0.93	no progression, followed 263.3 mo
TCGA-A2-A0EO	54	I	LumA	0.88	no progression, followed 80.3 mo
TCGA-E9-A1RA	48	I	LumA	0.87	no progression, followed 45.0 mo
TCGA-B6-A0X0	54	I	LumA	0.86	no progression, followed 129.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**55**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**TX**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-3C-AAAU-01** · Aneuploidy score: **19** · TMB (nonsyn., /Mb): **0.8** · MSIsensor score: **0.55** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

76th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

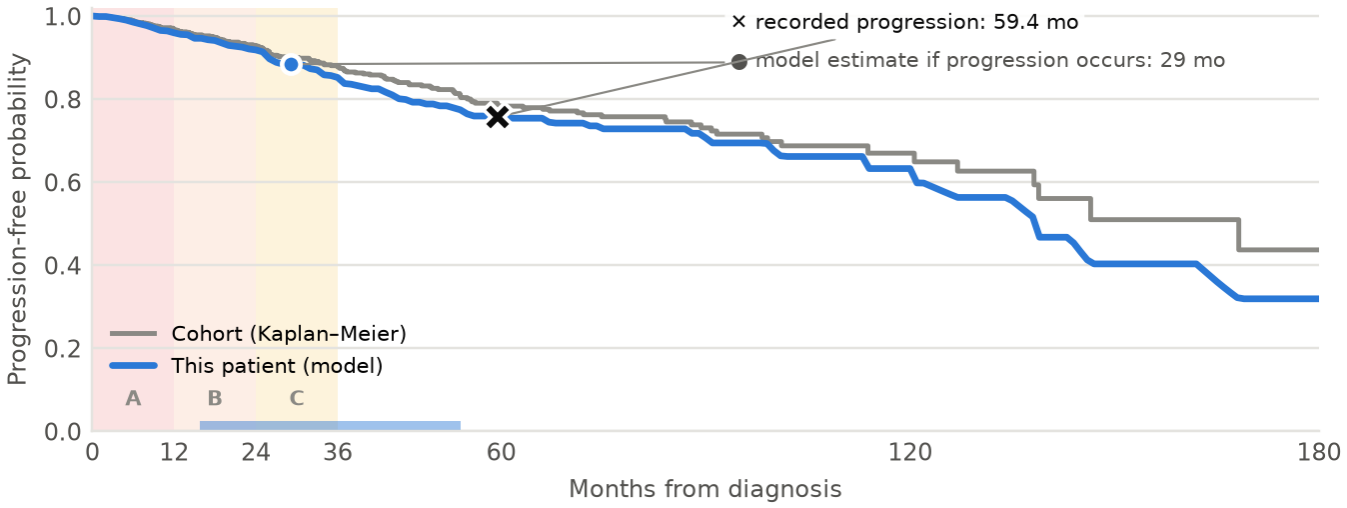
≈ 29 months

Window 24–36 months · IQR 16–54 mo
if progression occurs

Recorded progression

59.4 months

Group D · Late progression
✗ recorded window differs · |error| 30 mo



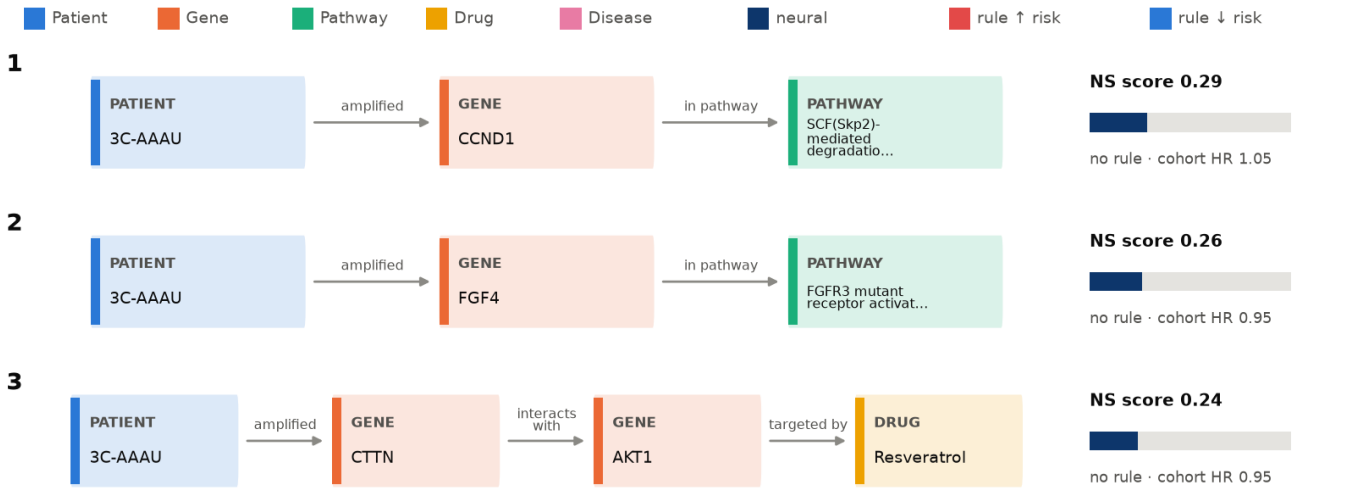
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 55-year-old patient has a LumA tumour, AJCC stage X (missing, missing, MX). The fused neuro-symbolic model places them in the **76th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **CCND1** to **SCF(Skp2)-mediated degradation of p27/p21**. Progression was recorded at 59.4 months; the model's estimate given progression was 29 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

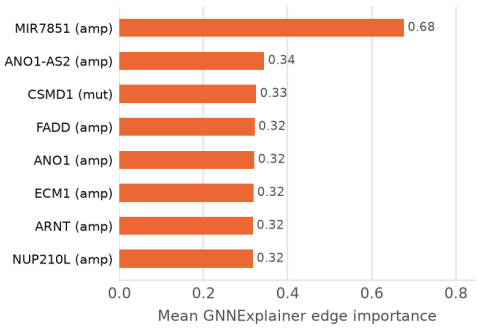
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-3C-AAAU** amplified → **CCND1** in pathway → **SCF(Skp2)-mediated degradation of p27/p21**	48	No learned formula fires on this path; its score is neural only. cohort: HR 1.05, n = 122 carriers	**0.29**
2	**TCGA-3C-AAAU** amplified → **FGF4** in pathway → **FGFR3 mutant receptor activation**	43	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 118 carriers	**0.26**
3	**TCGA-3C-AAAU** amplified → **CTTN** interacts with → **AKT1** targeted by → **Resveratrol**	40	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 61 carriers	**0.24**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
GATA3 · driver	Mutation	p.P408Afs*99	13	0.31
MIR7851	High-level amplification	GISTIC ±2	0	0.68
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.34
CSMD1	Mutation	p.X967_splice	0	0.33
FADD	High-level amplification	GISTIC ±2	14	0.32
ANO1	High-level amplification	GISTIC ±2	2	0.32
ECM1	High-level amplification	GISTIC ±2	1	0.32
ARNT	High-level amplification	GISTIC ±2	6	0.32
NUP210L	High-level amplification	GISTIC ±2	0	0.32
TARS2	High-level amplification	GISTIC ±2	2	0.32
FGF19	High-level amplification	GISTIC ±2	13	0.32



29 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); ANO1 → Crofelemer (DB04941); TARS2 → Threonine (DB00156); FGF19 → Heparin (DB01109); FGF4 → Pentosan polysulfate (DB00686); FGF4 → Heparin (DB01109); FGF4 → TP-508 (DB05309) … and 5 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-OL-A66O	39	II	LumA	0.59	no progression, followed 17.4 mo
TCGA-A1-A0SQ	45	II	LumA	0.57	no progression, followed 18.2 mo
TCGA-A8-A07E	81	X	LumA	0.56	no progression, followed 20.0 mo
TCGA-EW-A423	75	II	LumA	0.56	no progression, followed 17.5 mo
TCGA-AR-A1AK	70	I	LumA	0.56	no progression, followed 103.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

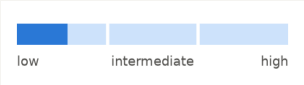
Sample: **TCGA-E2-A1IE-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.6** · MSIsensor score: **0.38** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

19th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

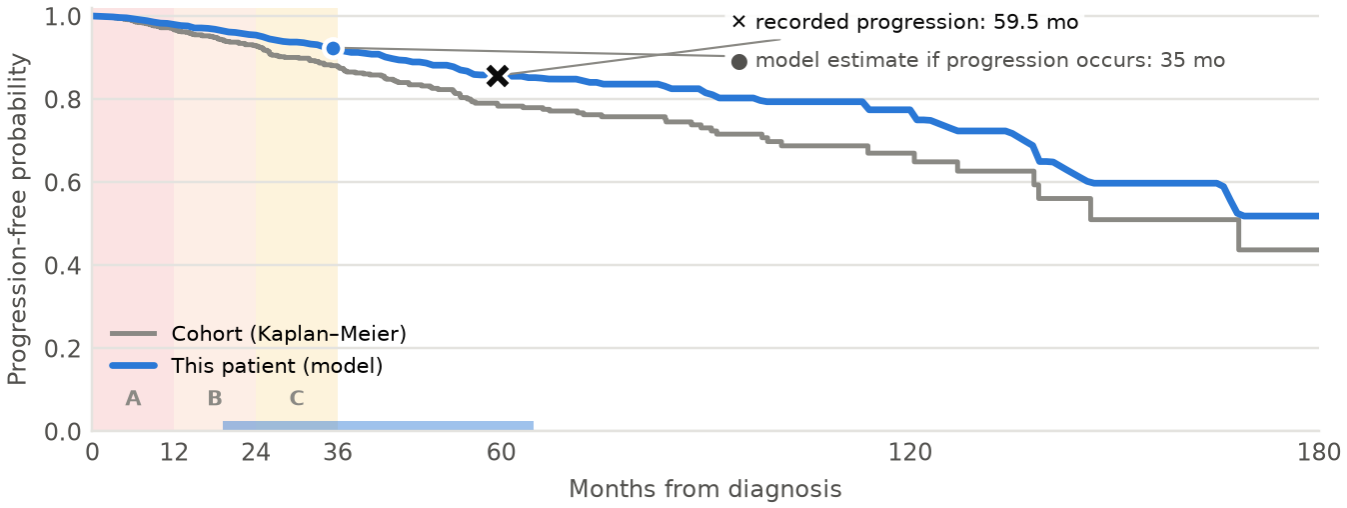
≈ 35 months

Window 24–36 months · IQR 19–65 mo
if progression occurs

Recorded progression

59.5 months

Group D · Late progression
✗ recorded window differs · |error| 24 mo



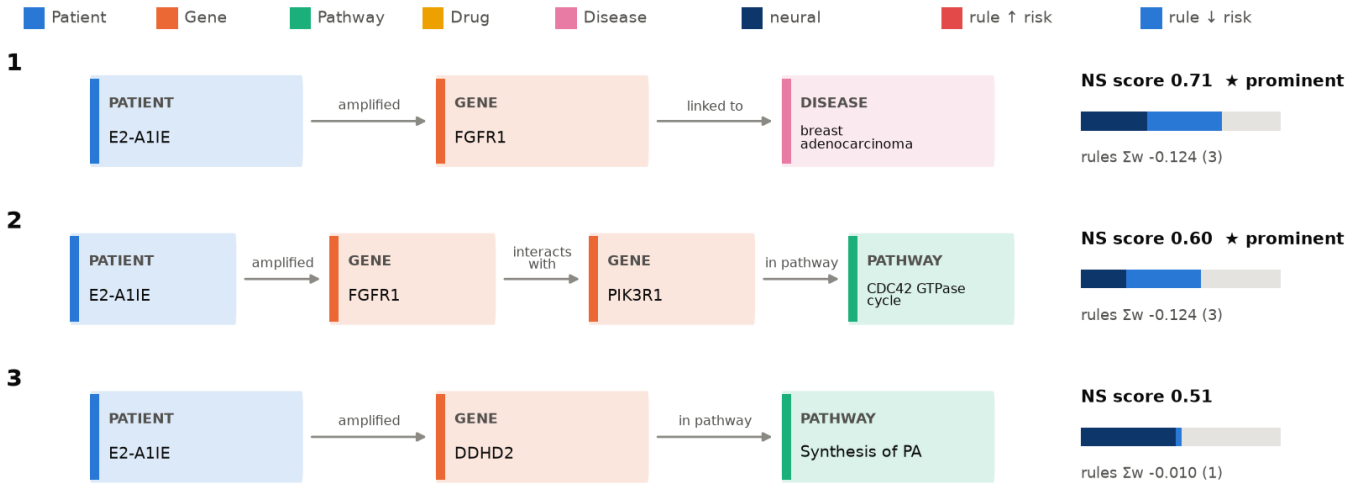
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **19th risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **FGFR1** to **breast adenocarcinoma**; the learned formulas it instantiates point to slower progression (Σw -0.124). Progression was recorded at 59.5 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

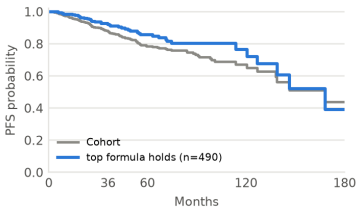


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E2-A1IE** amplified → **FGFR1** linked to → **breast adenocarcinoma**	55	↓ **slower progression** · Σw -0.124 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.95, n = 734 carriers	**0.71**
2 ★	**TCGA-E2-A1IE** amplified → **FGFR1** interacts with → **PIK3R1** in pathway → **CDC42 GTPase cycle**	38	↓ **slower progression** · Σw -0.124 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.86, n = 216 carriers	**0.60**
3	**TCGA-E2-A1IE** amplified → **DDHD2** in pathway → **Synthesis of PA**	79	↓ **slower progression** · Σw -0.010 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) cohort: HR 0.80, n = 189 carriers	**0.51**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

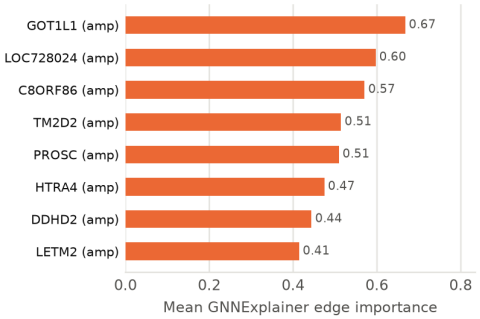
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.050 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Sensing of DNA Double Strand Breaks') ⇒ Progress(p) [KG]	single atom	40 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.034 ↓ risk	LowAneuploidy(p) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Deactivation of the beta-catenin transactivating complex') ⇒ Progress(p) [KG]	RL-proposed	41 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ATM · driver	Mutation	p.L2780R	29	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.67
LOC728024	High-level amplification	GISTIC ±2	0	0.60
C8ORF86	High-level amplification	GISTIC ±2	0	0.57
TM2D2	High-level amplification	GISTIC ±2	0	0.51
PROSC	High-level amplification	GISTIC ±2	0	0.51
HTRA4	High-level amplification	GISTIC ±2	0	0.47
DDHD2	High-level amplification	GISTIC ±2	1	0.44
LETM2	High-level amplification	GISTIC ±2	0	0.41
STAR	High-level amplification	GISTIC ±2	1	0.37
LSM1	High-level amplification	GISTIC ±2	1	0.34



34 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ATM → Caffeine (DB00201); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896) … and 47 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1PA	59	II	LumA	0.90	no progression, followed 18.9 mo
TCGA-B6-A0WT	61	II	LumA	0.88	progressed at 127.0 mo (group D)
TCGA-BH-A204	80	II	LumB	0.83	no progression, followed 83.3 mo
TCGA-WT-AB41	55	II	LumA	0.81	no progression, followed 53.0 mo
TCGA-A2-A0ES	52	II	LumA	0.80	no progression, followed 72.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**54**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

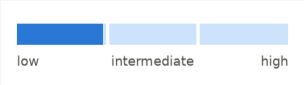
Sample: **TCGA-AO-A03P-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.33** · MSIsensor score: **1.02** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

32nd

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

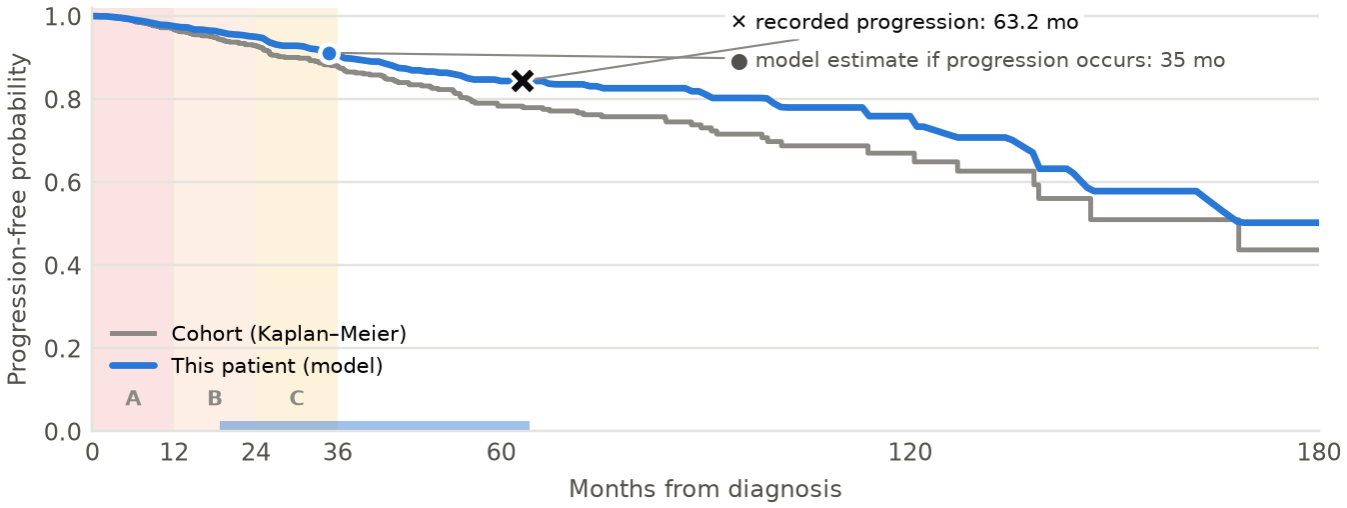
≈ 35 months

Window 24–36 months · IQR 19–64 mo
if progression occurs

Recorded progression

63.2 months

Group D · Late progression
✗ recorded window differs · |error| 29 mo



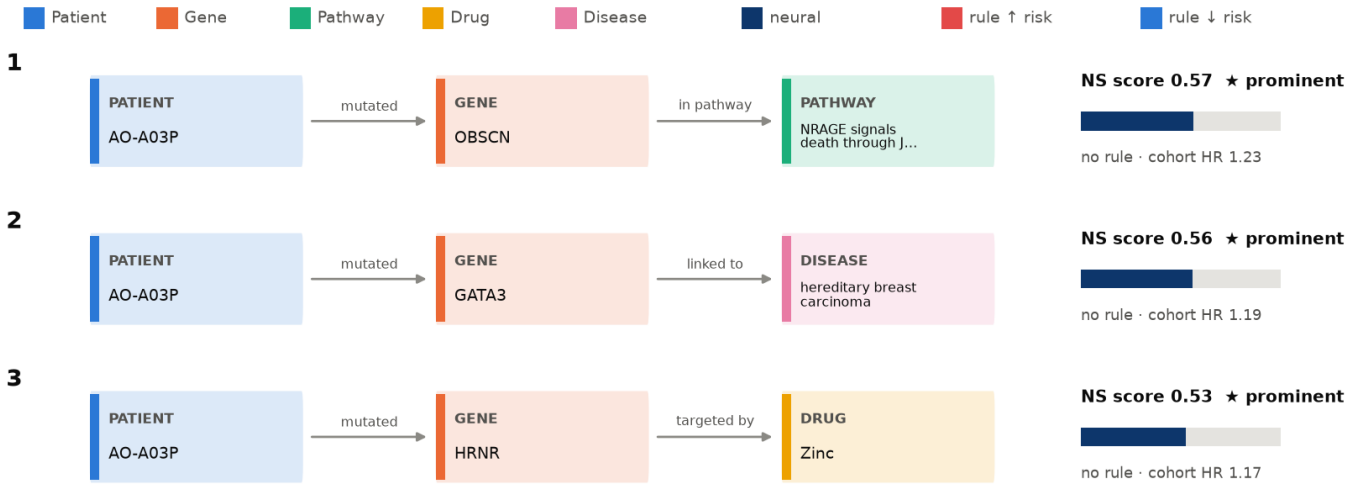
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 54-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **32nd risk percentile** (low tier). The highest-scoring knowledge-graph path runs from **OBSCN** to **NRAGE signals death through JNK**. Progression was recorded at 63.2 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

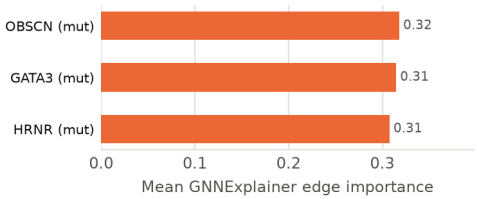
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AO-A03P** mutated → **OBSCN** in pathway → **NRAGE signals death through JNK**	94	No learned formula fires on this path; its score is neural only. cohort: HR 1.23, n = 121 carriers	**0.57**
2 ★	**TCGA-AO-A03P** mutated → **GATA3** linked to → **hereditary breast carcinoma**	94	No learned formula fires on this path; its score is neural only. cohort: HR 1.19, n = 797 carriers	**0.56**
3 ★	**TCGA-AO-A03P** mutated → **HRNR** targeted by → **Zinc**	88	No learned formula fires on this path; its score is neural only. cohort: HR 1.17, n = 379 carriers	**0.53**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.M442Ifs*65	13	0.31
OBSCN	Mutation	p.G3509V	8	0.32
HRNR	Mutation	p.S2027=	3	0.31



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

HRNR → Zinc (DB01593); HRNR → Zinc acetate (DB14487). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0C7	48	II	LumB	0.86	no progression, followed 91.0 mo
TCGA-D8-A1XR	56	II	LumB	0.85	no progression, followed 15.8 mo
TCGA-A8-A082	58	II	LumB	0.82	no progression, followed 18.0 mo
TCGA-B6-A1KC	67	II	LumB	0.81	no progression, followed 43.6 mo
TCGA-BH-A0HL	56	II	missing	0.80	no progression, followed 2.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0WY-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **1.03** · MSIsensor score: **0.08** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

75th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 24%

Model-estimated progression time

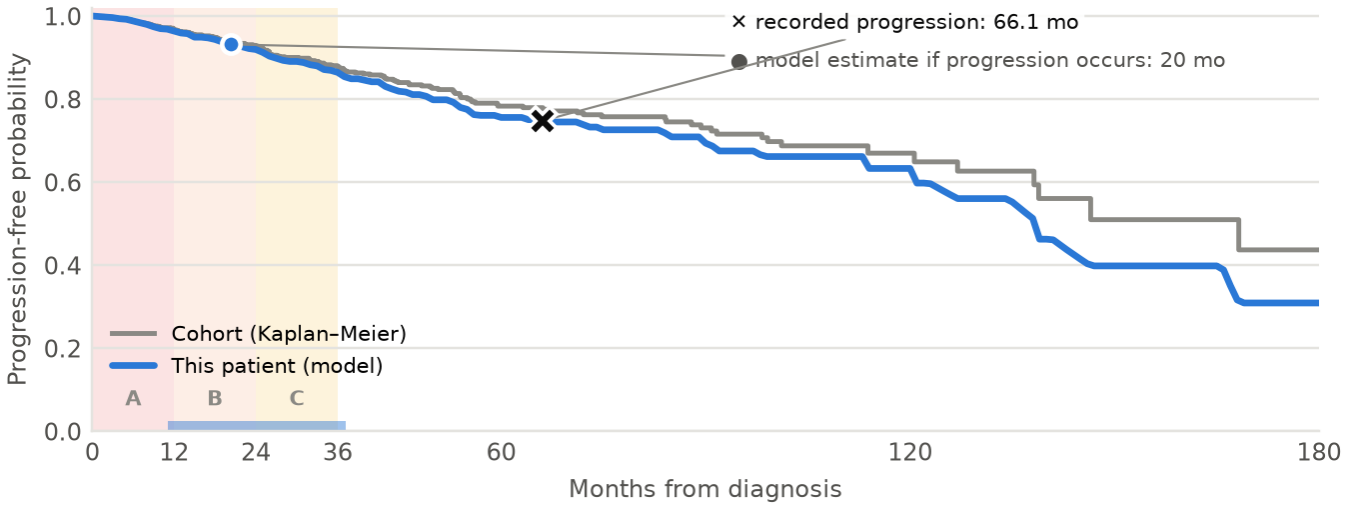
≈ 20 months

Window 12–24 months · IQR 11–37 mo
if progression occurs

Recorded progression

66.1 months

Group D · Late progression
✗ recorded window differs · |error| 46 mo



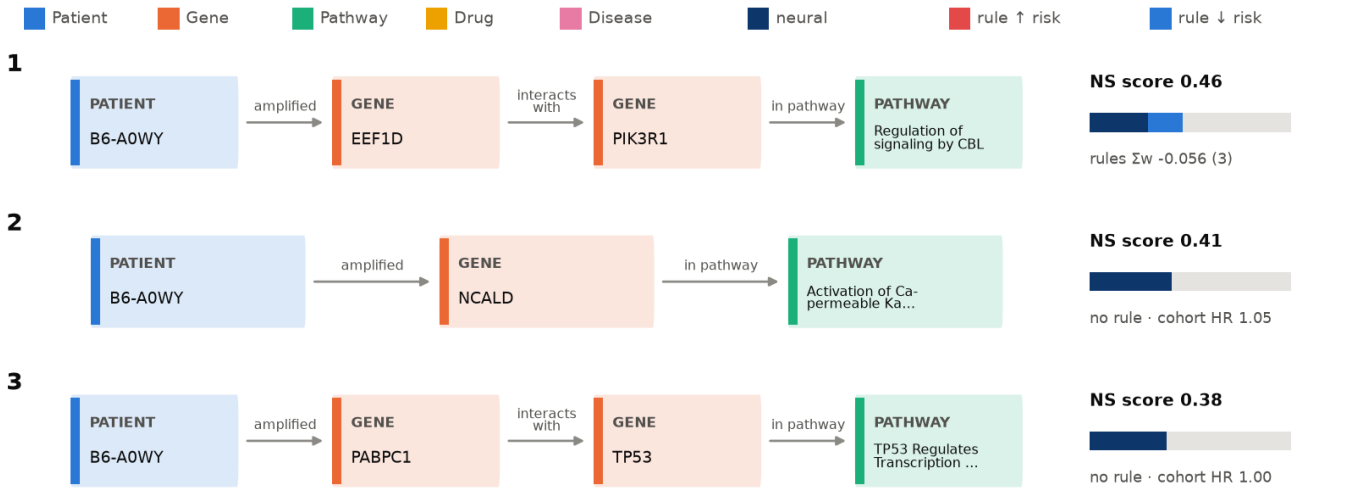
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a LumA tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **75th risk percentile** (high tier). The strongest learned formula that holds for them (*Subtype(p, LumA)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'RAF/MAP kinase cascade')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.048 per SD). The highest-scoring knowledge-graph path runs from **EEF1D** to **Regulation of signaling by CBL**; the learned formulas it instantiates point to slower progression (Σw -0.056). Progression was recorded at 66.1 months; the model's estimate given progression was 20 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

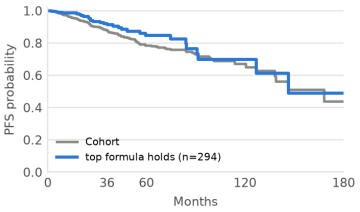


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0WY** amplified → **EEF1D** interacts with → **PIK3R1** in pathway → **Regulation of signaling by CBL**	49	↓ **slower progression** · Σw -0.056 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.86, n = 309 carriers	**0.46**
2	**TCGA-B6-A0WY** amplified → **NCALD** in pathway → **Activation of Ca-permeable Kainate Receptor**	68	No learned formula fires on this path; its score is neural only. cohort: HR 1.05, n = 100 carriers	**0.41**
3	**TCGA-B6-A0WY** amplified → **PABPC1** interacts with → **TP53** in pathway → **TP53 Regulates Transcription of Genes Involved in Cytochrome C Release**	64	No learned formula fires on this path; its score is neural only. cohort: HR 1.00, n = 366 carriers	**0.38**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

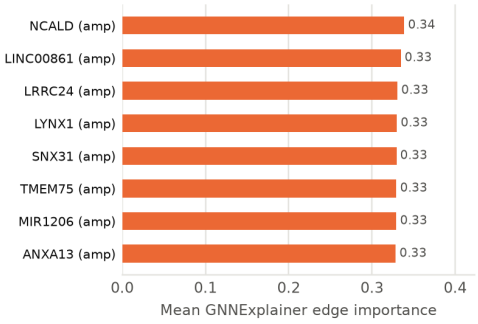
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.034 ↓ risk	Amplified(p, ZFHX4) ⇒ Progress(p)	single atom	55 pts
-0.008 ↓ risk	Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	134 pts
-0.000 ↓ risk	Amplified(p, STK3) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	RL-proposed	45 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
PIK3CA · driver	Mutation	p.E109_I112delinsD	70	0.32
PIK3R1 · driver	Mutation	p.I571_L573del	79	0.31
NCALD	High-level amplification	GISTIC ±2	1	0.34
LINC00861	High-level amplification	GISTIC ±2	0	0.33
LRRC24	High-level amplification	GISTIC ±2	0	0.33
LYNX1	High-level amplification	GISTIC ±2	0	0.33
SNX31	High-level amplification	GISTIC ±2	0	0.33
TMEM75	High-level amplification	GISTIC ±2	0	0.33
MIR1206	High-level amplification	GISTIC ±2	0	0.33
ANXA13	High-level amplification	GISTIC ±2	0	0.33
SCRT1	High-level amplification	GISTIC ±2	0	0.33



314 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PIK3R1 → Isoprenaline (DB01064) … and 93 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AN-A0XS	63	III	LumA	0.89	no progression, followed 0.3 mo
TCGA-A8-A06T	75	III	LumA	0.89	no progression, followed 53.1 mo
TCGA-AR-A24V	52	II	LumA	0.85	no progression, followed 105.3 mo
TCGA-C8-A12N	58	II	LumA	0.85	no progression, followed 11.8 mo
TCGA-A2-A04N	66	I	LumA	0.85	no progression, followed 143.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**38**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0RS-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **4.63** · MSIsensor score: **0.22** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

62nd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 21%

Model-estimated progression time

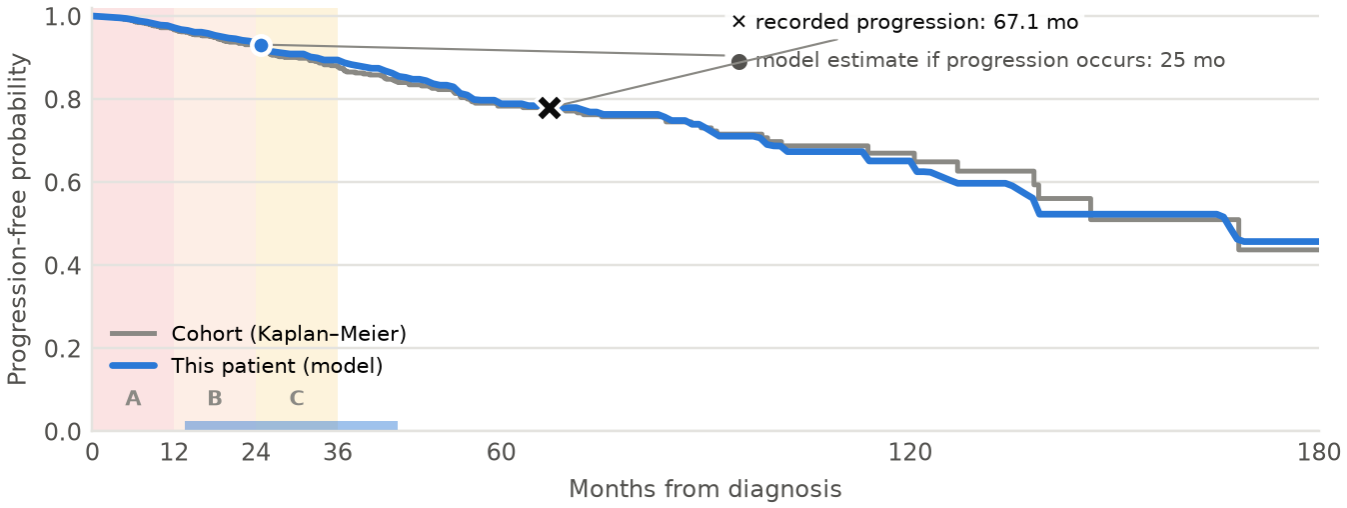
≈ 25 months

Window 24–36 months · IQR 14–45 mo
if progression occurs

Recorded progression

67.1 months

Group D · Late progression
✗ recorded window differs · |error| 42 mo



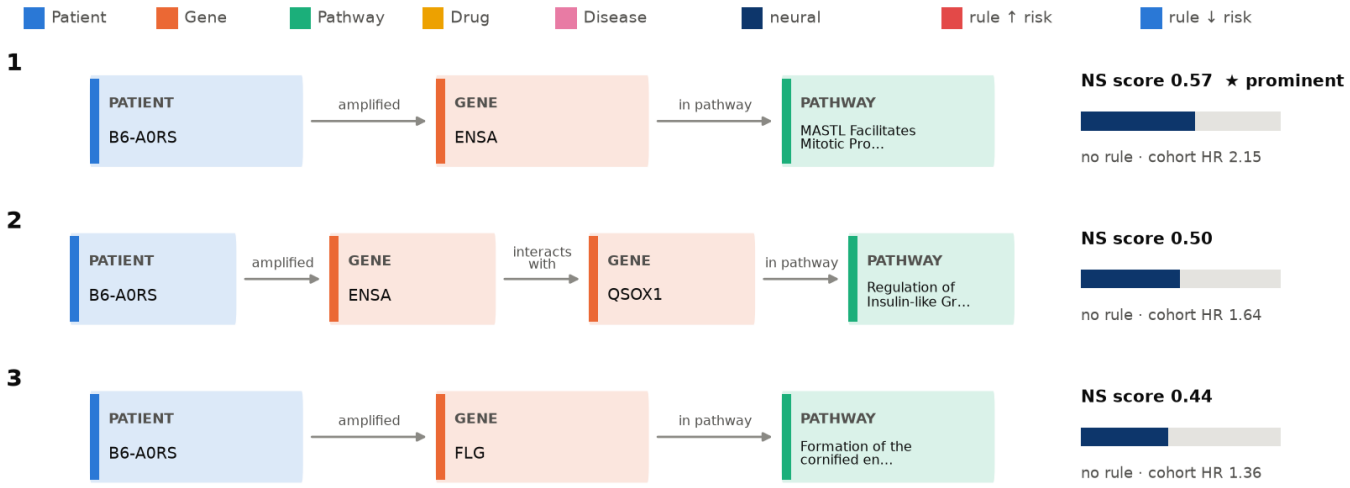
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 38-year-old patient has a Her2 tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **62nd risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.099 per SD). The highest-scoring knowledge-graph path runs from **ENSA** to **MASTL Facilitates Mitotic Progression**. Progression was recorded at 67.1 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

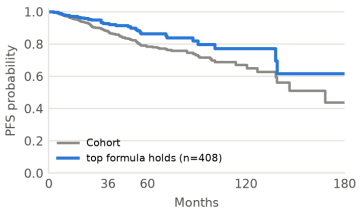


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A0RS** amplified → **ENSA** in pathway → **MASTL Facilitates Mitotic Progression**	95	No learned formula fires on this path; its score is neural only. cohort: HR 2.15, n = 91 carriers	**0.57**
2	**TCGA-B6-A0RS** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)**	83	No learned formula fires on this path; its score is neural only. cohort: HR 1.64, n = 111 carriers	**0.50**
3	**TCGA-B6-A0RS** amplified → **FLG** in pathway → **Formation of the cornified envelope**	73	No learned formula fires on this path; its score is neural only. cohort: HR 1.36, n = 158 carriers	**0.44**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

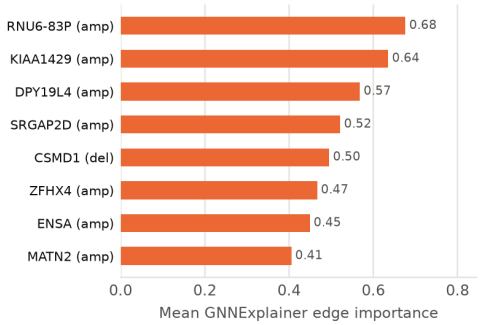
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.099 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	398 pts
+0.058 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Defective B3GALTL causes Peters-plus syndrome (PpS)') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	RL-proposed	44 pts
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ TargetedBy(g, 'ATP') ⇒ Progress(p) [KG]	mined pair	162 pts
-0.011 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.009 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	mined pair	212 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.31
TP53 · driver	Mutation	p.R273C	57	0.31
RNU6-83P	High-level amplification	GISTIC ±2	0	0.68
KIAA1429	High-level amplification	GISTIC ±2	0	0.64
DPY19L4	High-level amplification	GISTIC ±2	0	0.57
SRGAP2D	High-level amplification	GISTIC ±2	0	0.52
CSMD1	Homozygous deletion	GISTIC ±2	0	0.50
ZFHX4	High-level amplification	GISTIC ±2	0	0.47
ENSA	High-level amplification	GISTIC ±2	1	0.45
MATN2	High-level amplification	GISTIC ±2	0	0.41
HMCN1	High-level amplification	GISTIC ±2	0	0.37
FLG	High-level amplification	GISTIC ±2	1	0.37



54 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347) … and 30 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A248	51	II	Her2	0.71	no progression, followed 1.9 mo
TCGA-B6-A0RH	51	II	Her2	0.69	progressed at 71.2 mo (group D)
TCGA-AO-A0J6	61	II	Basal	0.67	no progression, followed 37.5 mo
TCGA-E9-A244	54	II	Basal	0.66	no progression, followed 0.7 mo
TCGA-AR-A24K	46	II	LumB	0.66	no progression, followed 50.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**51**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0RH-01** · Aneuploidy score: **10** · TMB (nonsyn., /Mb): **1.27** · MSIsensor score: **0.22** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

69th

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 4% · 24 mo 7% · 60 mo 22%

Model-estimated progression time

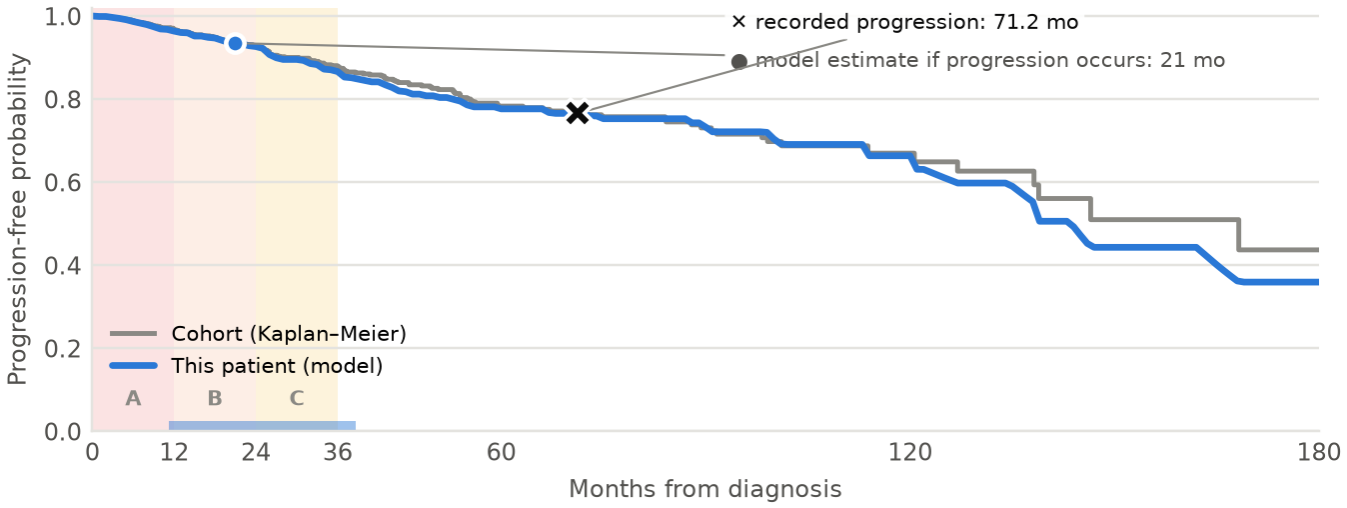
≈ 21 months

Window 12–24 months · IQR 11–39 mo
if progression occurs

Recorded progression

71.2 months

Group D · Late progression
✗ recorded window differs · |error| 50 mo



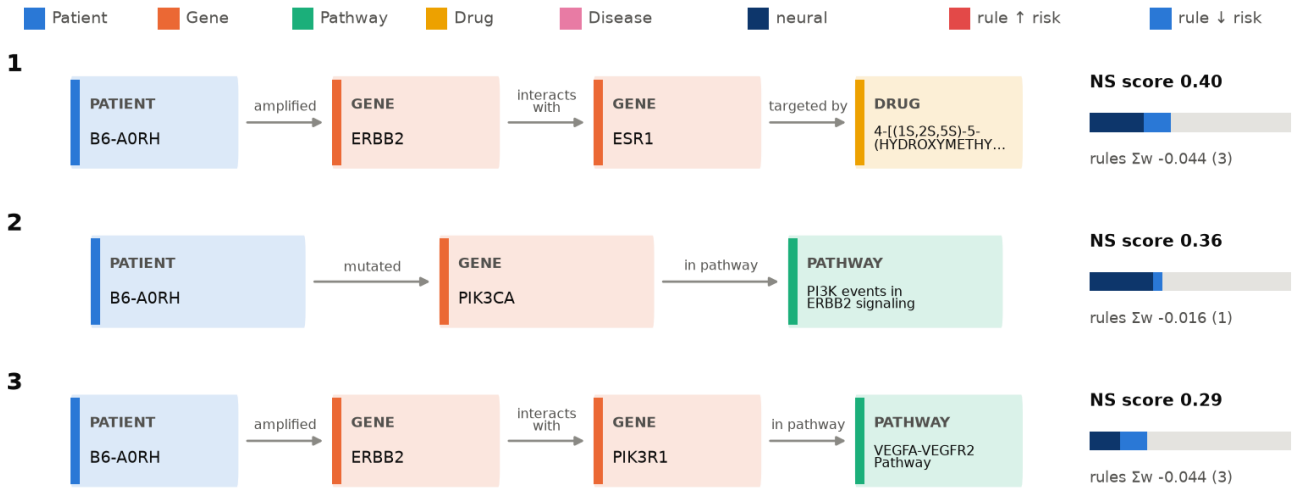
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 51-year-old patient has a Her2 tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **69th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **ERBB2** to **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**; the learned formulas it instantiates point to slower progression (Σw -0.044). Progression was recorded at 71.2 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

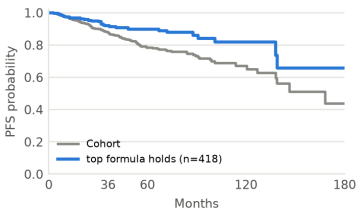


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0RH** amplified → **ERBB2** interacts with → **ESR1** targeted by → **4-[(1S,2S,5S)-5-(HYDROXYMETHYL)-6,8,9-TRIMETHYL-3-OXABICYCLO[3.3.1]NON-7-EN-2-YL]PHENOL**	45	↓ **slower progression** · Σw -0.044 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 2.93, n = 28 carriers	**0.40**
2	**TCGA-B6-A0RH** mutated → **PIK3CA** in pathway → **PI3K events in ERBB2 signaling**	53	↓ **slower progression** · Σw -0.016 over 1 formula(s) TStage(p, T2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K events in ERBB2 signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of PTEN gene transcription') ⇒ Progress(p) cohort: HR 1.02, n = 410 carriers	**0.36**
3	**TCGA-B6-A0RH** amplified → **ERBB2** interacts with → **PIK3R1** in pathway → **VEGFA-VEGFR2 Pathway**	25	↓ **slower progression** · Σw -0.044 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 0.85, n = 388 carriers	**0.29**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

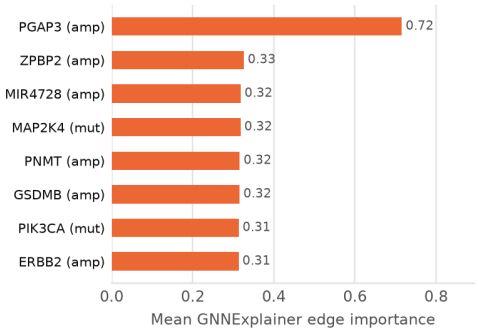
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
+0.103 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ⇒ Progress(p) [KG]	mined pair	32 pts
-0.021 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) [KG]	RL-proposed	73 pts
-0.016 ↓ risk	TStage(p, T2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K events in ERBB2 signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of PTEN gene transcription') ⇒ Progress(p) [KG]	RL-proposed	98 pts
-0.008 ↓ risk	Stage(p, II) ∧ Mutated(p, TP53) ∧ Altered(p, g) ∧ InPathway(g, 'Downregulation of ERBB2:ERBB3 signaling') ⇒ Progress(p) [KG]	RL-proposed	50 pts
-0.000 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Pregnenolone biosynthesis') ⇒ Progress(p) [KG]	mined pair	69 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP2K4 · driver	Mutation	p.X211_splice	12	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.31
PIK3CA · driver	Mutation	p.E545K	70	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
TP53 · driver	Mutation	p.Y220S	57	0.31
PGAP3	High-level amplification	GISTIC ±2	0	0.72
ZPBP2	High-level amplification	GISTIC ±2	0	0.33
MIR4728	High-level amplification	GISTIC ±2	0	0.32
PNMT	High-level amplification	GISTIC ±2	16	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
GRB7	High-level amplification	GISTIC ±2	12	0.31
ECM1	Mutation	p.*15*	1	0.31



22 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241) … and 42 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XV	46	II	Her2	0.89	no progression, followed 32.7 mo
TCGA-D8-A1JG	62	II	Her2	0.89	no progression, followed 53.0 mo
TCGA-C8-A1HF	48	II	Her2	0.88	no progression, followed 10.9 mo
TCGA-AN-A0FV	58	II	Her2	0.88	no progression, followed 0.3 mo
TCGA-EW-A1OZ	56	II	LumB	0.83	no progression, followed 40.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1FN · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**34**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

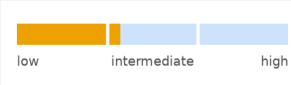
Sample: **TCGA-BH-A1FN-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **3.27** · MSIsensor score: **0.72** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

38th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 17%

Model-estimated progression time

≈ 51 months

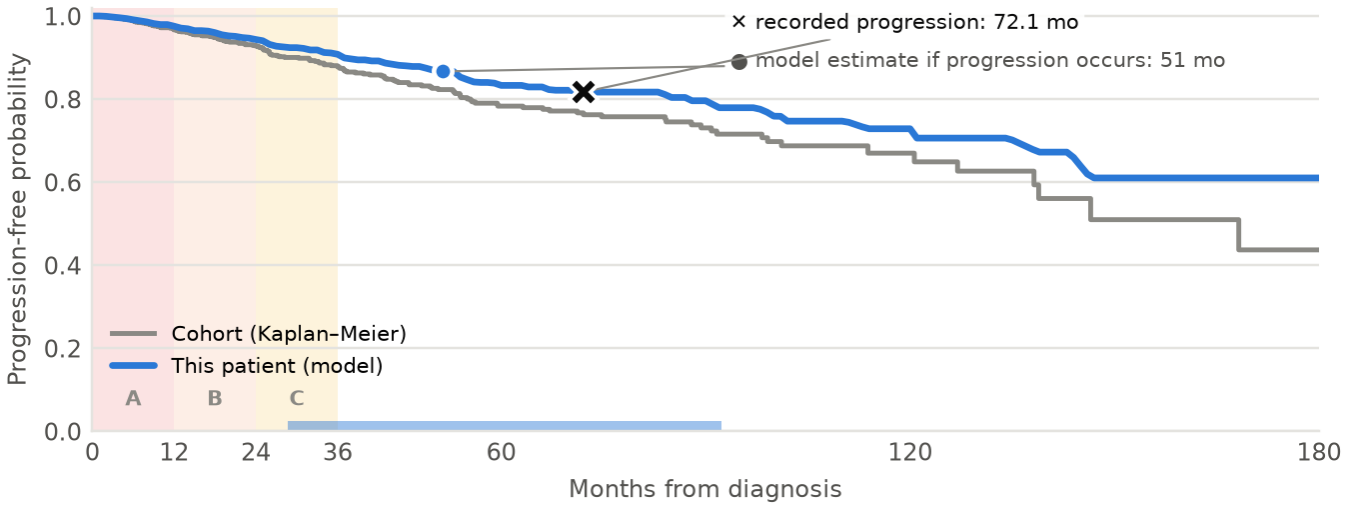
Window > 36 months · IQR 29–92 mo if progression occurs

Recorded progression

72.1 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 21 mo



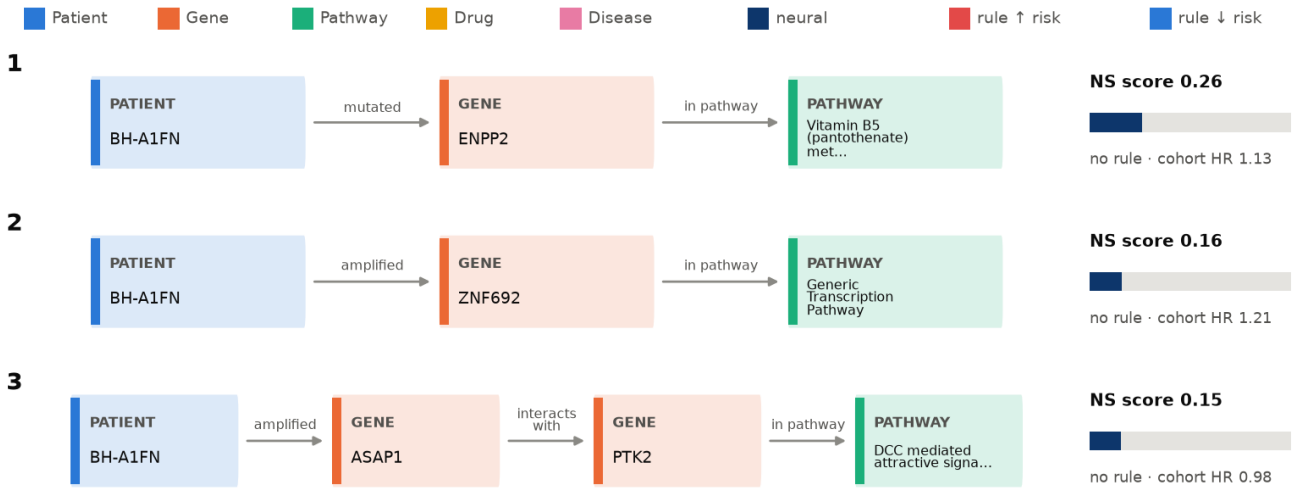
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 34-year-old patient has a LumB tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **38th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.095 per SD). The highest-scoring knowledge-graph path runs from **ENPP2** to **Vitamin B5 (pantothenate) metabolism**. Progression was recorded at 72.1 months; the model's estimate given progression was 51 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

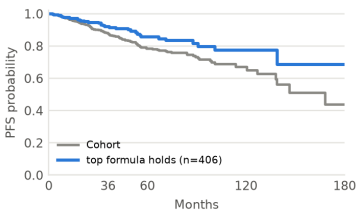


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A1FN** mutated → **ENPP2** in pathway → **Vitamin B5 (pantothenate) metabolism**	44	No learned formula fires on this path; its score is neural only. cohort: HR 1.13, n = 120 carriers	**0.26**
2	**TCGA-BH-A1FN** amplified → **ZNF692** in pathway → **Generic Transcription Pathway**	27	No learned formula fires on this path; its score is neural only. cohort: HR 1.21, n = 315 carriers	**0.16**
3	**TCGA-BH-A1FN** amplified → **ASAP1** interacts with → **PTK2** in pathway → **DCC mediated attractive signaling**	26	No learned formula fires on this path; its score is neural only. cohort: HR 0.98, n = 109 carriers	**0.15**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

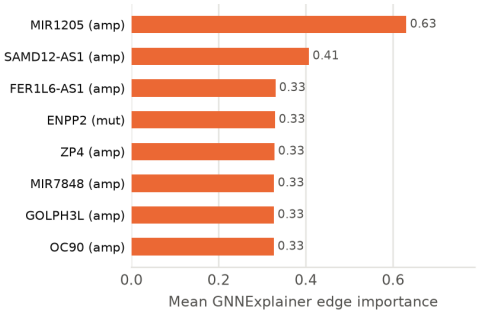
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.095 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	397 pts
+0.058 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ⇒ Progress(p) [KG]	mined pair	55 pts
+0.025 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	56 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts
+0.005 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	mined pair	48 pts
+0.004 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Defective B3GALTL causes Peters-plus syndrome (PpS)') ⇒ Progress(p) [KG]	single atom	94 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	High-level amplification	GISTIC ±2	22	0.32
PTEN · driver	Mutation	p.X154_splice	22	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
RB1 · driver	Mutation	p.Q93*	25	0.31
TP53 · driver	Mutation	p.R280T	57	0.31
MIR1205	High-level amplification	GISTIC ±2	0	0.63
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.41
FER1L6-AS1	High-level amplification	GISTIC ±2	0	0.33
ENPP2	High-level amplification	GISTIC ±2	1	0.33
ENPP2	Mutation	p.E371Q	1	0.33
ZP4	High-level amplification	GISTIC ±2	1	0.33
MIR7848	High-level amplification	GISTIC ±2	0	0.33



203 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); PTEN → Phosphatidylethanolamine (DB04327); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404) … and 43 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A13D	46	II	Basal	0.87	no progression, followed 31.7 mo
TCGA-AR-A0TP	43	II	Basal	0.84	no progression, followed 140.5 mo
TCGA-A2-A04T	62	II	Basal	0.80	no progression, followed 73.8 mo
TCGA-B6-A0I1	73	II	Basal	0.79	progressed at 36.8 mo (group D)
TCGA-A7-A4SE	54	II	Basal	0.78	no progression, followed 21.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**31**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

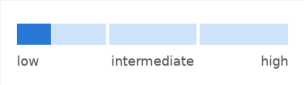
Sample: **TCGA-BH-A1FE-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

13th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

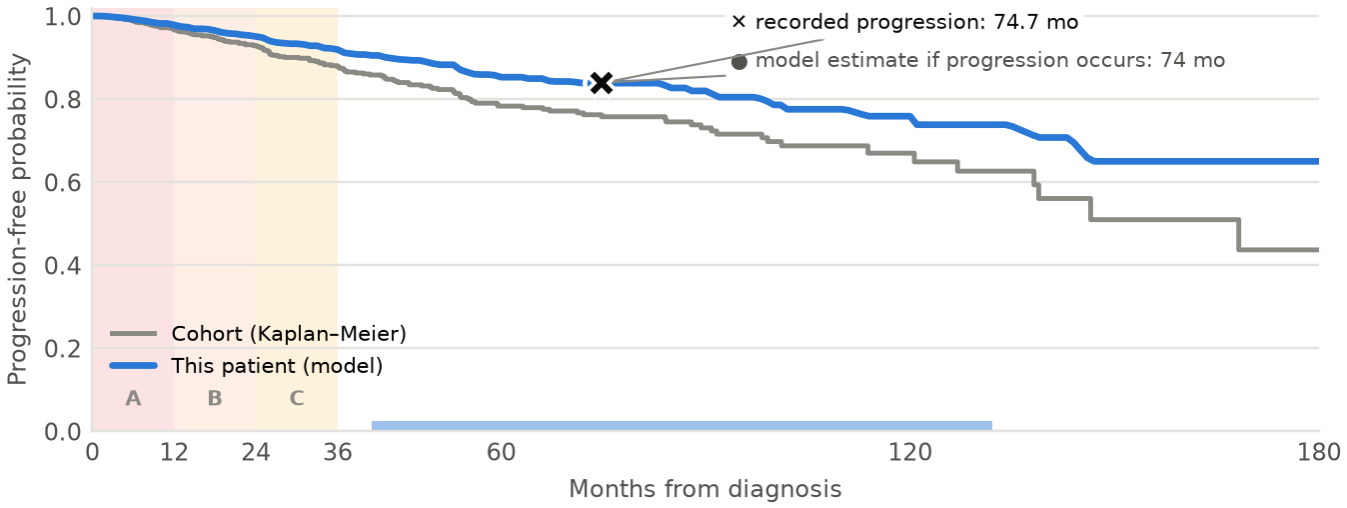
≈ 74 months

Window > 36 months · IQR 41–132 mo
if progression occurs

Recorded progression

74.7 months

Group D · Late progression
✓ estimate in recorded window · |error| 1 mo



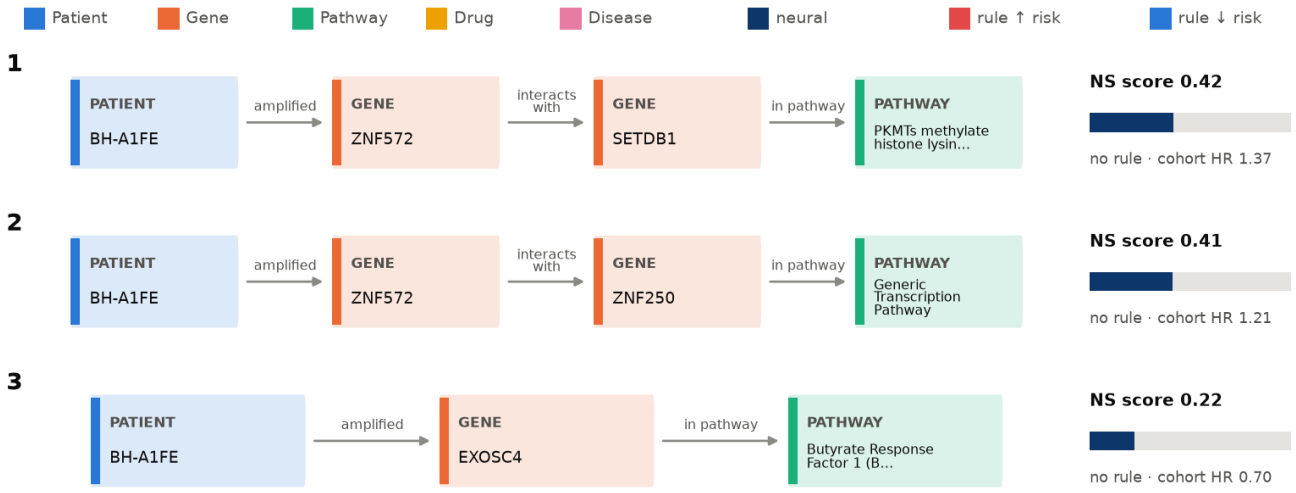
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 31-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **13th risk percentile** (low tier). The strongest learned formula that holds for them (*Age(p) < 40 ⇒ Progress(p)*) raises the progression hazard (weight +0.025 per SD). The highest-scoring knowledge-graph path runs from **ZNF572** to **PKMTs methylate histone lysines**. Progression was recorded at 74.7 months; the model's estimate given progression was 74 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

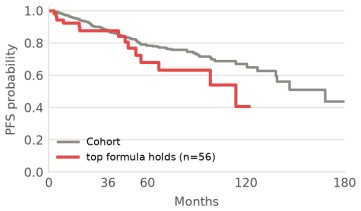


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A1FE** amplified → **ZNF572** interacts with → **SETDB1** in pathway → **PKMTs methylate histone lysines**	69	No learned formula fires on this path; its score is neural only. cohort: HR 1.37, n = 323 carriers	**0.42**
2	**TCGA-BH-A1FE** amplified → **ZNF572** interacts with → **ZNF250** in pathway → **Generic Transcription Pathway**	69	No learned formula fires on this path; its score is neural only. cohort: HR 1.21, n = 315 carriers	**0.41**
3	**TCGA-BH-A1FE** amplified → **EXOSC4** in pathway → **Butyrate Response Factor 1 (BRF1) binds and destabilizes mRNA**	37	No learned formula fires on this path; its score is neural only. cohort: HR 0.70, n = 126 carriers	**0.22**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

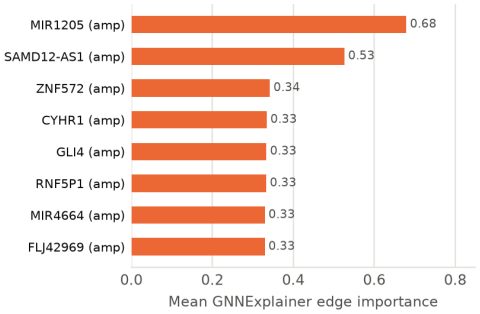
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.025 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	56 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	Mutation	p.D769Y	55	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
PIK3CA · driver	Mutation	p.K111E	70	0.31
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.31
TP53 · driver	Mutation	p.N247I	57	0.31
MIR1205	High-level amplification	GISTIC ±2	0	0.68
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.53
ZNF572	High-level amplification	GISTIC ±2	0	0.34
CYHR1	High-level amplification	GISTIC ±2	0	0.33
GLI4	High-level amplification	GISTIC ±2	0	0.33
RNF5P1	High-level amplification	GISTIC ±2	0	0.33
MIR4664	High-level amplification	GISTIC ±2	0	0.33



295 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 106 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A15E	40	II	LumA	0.88	no progression, followed 20.7 mo
TCGA-C8-A135	64	II	Her2	0.87	no progression, followed 12.9 mo
TCGA-AR-A24U	47	II	Her2	0.86	no progression, followed 102.8 mo
TCGA-LL-A5YN	46	II	LumA	0.86	no progression, followed 14.7 mo
TCGA-E2-A14Y	35	II	Basal	0.86	no progression, followed 69.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A8-A08T · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**64**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A8-A08T-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **1.3** · MSIsensor score: **0.24** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

66% by 36 mo

12 mo 24% · 24 mo 43% · 60 mo 85%

Model-estimated progression time

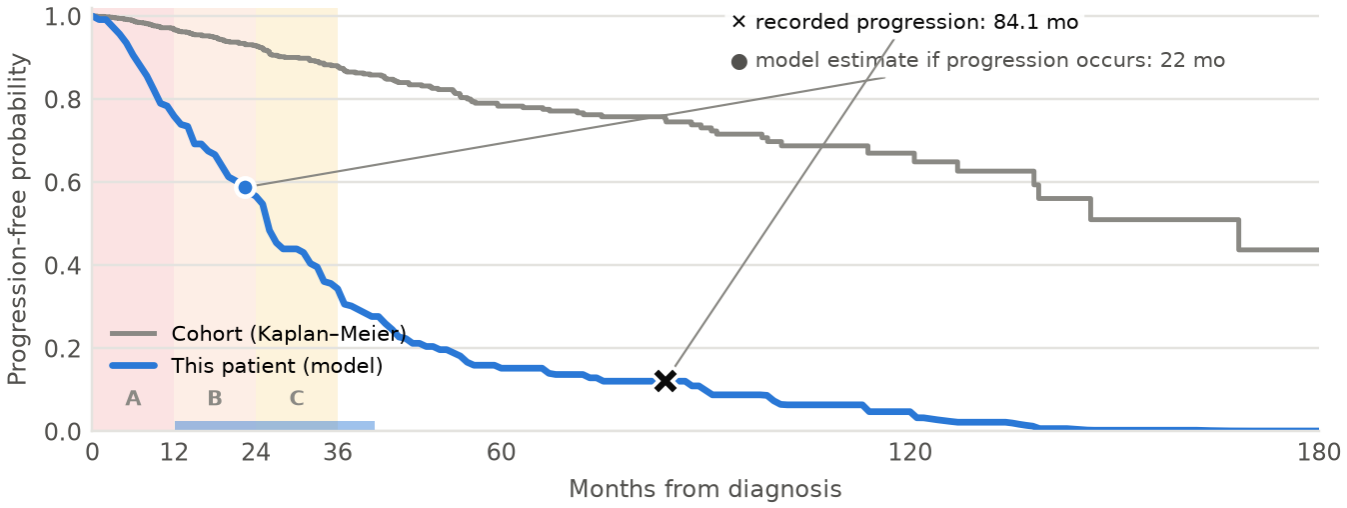
≈ 22 months

Window 12–24 months · IQR 12–41 mo
if progression occurs

Recorded progression

84.1 months

Group D · Late progression
✗ recorded window differs · |error| 62 mo



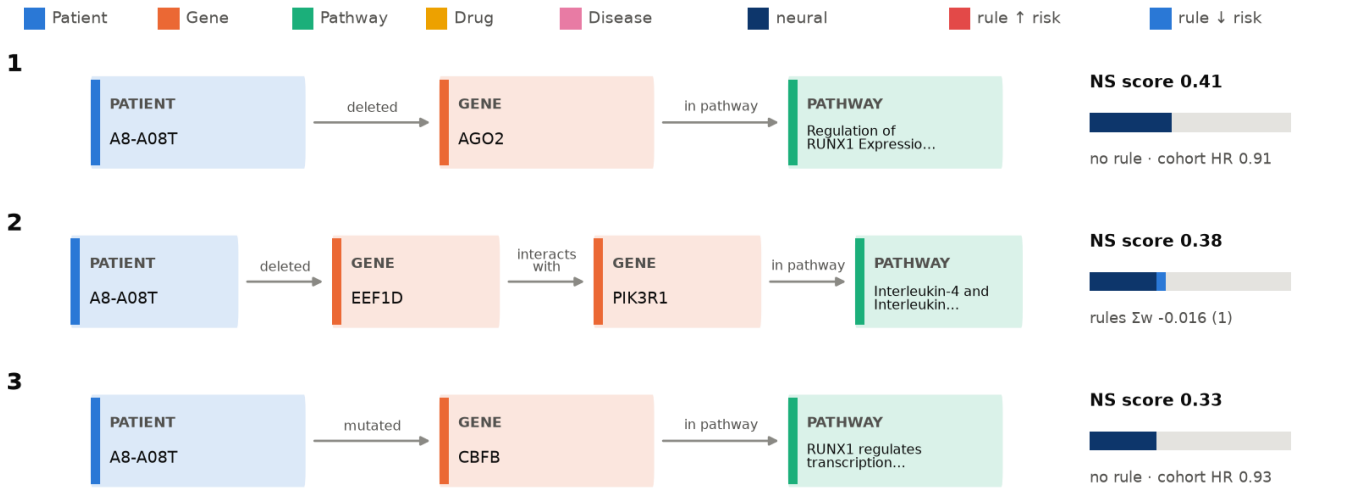
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 64-year-old patient has a LumA tumour, AJCC stage IV (T2, N1, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T2)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'PI3K events in ERBB2 signaling')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Regulation of PTEN gene transcription')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.016 per SD). The highest-scoring knowledge-graph path runs from **AGO2** to **Regulation of RUNX1 Expression and Activity**. Progression was recorded at 84.1 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

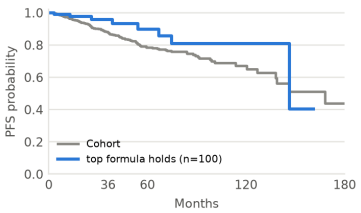


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A8-A08T** deleted → **AGO2** in pathway → **Regulation of RUNX1 Expression and Activity**	68	No learned formula fires on this path; its score is neural only. cohort: HR 0.91, n = 277 carriers	**0.41**
2	**TCGA-A8-A08T** deleted → **EEF1D** interacts with → **PIK3R1** in pathway → **Interleukin-4 and Interleukin-13 signaling**	55	↓ **slower progression** · Σw -0.016 over 1 formula(s) TStage(p, T2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K events in ERBB2 signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of PTEN gene transcription') ⇒ Progress(p) cohort: HR 0.90, n = 509 carriers	**0.38**
3	**TCGA-A8-A08T** mutated → **CBFB** in pathway → **RUNX1 regulates transcription of genes involved in differentiation of HSCs**	55	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 194 carriers	**0.33**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

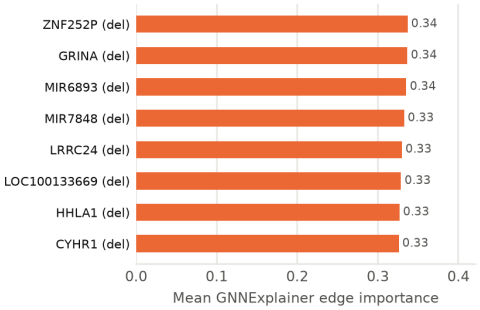
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.016 ↓ risk	TStage(p, T2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K events in ERBB2 signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of PTEN gene transcription') ⇒ Progress(p) [KG]	RL-proposed	98 pts
-0.002 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'MyD88:MAL(TIRAP) cascade initiated on plasma membrane') ⇒ Progress(p) [KG]	mined pair	57 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3R1 · driver	Homozygous deletion	GISTIC ±2	79	0.31
PIK3CA · driver	Mutation	p.E545K	70	0.31
CBFB · driver	Mutation	p.Q74K, p.W73*	29	0.31
MAP3K1 · driver	Homozygous deletion	GISTIC ±2	13	0.31
ZNF252P	Homozygous deletion	GISTIC ±2	0	0.34
GRINA	Homozygous deletion	GISTIC ±2	0	0.34
MIR6893	Homozygous deletion	GISTIC ±2	0	0.34
MIR7848	Homozygous deletion	GISTIC ±2	0	0.33
LRRC24	Homozygous deletion	GISTIC ±2	0	0.33
LOC100133669	Homozygous deletion	GISTIC ±2	0	0.33
HHLA1	Homozygous deletion	GISTIC ±2	0	0.33
CYHR1	Homozygous deletion	GISTIC ±2	0	0.33



155 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486); PIK3R1 → Wortmannin (DB08059); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 77 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A203	78	II	Her2	0.68	progressed at 38.6 mo (group D)
TCGA-B6-A0RM	57	X	LumA	0.65	no progression, followed 78.0 mo
TCGA-BH-A1FE	31	II	LumA	0.65	progressed at 74.7 mo (group D)
TCGA-BH-A0E7	79	II	LumA	0.64	no progression, followed 44.8 mo
TCGA-AN-A0FZ	45	III	LumA	0.63	no progression, followed 0.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**43**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

Sample: **TCGA-AR-A24O-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **0.08** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

81st

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 26%

Model-estimated progression time

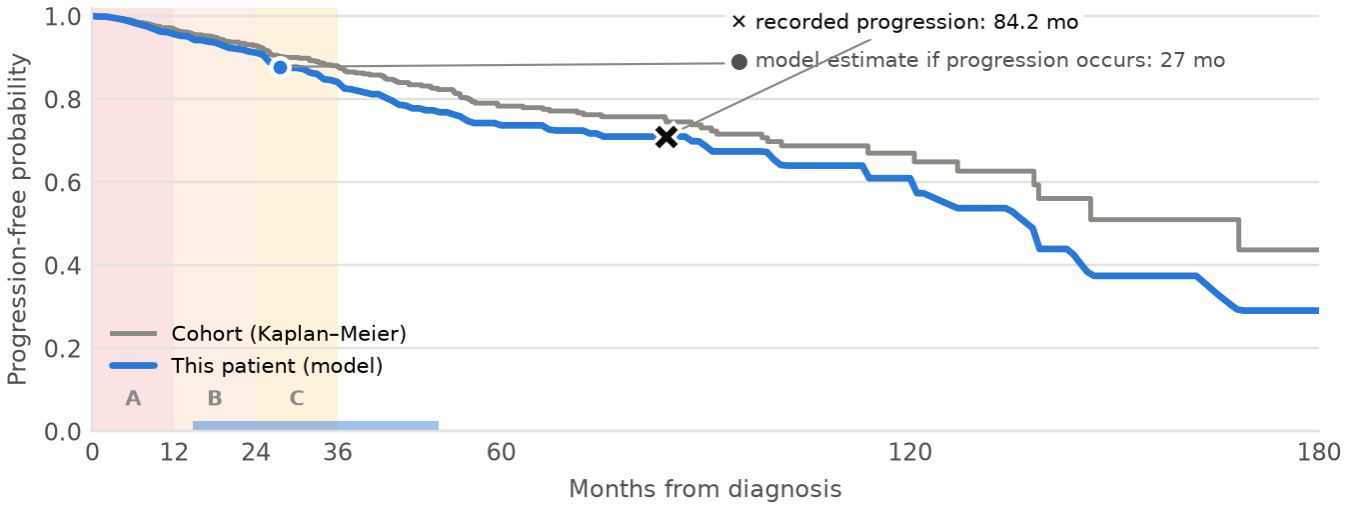
≈ 27 months

Window 24–36 months · IQR 15–51 mo
if progression occurs

Recorded progression

84.2 months

Group D · Late progression
✗ recorded window differs · |error| 57 mo



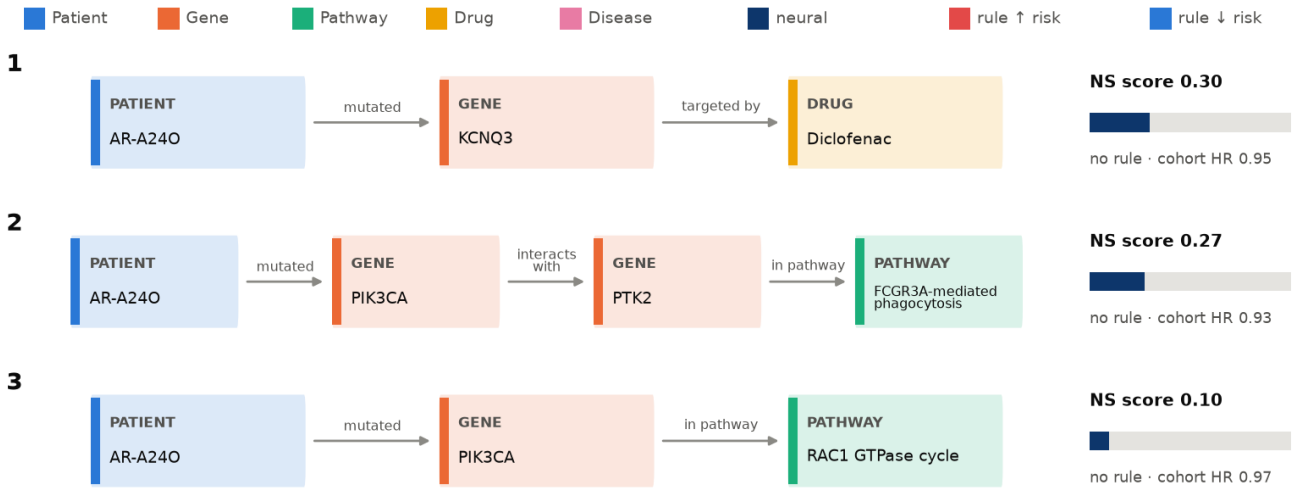
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 43-year-old patient has a LumA tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **81st risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T3)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.026 per SD). The highest-scoring knowledge-graph path runs from **KCNQ3** to **Diclofenac**. Progression was recorded at 84.2 months; the model's estimate given progression was 27 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

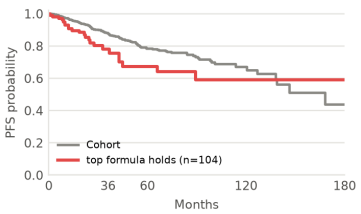


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A24O** mutated → **KCNQ3** targeted by → **Diclofenac**	50	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 107 carriers	**0.30**
2	**TCGA-AR-A24O** mutated → **PIK3CA** interacts with → **PTK2** in pathway → **FCGR3A-mediated phagocytosis**	45	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 100 carriers	**0.27**
3	**TCGA-AR-A24O** mutated → **PIK3CA** in pathway → **RAC1 GTPase cycle**	16	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 428 carriers	**0.10**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

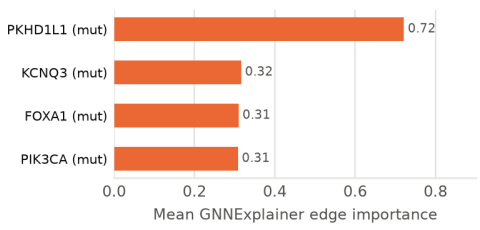
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.026 ↑ risk	TStage(p, T3) ⇒ Progress(p)	single atom	102 pts
+0.008 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.I176V	9	0.31
PIK3CA · driver	Mutation	p.H1047L	70	0.31
PKHD1L1	Mutation	p.Y1437=	0	0.72
KCNQ3	Mutation	p.T655M	8	0.32



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); KCNQ3 → Amitriptyline (DB00321); KCNQ3 → Diclofenac (DB00586); KCNQ3 → Meclofenamic acid (DB00939) … and 3 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YL	48	III	LumA	0.74	no progression, followed 48.5 mo
TCGA-AR-A24M	38	III	LumA	0.74	no progression, followed 120.3 mo
TCGA-B6-A0RV	42	III	LumA	0.73	no progression, followed 169.5 mo
TCGA-C8-A133	65	III	LumA	0.73	no progression, followed 0.0 mo
TCGA-B6-A0IE	38	III	LumA	0.72	progressed at 27.4 mo (group C)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**—**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

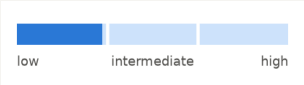
Sample: **TCGA-BH-A0C3-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **0.37** · MSIsensor score: **0.05** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

31st

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

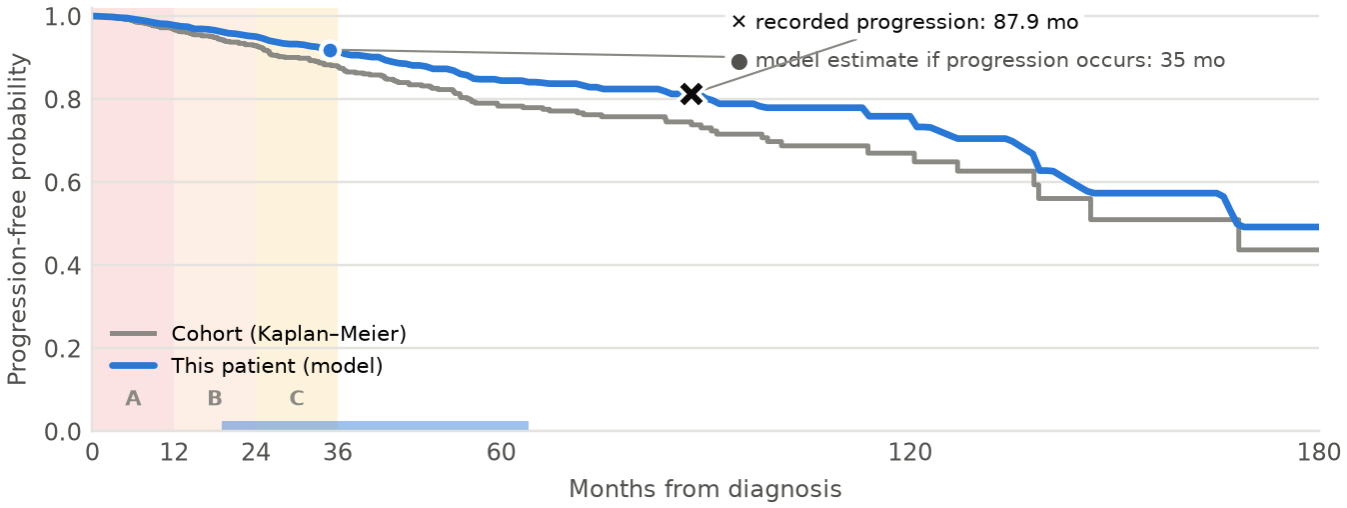
≈ 35 months

Window 24–36 months · IQR 19–64 mo
if progression occurs

Recorded progression

87.9 months

Group D · Late progression
✗ recorded window differs · |error| 53 mo



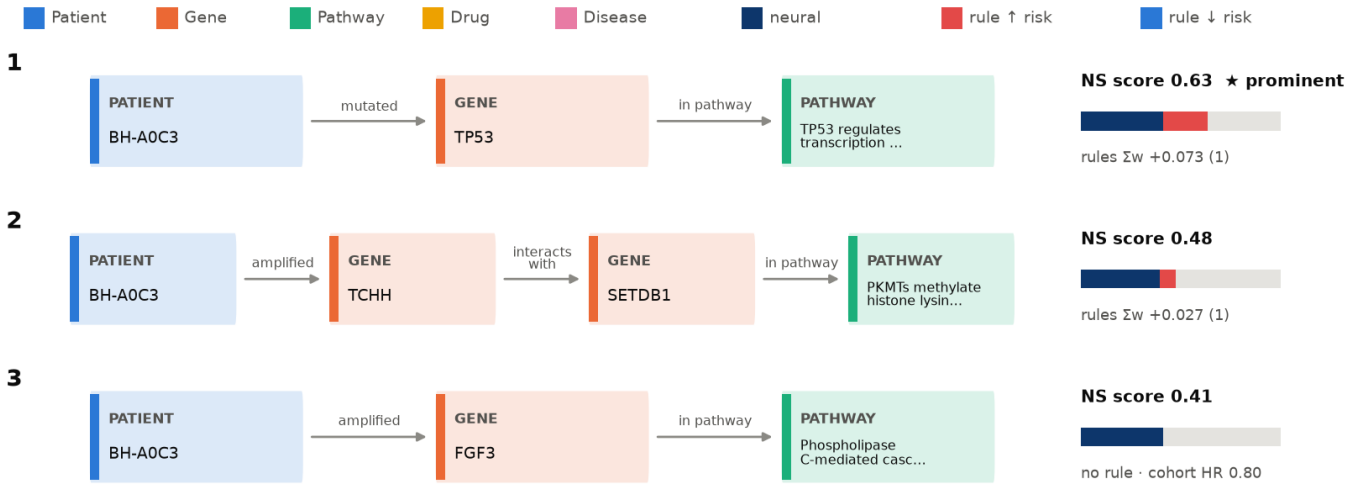
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a LumB tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **31st risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.117 per SD). The highest-scoring knowledge-graph path runs from **TP53** to **TP53 regulates transcription of additional cell cycle genes whose exact role in the p53 pathway remain uncertain**; the learned formulas it instantiates point to faster progression (Σw +0.073). Progression was recorded at 87.9 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

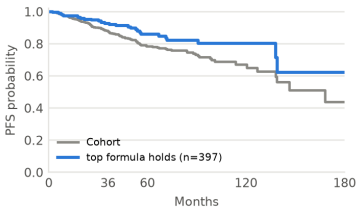


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A0C3** mutated → **TP53** in pathway → **TP53 regulates transcription of additional cell cycle genes whose exact role in the p53 pathway remain uncertain**	69	↑ **faster progression** · Σw +0.073 over 1 formula(s) LowTmb(p) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 0.95, n = 283 carriers	**0.63**
2	**TCGA-BH-A0C3** amplified → **TCHH** interacts with → **SETDB1** in pathway → **PKMTs methylate histone lysines**	66	↑ **faster progression** · Σw +0.027 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) cohort: HR 1.01, n = 324 carriers	**0.48**
3	**TCGA-BH-A0C3** amplified → **FGF3** in pathway → **Phospholipase C-mediated cascade; FGFR2**	69	No learned formula fires on this path; its score is neural only. cohort: HR 0.80, n = 126 carriers	**0.41**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

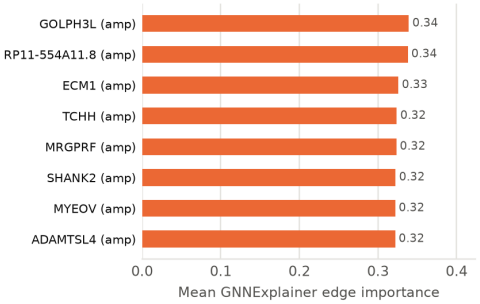
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
+0.073 ↑ risk	LowTmb(p) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	73 pts
-0.051 ↓ risk	Stage(p, I) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of TNFR1 signaling') ⇒ Progress(p) [KG]	mined pair	26 pts
+0.027 ↑ risk	Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	RL-proposed	95 pts
+0.019 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts
-0.008 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	132 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.G187S	57	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
RUNX1 · driver	Mutation	p.D198G	18	0.31
GOLPH3L	High-level amplification	GISTIC ±2	0	0.34
RP11-554A11.8	High-level amplification	GISTIC ±2	0	0.34
ECM1	High-level amplification	GISTIC ±2	1	0.33
TCHH	High-level amplification	GISTIC ±2	1	0.32
MRGPRF	High-level amplification	GISTIC ±2	0	0.32
SHANK2	High-level amplification	GISTIC ±2	1	0.32
MYEOV	High-level amplification	GISTIC ±2	0	0.32
ADAMTSL4	High-level amplification	GISTIC ±2	2	0.32
CA14	High-level amplification	GISTIC ±2	6	0.32



35 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169) … and 15 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-Z7-A8R6	46	I	LumB	0.77	no progression, followed 107.0 mo
TCGA-AR-A1AP	80	I	LumA	0.74	no progression, followed 93.9 mo
TCGA-GM-A2DL	50	I	LumA	0.69	no progression, followed 115.7 mo
TCGA-A2-A0YF	67	I	LumA	0.68	no progression, followed 50.5 mo
TCGA-A1-A0SE	56	I	LumA	0.68	no progression, followed 43.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0IB · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**64**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

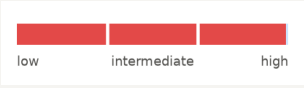
Sample: **TCGA-B6-A0IB-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **2.2** · MSIsensor score: **0.25** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

68% by 36 mo

12 mo 25% · 24 mo 49% · 60 mo 88%

Model-estimated progression time

≈ 8 months

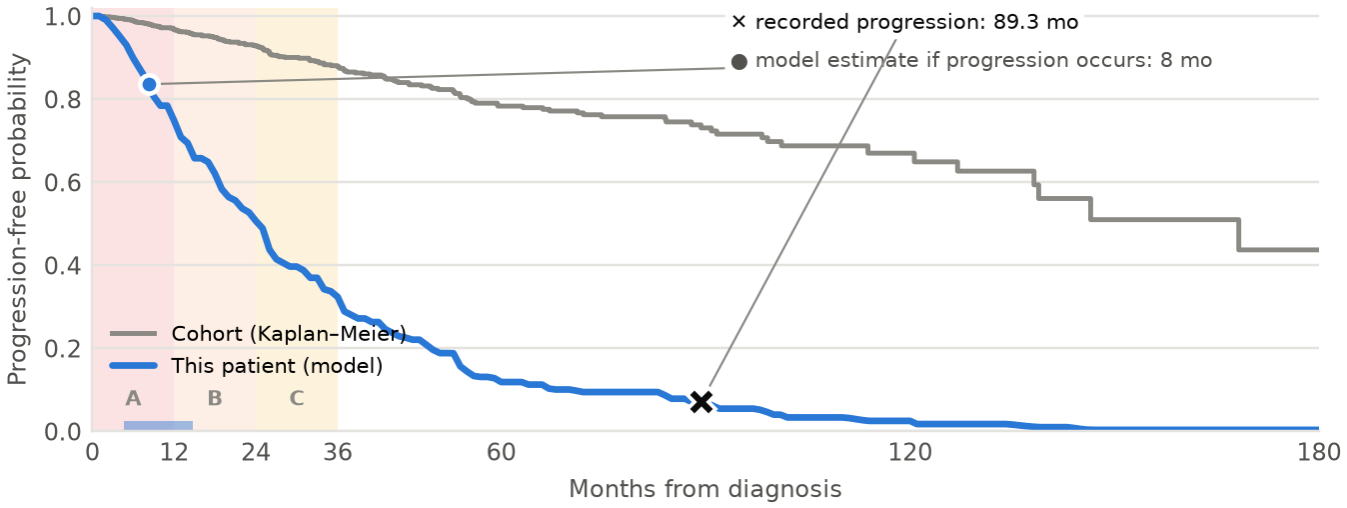
Window ≤ 12 months · IQR 5–15 mo if progression occurs

Recorded progression

89.3 months

Group D · Late progression

✗ **recorded window differs** · |error| 81 mo



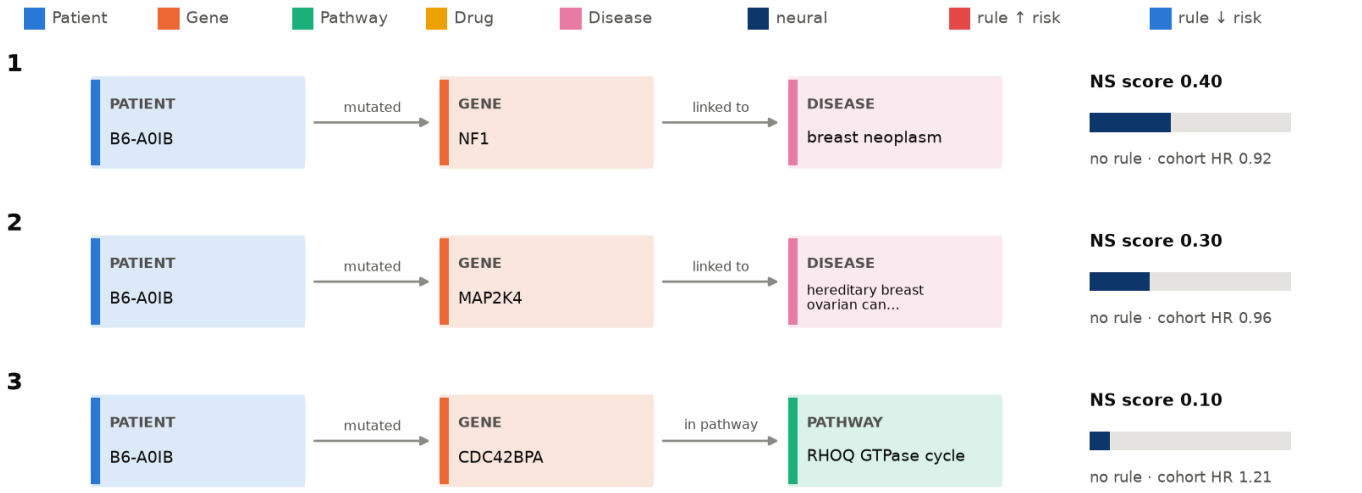
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 64-year-old patient has a LumB tumour, AJCC stage IV (T3, N3, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N3) ⇒ Progress(p)*) raises the progression hazard (weight +0.098 per SD). The highest-scoring knowledge-graph path runs from **NF1** to **breast neoplasm**. Progression was recorded at 89.3 months; the model's estimate given progression was 8 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

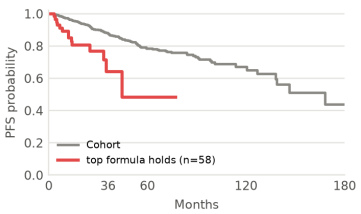


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0IB** mutated → **NF1** linked to → **breast neoplasm**	67	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 793 carriers	**0.40**
2	**TCGA-B6-A0IB** mutated → **MAP2K4** linked to → **hereditary breast ovarian cancer syndrome**	50	No learned formula fires on this path; its score is neural only. cohort: HR 0.96, n = 798 carriers	**0.30**
3	**TCGA-B6-A0IB** mutated → **CDC42BPA** in pathway → **RHOQ GTPase cycle**	17	No learned formula fires on this path; its score is neural only. cohort: HR 1.21, n = 217 carriers	**0.10**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

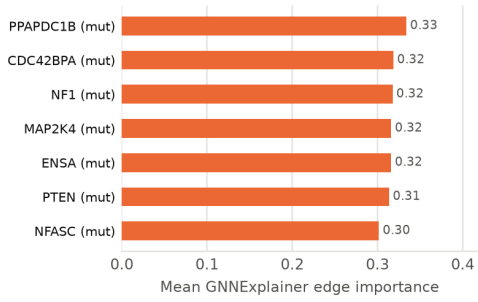
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.098 ↑ risk	NStage(p, N3) ⇒ Progress(p)	single atom	57 pts
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
NF1 · driver	Mutation	p.E1894Q	6	0.32
MAP2K4 · driver	Mutation	p.X73_splice	12	0.32
PTEN · driver	Mutation	p.X165_splice	22	0.31
PPAPDC1B	Mutation	p.D82=	0	0.33
CDC42BPA	Mutation	p.R1708W	4	0.32
ENSA	Mutation	p.L37V	1	0.32
NFASC	Mutation	p.K321R	3	0.30



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CS	73	IV	LumA	0.59	no progression, followed 77.2 mo
TCGA-A2-A0T2	66	IV	Basal	0.58	progressed at 8.4 mo (group A)
TCGA-A8-A07W	76	IV	LumB	0.54	no progression, followed 10.0 mo
TCGA-A2-A0SV	63	IV	LumB	0.50	progressed at 27.1 mo (group C)
TCGA-A2-A0SW	82	IV	LumB	0.50	progressed at 21.5 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**46**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

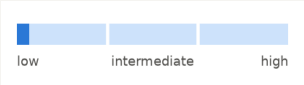
Sample: **TCGA-BH-A18K-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **1.67** · MSIsensor score: **0.3** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

5th

Low risk tier · fused neuro-symbolic model



Probability of progression

6% by 36 mo

12 mo 1% · 24 mo 3% · 60 mo 13%

Model-estimated progression time

≈ 36 months

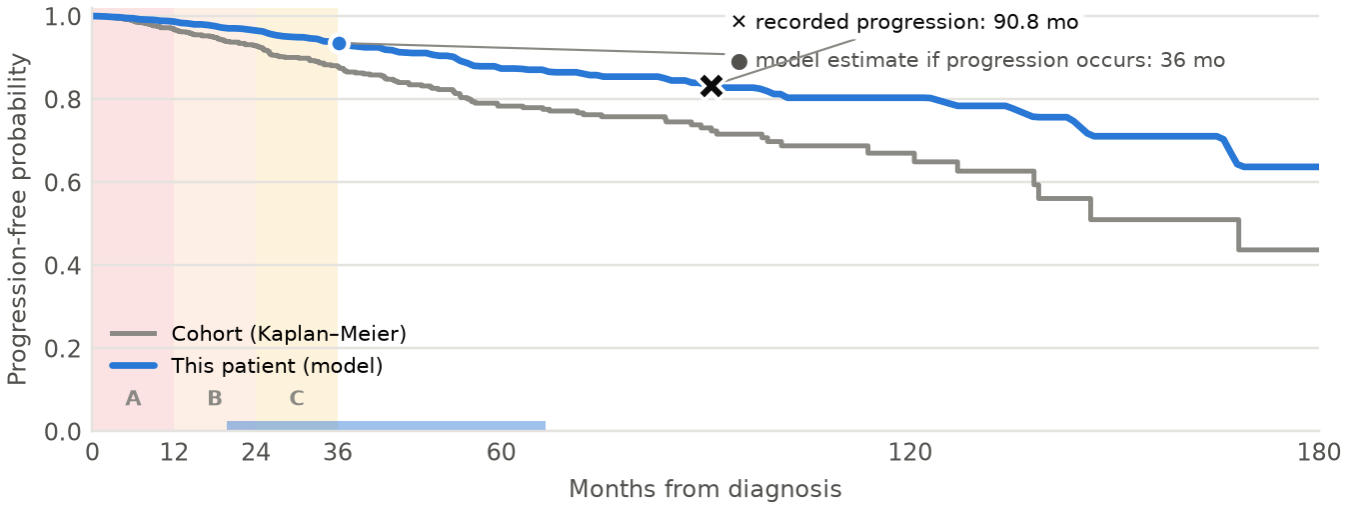
Window > 36 months · IQR 20–67 mo if progression occurs

Recorded progression

90.8 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 55 mo



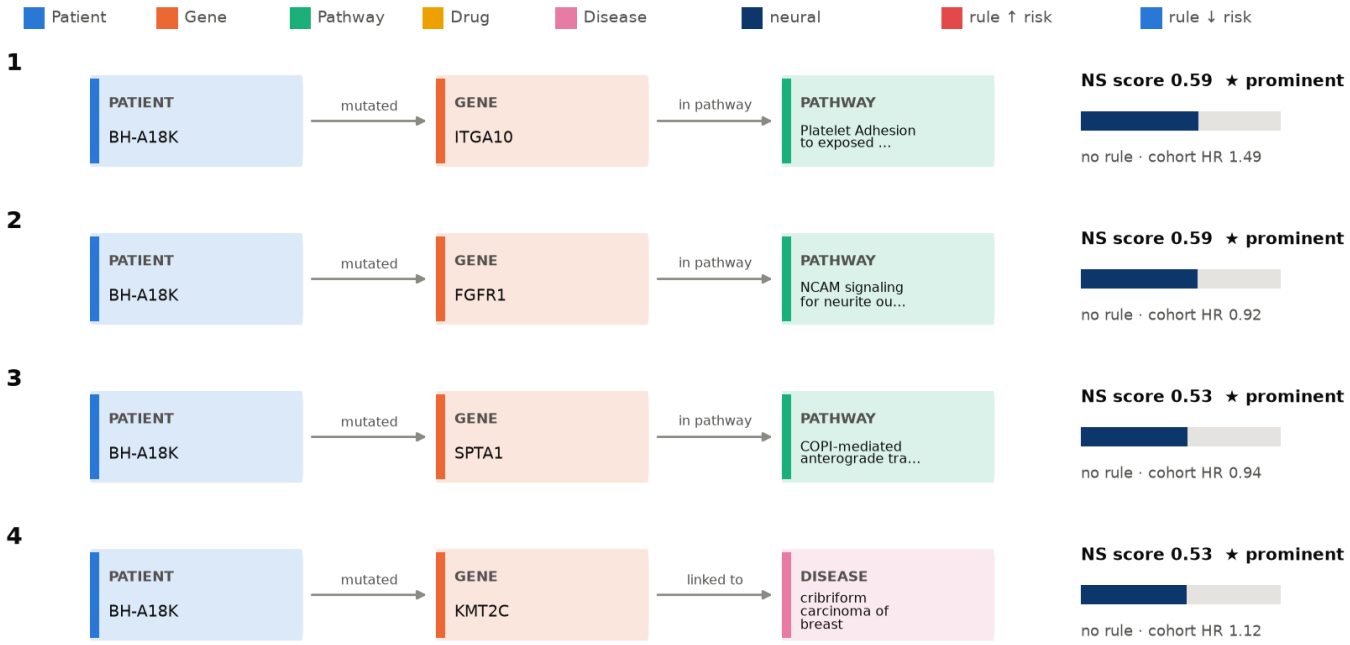
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 46-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **5th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **ITGA10** to **Platelet Adhesion to exposed collagen**. Progression was recorded at 90.8 months; the model's estimate given progression was 36 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

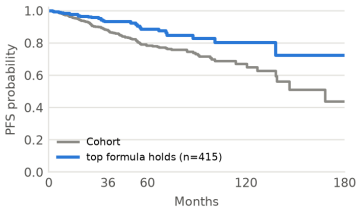


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A18K** mutated → **ITGA10** in pathway → **Platelet Adhesion to exposed collagen**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.49, n = 93 carriers	**0.59**
2 ★	**TCGA-BH-A18K** mutated → **FGFR1** in pathway → **NCAM signaling for neurite out-growth**	98	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 282 carriers	**0.59**
3 ★	**TCGA-BH-A18K** mutated → **SPTA1** in pathway → **COPI-mediated anterograde transport**	89	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 196 carriers	**0.53**
4 ★	**TCGA-BH-A18K** mutated → **KMT2C** linked to → **cribriform carcinoma of breast**	89	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 662 carriers	**0.53**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

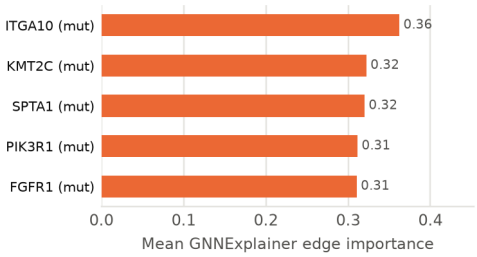
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
-0.069 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	133 pts
-0.032 ↓ risk	NStage(p, N0) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	194 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.V3284F	4	0.32
PIK3R1 · driver	Mutation	p.D569dup	79	0.31
FGFR1 · driver	Mutation	p.I788=	43	0.31
ITGA10	Mutation	p.L990=	3	0.36
SPTA1	Mutation	p.D420=	4	0.32



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486); PIK3R1 → Wortmannin (DB08059); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525) … and 10 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0EM	73	I	LumA	0.77	no progression, followed 101.7 mo
TCGA-BH-A0BO	54	I	LumA	0.77	no progression, followed 72.2 mo
TCGA-E2-A1IO	37	I	LumA	0.76	no progression, followed 61.0 mo
TCGA-BH-A1ET	55	I	LumA	0.76	no progression, followed 82.8 mo
TCGA-BH-A0EB	69	I	LumA	0.76	no progression, followed 24.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

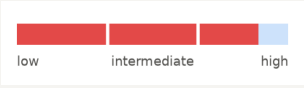
Sample: **TCGA-B6-A0RP-01** · Aneuploidy score: **20** · TMB (nonsyn., /Mb): **0.7** · MSIsensor score: **0.01** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

89th

High risk tier · fused neuro-symbolic model



Probability of progression

19% by 36 mo

12 mo 5% · 24 mo 10% · 60 mo 31%

Model-estimated progression time

≈ 14 months

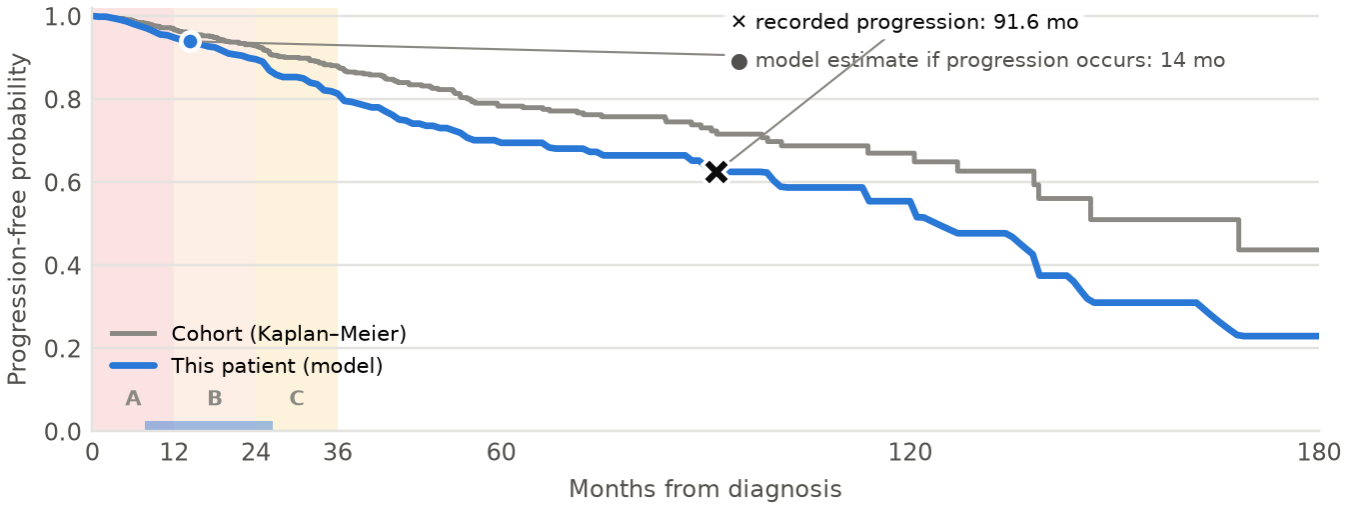
Window 12–24 months · IQR 8–26 mo if progression occurs

Recorded progression

91.6 months

Group D · Late progression

✗ recorded window differs · |error| 77 mo



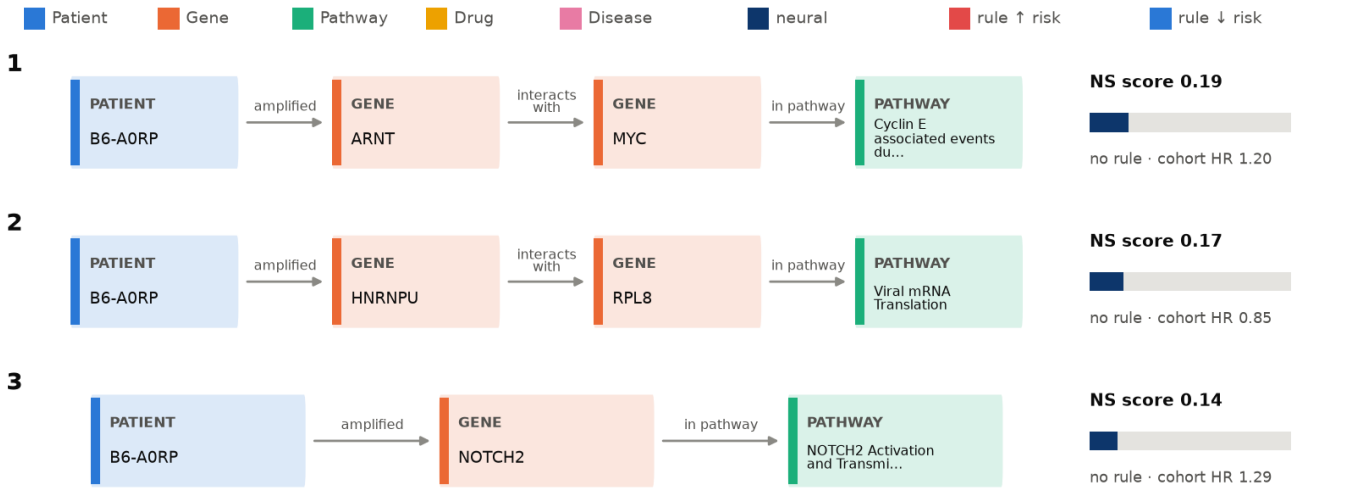
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a LumA tumour, AJCC stage III (T2, N1, M0). The fused neuro-symbolic model places them in the **89th risk percentile** (high tier). The strongest learned formula that holds for them (*Stage(p, III)* ∧ *Amplified(p, FAM72C)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.098 per SD). The highest-scoring knowledge-graph path runs from **ARNT** to **Cyclin E associated events during G1/S transition** . Progression was recorded at 91.6 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

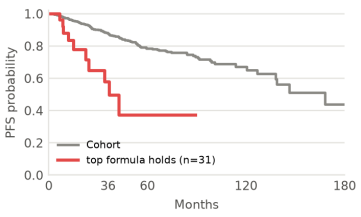


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0RP** amplified → **ARNT** interacts with → **MYC** in pathway → **Cyclin E associated events during G1/S transition**	32	No learned formula fires on this path; its score is neural only. cohort: HR 1.20, n = 260 carriers	**0.19**
2	**TCGA-B6-A0RP** amplified → **HNRNPU** interacts with → **RPL8** in pathway → **Viral mRNA Translation**	28	No learned formula fires on this path; its score is neural only. cohort: HR 0.85, n = 89 carriers	**0.17**
3	**TCGA-B6-A0RP** amplified → **NOTCH2** in pathway → **NOTCH2 Activation and Transmission of Signal to the Nucleus**	23	No learned formula fires on this path; its score is neural only. cohort: HR 1.29, n = 104 carriers	**0.14**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

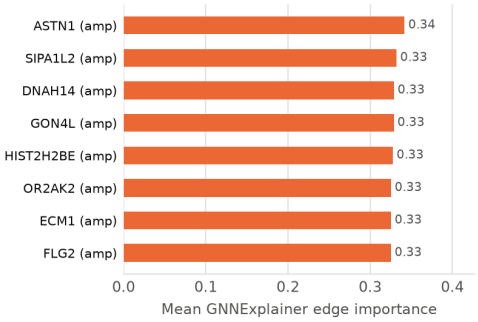
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.098 ↑ risk	Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	mined pair	31 pts
+0.012 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	mined pair	45 pts
+0.008 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts
-0.001 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	147 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.D4425Y	4	0.32
PIK3CA · driver	Mutation	p.N345K	70	0.32
ASTN1	High-level amplification	GISTIC ±2	0	0.34
SIPA1L2	High-level amplification	GISTIC ±2	0	0.33
DNAH14	High-level amplification	GISTIC ±2	0	0.33
GON4L	High-level amplification	GISTIC ±2	0	0.33
HIST2H2BE	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
ECM1	High-level amplification	GISTIC ±2	1	0.33
FLG2	High-level amplification	GISTIC ±2	1	0.33
FAM72C	High-level amplification	GISTIC ±2	0	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32



111 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); CACNA1E → Ergocalciferol (DB00153); CACNA1E → Enflurane (DB00228); CACNA1E → Topiramate (DB00273) … and 55 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AN-A0XP	69	III	LumA	0.91	no progression, followed 0.3 mo
TCGA-A8-A09B	58	III	LumA	0.87	no progression, followed 12.0 mo
TCGA-BH-A0BA	51	III	LumA	0.87	no progression, followed 37.2 mo
TCGA-D8-A1XG	86	III	LumA	0.86	no progression, followed 14.7 mo
TCGA-A8-A07G	65	II	LumA	0.85	no progression, followed 19.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**34**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A25E-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.7** · MSIsensor score: **1.75** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

75th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 25%

Model-estimated progression time

≈ 42 months

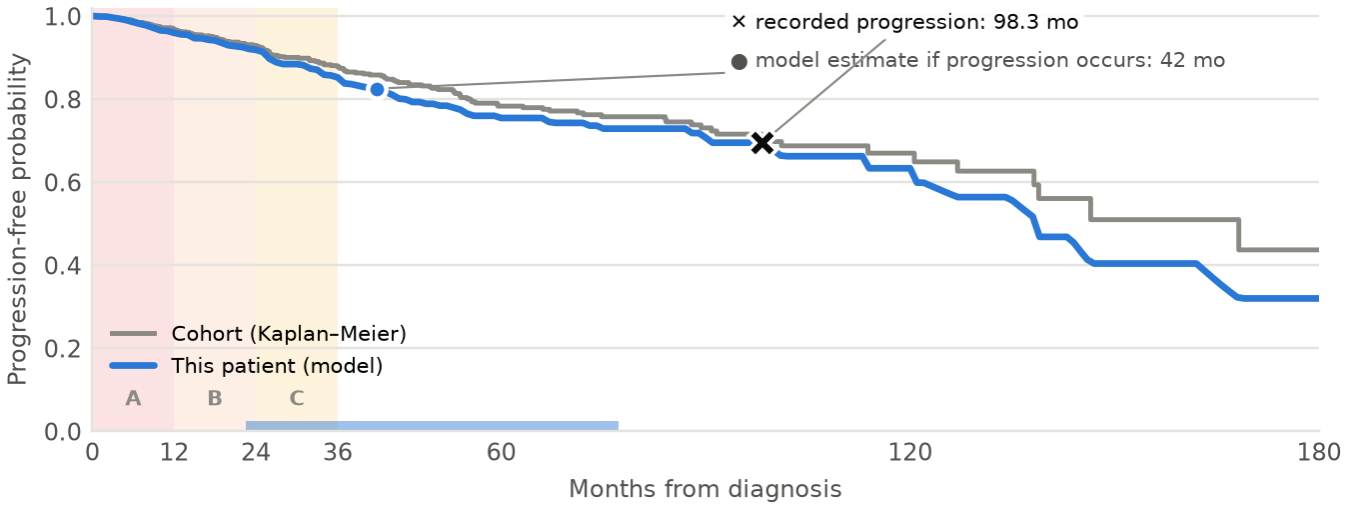
Window > 36 months · IQR 23–77 mo if progression occurs

Recorded progression

98.3 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 57 mo



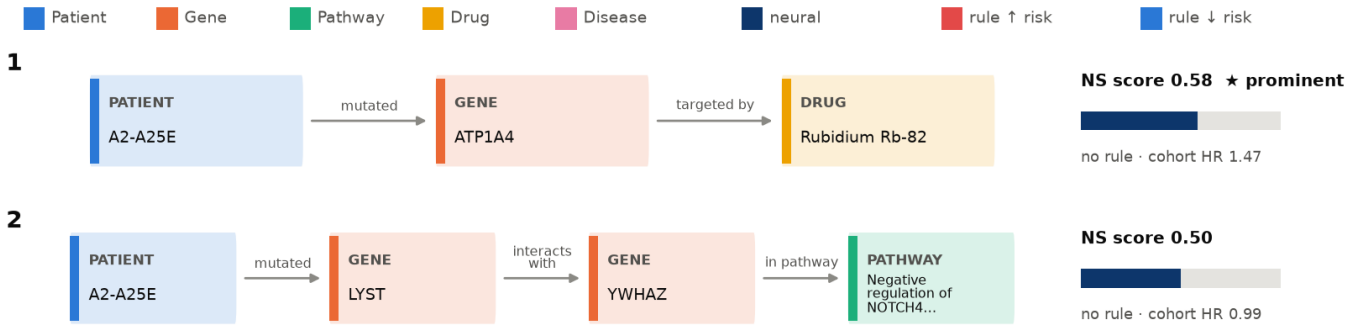
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 34-year-old patient has a LumB tumour, AJCC stage III (T2, N2, M0). The fused neuro-symbolic model places them in the **75th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **ATP1A4** to **Rubidium Rb-82**. Progression was recorded at 98.3 months; the model's estimate given progression was 42 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 1 of the 19 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

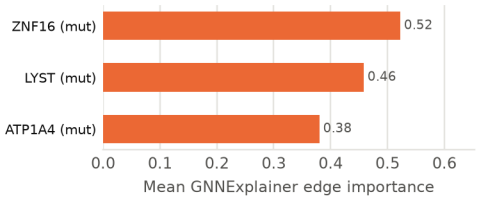
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A25E** mutated → **ATP1A4** targeted by → **Rubidium Rb-82**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.47, n = 91 carriers	**0.58**
2	**TCGA-A2-A25E** mutated → **LYST** interacts with → **YWHAZ** in pathway → **Negative regulation of NOTCH4 signaling**	83	No learned formula fires on this path; its score is neural only. cohort: HR 0.99, n = 125 carriers	**0.50**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ZNF16	Mutation	p.D184N	0	0.52
LYST	Mutation	p.N3046D	0	0.46
ATP1A4	Mutation	p.A328G	4	0.38



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ATP1A4 → Rubidium Rb-82 (DB09479). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A0TT	53	III	LumB	0.88	no progression, followed 109.0 mo
TCGA-D8-A27N	36	III	LumB	0.79	no progression, followed 17.1 mo
TCGA-A2-A1G6	50	III	Normal	0.79	no progression, followed 16.5 mo
TCGA-AR-A1AU	39	III	LumA	0.76	no progression, followed 94.3 mo
TCGA-C8-A1HI	40	III	LumA	0.74	no progression, followed 11.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**46**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

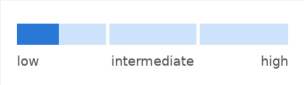
Sample: **TCGA-GM-A2DA-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **0.08** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

16th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 14%

Model-estimated progression time

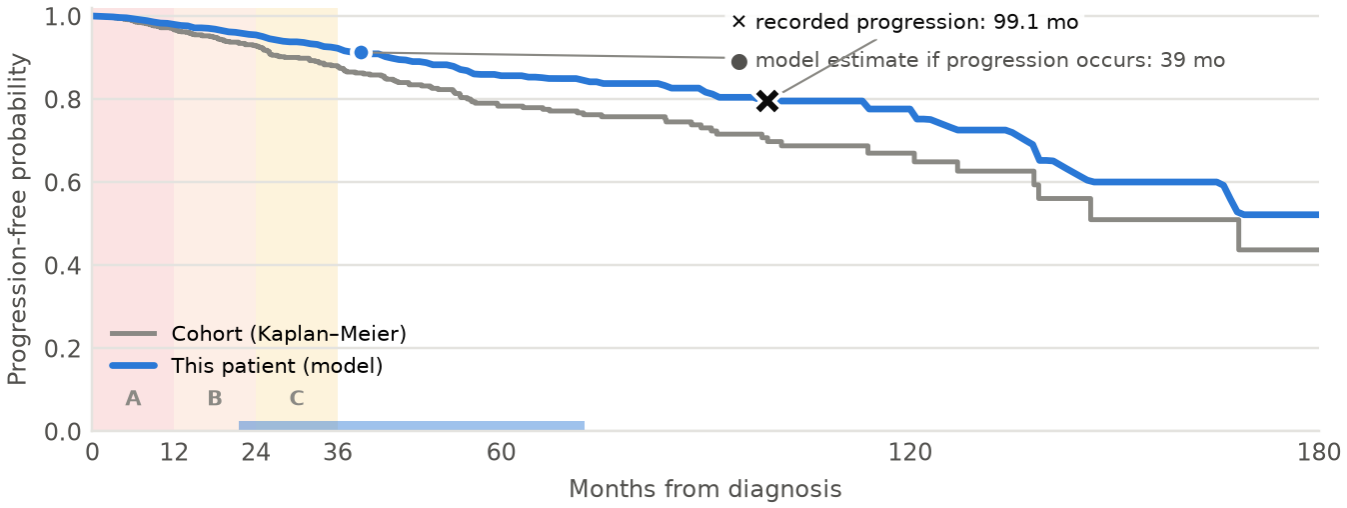
≈ 39 months

Window > 36 months · IQR 21–72 mo if progression occurs

Recorded progression

99.1 months

Group D · Late progression
✓ estimate in recorded window · |error| 60 mo



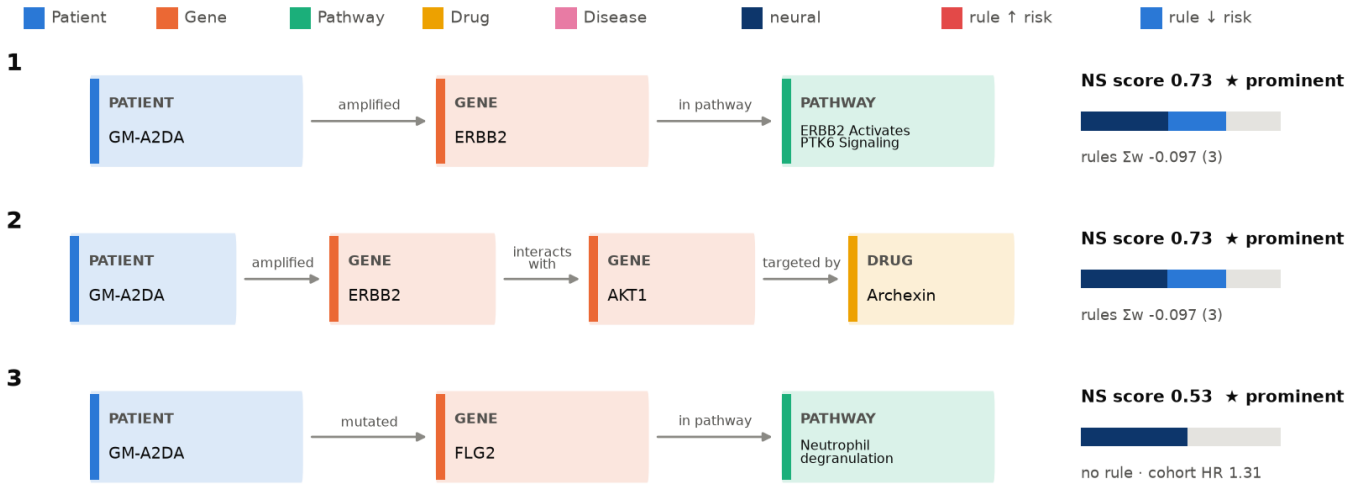
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 46-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **16th risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **ERBB2** to **ERBB2 Activates PTK6 Signaling**; the learned formulas it instantiates point to slower progression (Σw -0.097). Progression was recorded at 99.1 months; the model's estimate given progression was 39 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

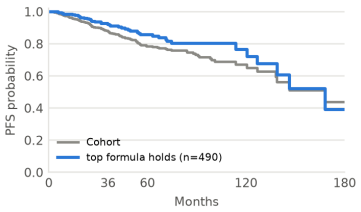


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-GM-A2DA** amplified → **ERBB2** in pathway → **ERBB2 Activates PTK6 Signaling**	73	↓ **slower progression** · Σw -0.097 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 1.00, n = 154 carriers	**0.73**
2 ★	**TCGA-GM-A2DA** amplified → **ERBB2** interacts with → **AKT1** targeted by → **Archexin**	72	↓ **slower progression** · Σw -0.097 over 3 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.19, n = 35 carriers	**0.73**
3 ★	**TCGA-GM-A2DA** mutated → **FLG2** in pathway → **Neutrophil degranulation**	89	No learned formula fires on this path; its score is neural only. cohort: HR 1.31, n = 320 carriers	**0.53**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

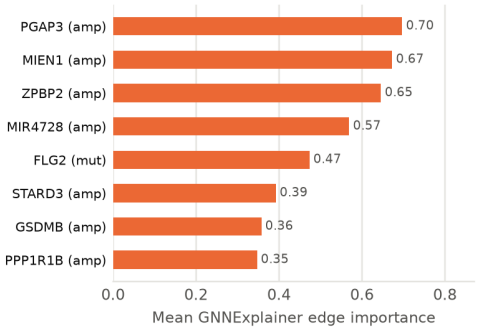
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
-0.007 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	306 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
PGAP3	High-level amplification	GISTIC ±2	0	0.70
MIEN1	High-level amplification	GISTIC ±2	0	0.67
ZPBP2	High-level amplification	GISTIC ±2	0	0.65
MIR4728	High-level amplification	GISTIC ±2	0	0.57
FLG2	Mutation	p.G1757A	1	0.47
STARD3	High-level amplification	GISTIC ±2	1	0.39
GSDMB	High-level amplification	GISTIC ±2	4	0.36
PPP1R1B	High-level amplification	GISTIC ±2	1	0.35
MUC16	Mutation	p.G14473S	8	0.32
GRB7	High-level amplification	GISTIC ±2	12	0.32
CDH17	Mutation	p.V294=	1	0.32



17 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 20 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A132	56	II	LumA	0.94	no progression, followed 12.6 mo
TCGA-E2-A1LB	41	II	Her2	0.87	no progression, followed 75.8 mo
TCGA-A8-A07P	68	II	LumA	0.83	no progression, followed 11.0 mo
TCGA-E2-A15H	38	II	LumA	0.78	no progression, followed 12.9 mo
TCGA-E9-A245	47	II	LumA	0.78	no progression, followed 0.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8523/3**

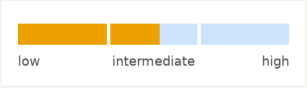
Sample: **TCGA-B6-A0RU-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **1.53** · MSIsensor score: **0.25** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

52nd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 18%

Model-estimated progression time

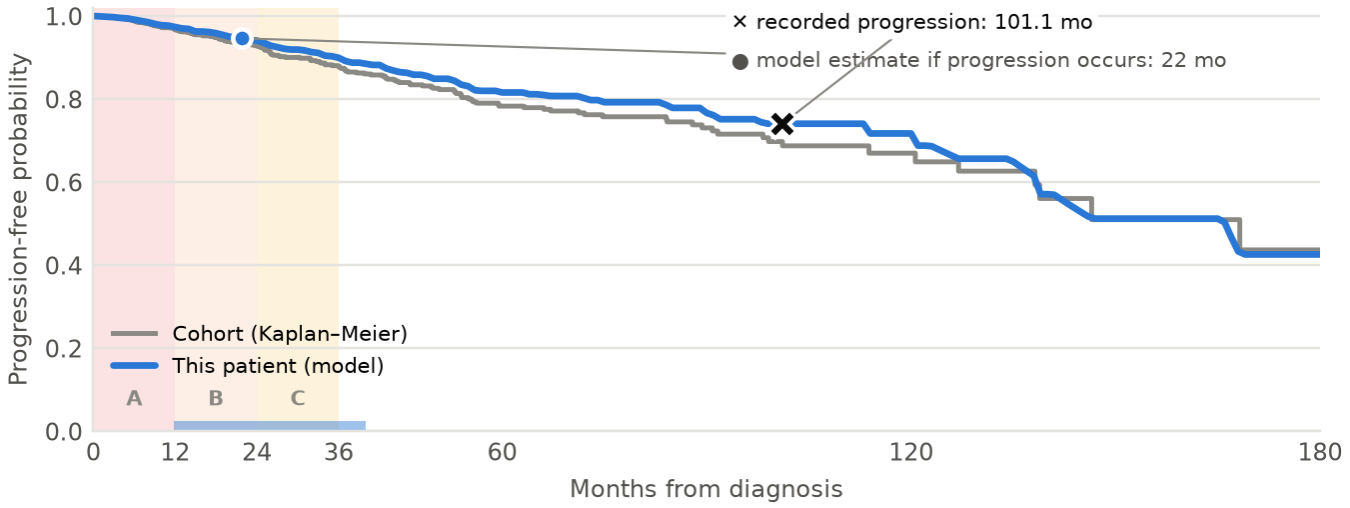
≈ 22 months

Window 12–24 months · IQR 12–40 mo
if progression occurs

Recorded progression

101.1 months

Group D · Late progression
✗ recorded window differs · |error| 79 mo



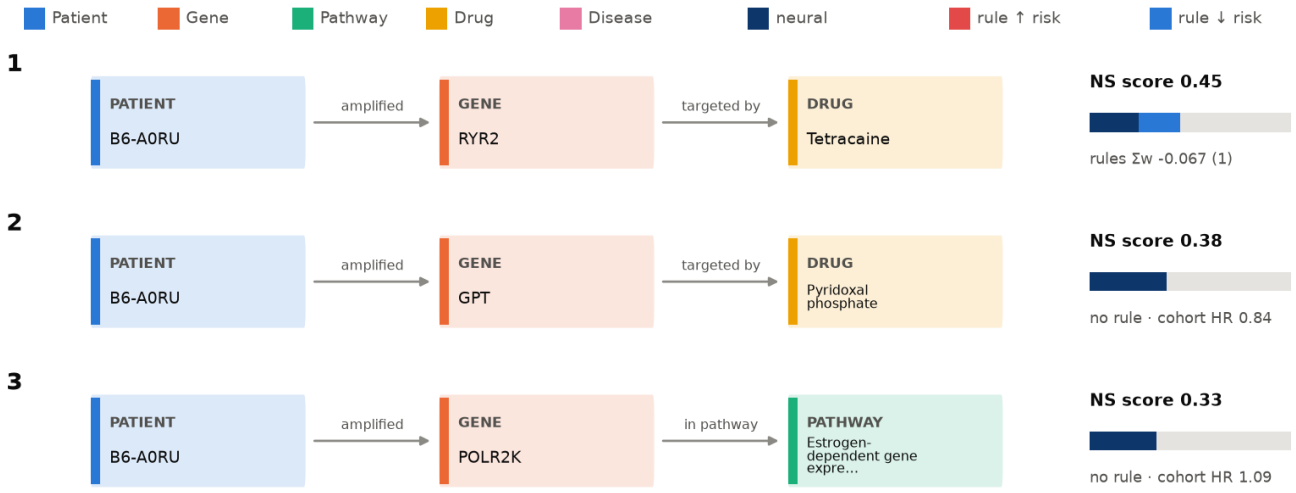
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a Basal tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **52nd risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.117 per SD). The highest-scoring knowledge-graph path runs from **RYR2** to **Tetracaine**; the learned formulas it instantiates point to slower progression (Σw -0.067). Progression was recorded at 101.1 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

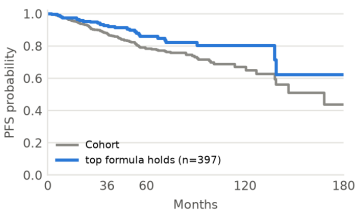


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0RU** amplified → **RYR2** targeted by → **Tetracaine**	41	↓ **slower progression** · Σw -0.067 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) cohort: HR 0.83, n = 146 carriers	**0.45**
2	**TCGA-B6-A0RU** amplified → **GPT** targeted by → **Pyridoxal phosphate**	64	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 97 carriers	**0.38**
3	**TCGA-B6-A0RU** amplified → **POLR2K** in pathway → **Estrogen-dependent gene expression**	55	No learned formula fires on this path; its score is neural only. cohort: HR 1.09, n = 460 carriers	**0.33**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

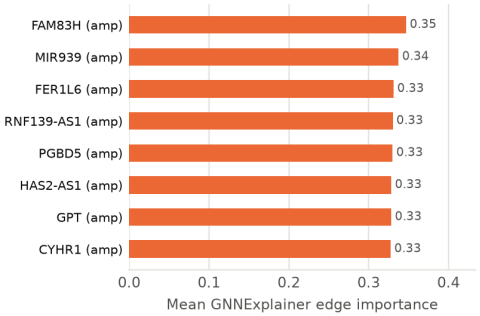
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
+0.079 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Formation of the beta-catenin:TCF transactivating complex') ∧ Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'Purinergic signaling in leishmaniasis infection') ⇒ Progress(p) [KG]	RL-proposed	49 pts
-0.067 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	143 pts
-0.051 ↓ risk	Stage(p, I) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of TNFR1 signaling') ⇒ Progress(p) [KG]	mined pair	26 pts
+0.019 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts
-0.010 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Synthesis of PA') ⇒ Progress(p) [KG]	single atom	187 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R196*	57	0.32
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
FAM83H	High-level amplification	GISTIC ±2	0	0.35
MIR939	High-level amplification	GISTIC ±2	0	0.34
FER1L6	High-level amplification	GISTIC ±2	0	0.33
RNF139-AS1	High-level amplification	GISTIC ±2	0	0.33
PGBD5	High-level amplification	GISTIC ±2	0	0.33
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.33
GPT	High-level amplification	GISTIC ±2	5	0.33
CYHR1	High-level amplification	GISTIC ±2	0	0.33
RNU6-83P	High-level amplification	GISTIC ±2	0	0.33



328 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327); MYC → Acetylsalicylic acid (DB00945) … and 108 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A0J4	41	I	Basal	0.87	no progression, followed 52.2 mo
TCGA-C8-A27B	48	II	Basal	0.82	no progression, followed 14.4 mo
TCGA-B6-A0I1	73	II	Basal	0.81	progressed at 36.8 mo (group D)
TCGA-BH-A0BW	71	I	Basal	0.81	no progression, followed 77.9 mo
TCGA-D8-A27M	59	I	Basal	0.80	no progression, followed 13.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**35**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

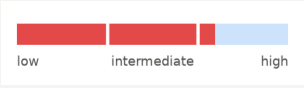
Sample: **TCGA-BH-A1ES-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.25** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

73rd

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 24%

Model-estimated progression time

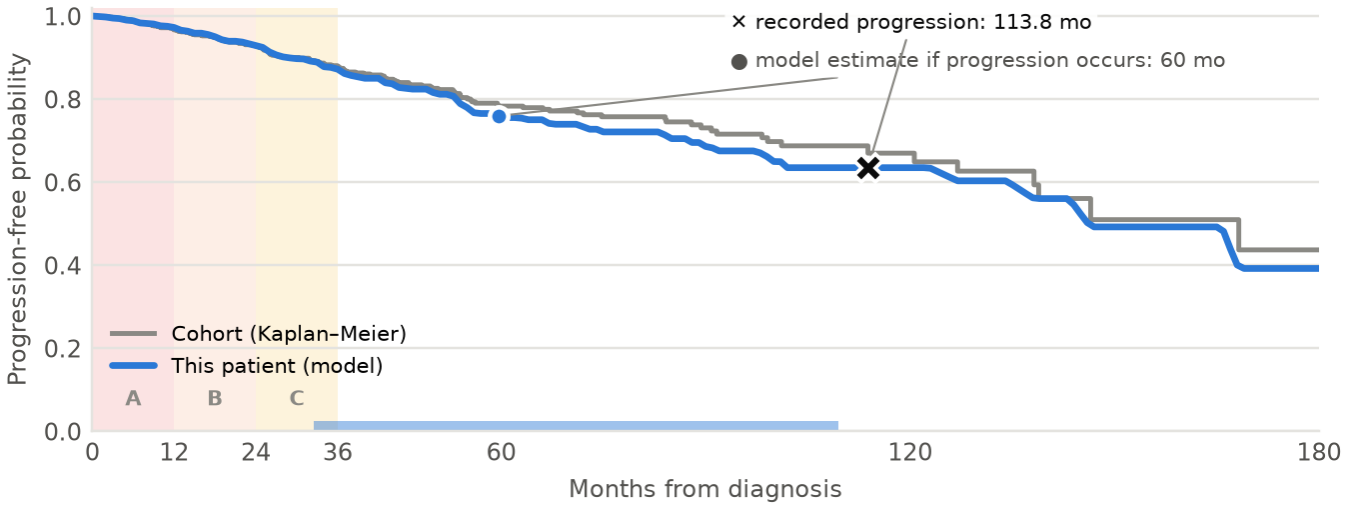
≈ 60 months

Window > 36 months · IQR 32–110 mo
if progression occurs

Recorded progression

113.8 months

Group D · Late progression
✓ **estimate in recorded window** · |error| 54 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 35-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **73rd risk percentile** (high tier). The strongest learned formula that holds for them (*Age(p) < 40 ⇒ Progress(p)*) raises the progression hazard (weight +0.084 per SD). The highest-scoring knowledge-graph path runs from **DDHD2** to **Synthesis of PA**. Progression was recorded at 113.8 months; the model's estimate given progression was 60 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

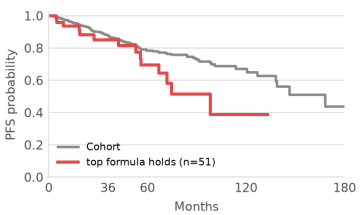


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1ES** amplified → **DDHD2** in pathway → **Synthesis of PA**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 192 carriers	**0.60**
2 ★	**TCGA-BH-A1ES** amplified → **STAR** in pathway → **Pregnenolone biosynthesis**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 196 carriers	**0.59**
3 ★	**TCGA-BH-A1ES** amplified → **FGFR1** targeted by → **Lenvatinib**	98	No learned formula fires on this path; its score is neural only. cohort: HR 0.88, n = 114 carriers	**0.59**
4 ★	**TCGA-BH-A1ES** amplified → **ADAM2** in pathway → **Interaction With Cumulus Cells And The Zona Pellucida**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.28, n = 183 carriers	**0.58**
5 ★	**TCGA-BH-A1ES** amplified → **KCNU1** in pathway → **Sperm Motility And Taxes**	87	No learned formula fires on this path; its score is neural only. cohort: HR 0.82, n = 108 carriers	**0.52**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

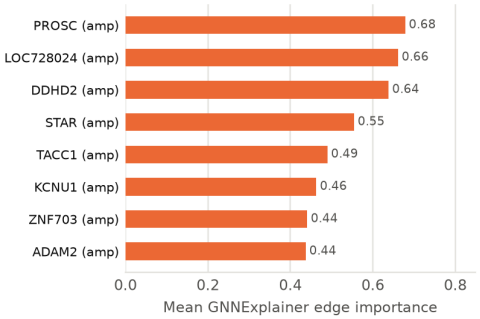
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.084 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	51 pts
+0.072 ↑ risk	LowTmb(p) ∧ NStage(p, N1) ⇒ Progress(p)	mined pair	83 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.42
PROSC	High-level amplification	GISTIC ±2	0	0.68
LOC728024	High-level amplification	GISTIC ±2	0	0.66
DDHD2	High-level amplification	GISTIC ±2	1	0.64
STAR	High-level amplification	GISTIC ±2	1	0.55
TACC1	High-level amplification	GISTIC ±2	0	0.49
KCNU1	High-level amplification	GISTIC ±2	1	0.46
ZNF703	High-level amplification	GISTIC ±2	1	0.44
ADAM2	High-level amplification	GISTIC ±2	1	0.44
RAB11FIP1	High-level amplification	GISTIC ±2	0	0.41
CSMD1	High-level amplification	GISTIC ±2	0	0.40
HTRA4	High-level amplification	GISTIC ±2	0	0.37



31 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 45 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XY	49	II	missing	0.80	progressed at 26.4 mo (group C)
TCGA-HN-A2OB	45	II	LumA	0.80	progressed at 50.7 mo (group D)
TCGA-E2-A1IE	61	II	LumA	0.78	progressed at 59.5 mo (group D)
TCGA-BH-A0EI	51	II	LumA	0.77	no progression, followed 63.3 mo
TCGA-B6-A0WT	61	II	LumA	0.75	progressed at 127.0 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1FB-01** · Aneuploidy score: **—** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **—** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

67th

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 23%

Model-estimated progression time

≈ 40 months

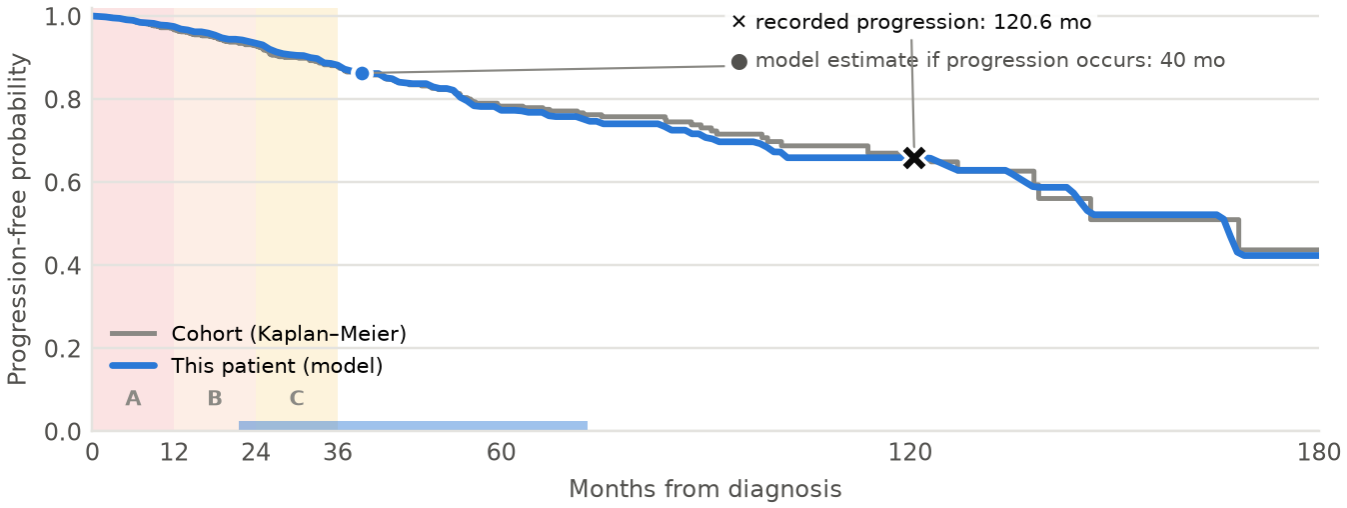
Window > 36 months · IQR 22–73 mo if progression occurs

Recorded progression

120.6 months

Group D · Late progression

✓ **estimate in recorded window** · |error| 81 mo



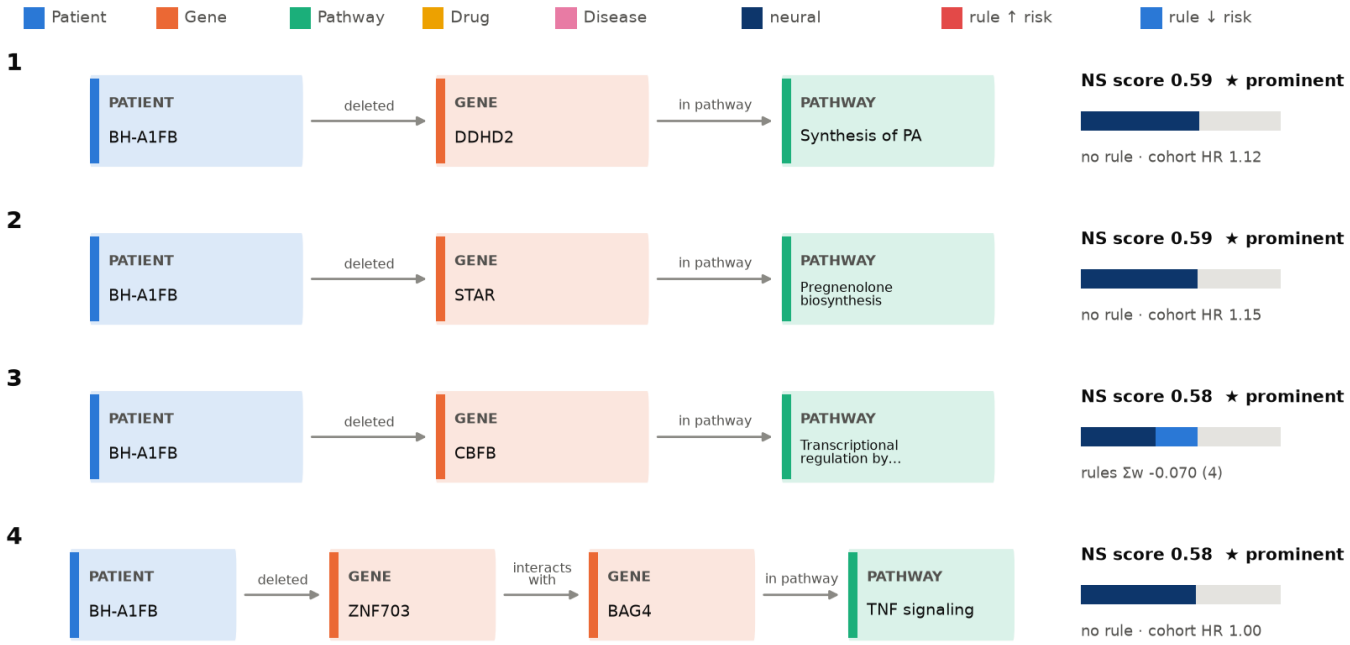
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **67th risk percentile** (high tier). The strongest learned formula that holds for them ($LowTmb(p) \wedge NStage(p, N1) \Rightarrow Progress(p)$) raises the progression hazard (weight +0.072 per SD). The highest-scoring knowledge-graph path runs from **DDHD2** to **Synthesis of PA**. Progression was recorded at 120.6 months; the model's estimate given progression was 40 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

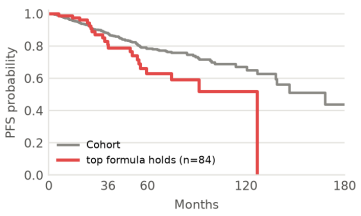


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1FB** deleted → **DDHD2** in pathway → **Synthesis of PA**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 192 carriers	**0.59**
2 ★	**TCGA-BH-A1FB** deleted → **STAR** in pathway → **Pregnenolone biosynthesis**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 196 carriers	**0.59**
3 ★	**TCGA-BH-A1FB** deleted → **CBFB** in pathway → **Transcriptional regulation by RUNX2**	62	↓ **slower progression** · Σw -0.070 over 4 formula(s) Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX2 expression and activity') ⇒ Progress(p) cohort: HR 0.86, n = 164 carriers	**0.58**
4 ★	**TCGA-BH-A1FB** deleted → **ZNF703** interacts with → **BAG4** in pathway → **TNF signaling**	96	No learned formula fires on this path; its score is neural only. cohort: HR 1.00, n = 111 carriers	**0.58**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

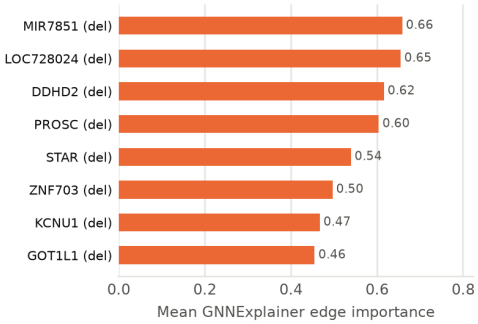
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.072 ↑ risk	LowTmb(p) ∧ NStage(p, N1) ⇒ Progress(p)	mined pair	83 pts
+0.054 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX2 expression and activity') ⇒ Progress(p) [KG]	single atom	57 pts
-0.053 ↓ risk	NStage(p, N1) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX1 Expression and Activity') ⇒ Progress(p) [KG]	mined pair	89 pts
-0.048 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known') ⇒ Progress(p) [KG]	mined pair	68 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
-0.023 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX1 Expression and Activity') ⇒ Progress(p) [KG]	mined pair	162 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CBFB · driver	Homozygous deletion	GISTIC ±2	29	0.32
CDH1 · driver	Homozygous deletion	GISTIC ±2	17	0.31
MIR7851	Homozygous deletion	GISTIC ±2	0	0.66
LOC728024	Homozygous deletion	GISTIC ±2	0	0.65
DDHD2	Homozygous deletion	GISTIC ±2	1	0.62
PROSC	Homozygous deletion	GISTIC ±2	0	0.60
STAR	Homozygous deletion	GISTIC ±2	1	0.54
ZNF703	Homozygous deletion	GISTIC ±2	1	0.50
KCNU1	Homozygous deletion	GISTIC ±2	1	0.47
GOT1L1	Homozygous deletion	GISTIC ±2	0	0.46
BAG4	Homozygous deletion	GISTIC ±2	4	0.42
RAB11FIP1	Homozygous deletion	GISTIC ±2	0	0.38



22 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ADRB3 → Amphetamine (DB00182); ADRB3 → Bethanidine (DB00217); ADRB3 → Olanzapine (DB00334); ADRB3 → Norepinephrine (DB00368); ADRB3 → Nortriptyline (DB00540); ADRB3 → Propranolol (DB00571); ADRB3 → Paroxetine (DB00715); ADRB3 → Trimipramine (DB00726); ADRB3 → Cryptenamine (DB00785); ADRB3 → Alprenolol (DB00866) … and 28 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1FL	69	II	LumA	0.70	progressed at 55.0 mo (group D)
TCGA-BH-A1ES	35	II	missing	0.70	progressed at 113.8 mo (group D)
TCGA-LL-A442	56	II	missing	0.69	no progression, followed 29.2 mo
TCGA-A2-A3XY	49	II	missing	0.68	progressed at 26.4 mo (group C)
TCGA-A1-A0SG	61	II	LumA	0.67	no progression, followed 14.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0WT · Group D

Report date 31 Oct 2026

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

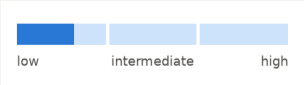
Sample: **TCGA-B6-A0WT-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **0.47** · MSIsensor score: **0.05** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

21st

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

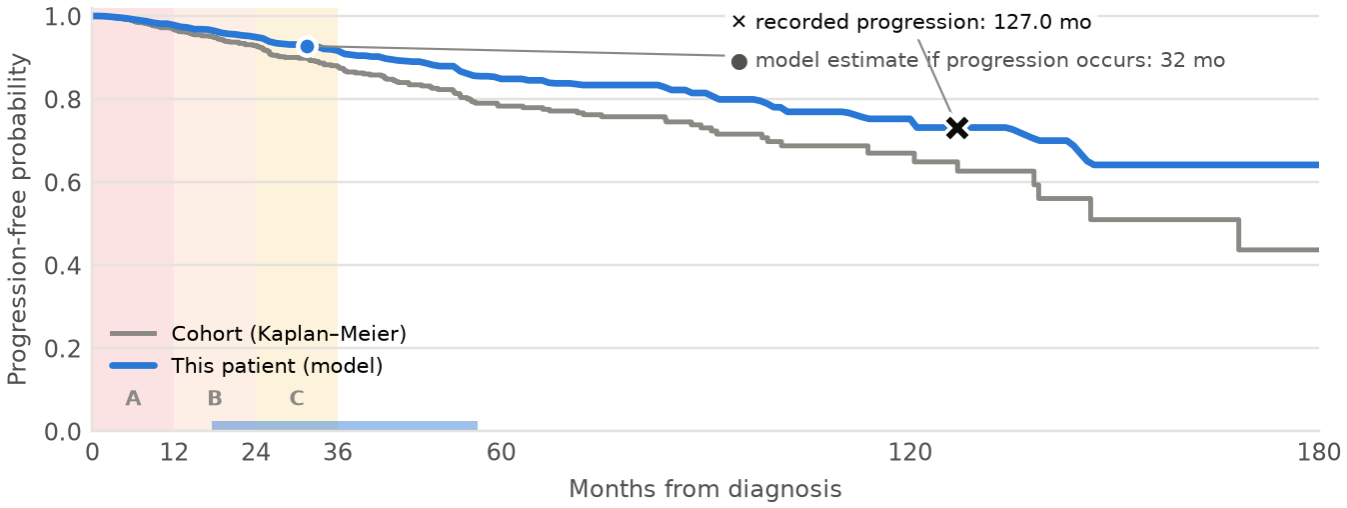
≈ 32 months

Window 24–36 months · IQR 18–57 mo
if progression occurs

Recorded progression

127.0 months

Group D · Late progression
✗ recorded window differs · |error| 95 mo



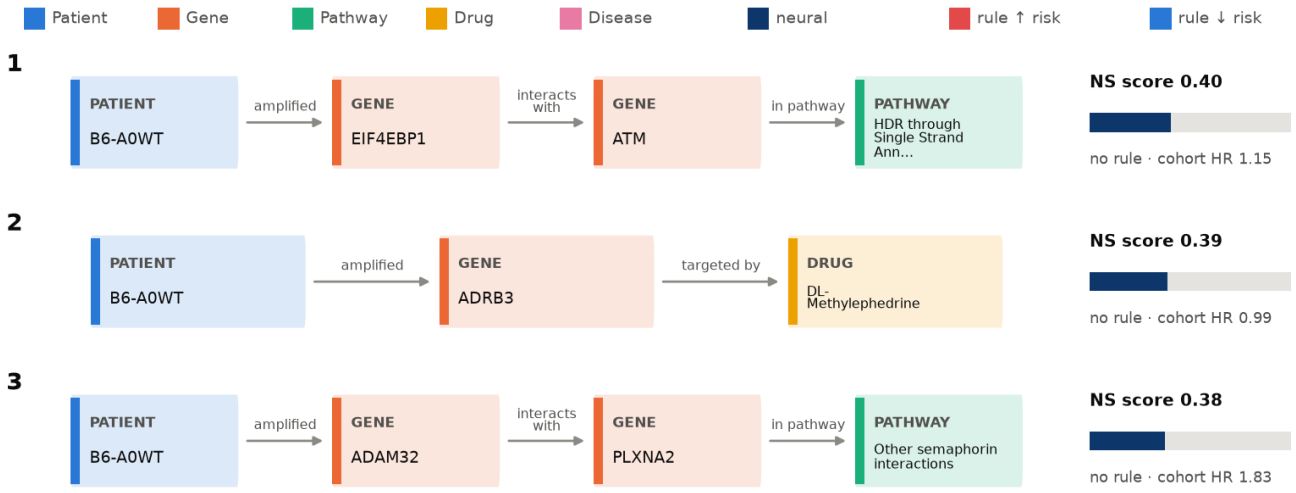
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **21st risk percentile** (low tier). The strongest learned formula that holds for them (*Subtype(p, LumA) ⇒ Progress(p)*) lowers the progression hazard (weight -0.010 per SD). The highest-scoring knowledge-graph path runs from **EIF4EBP1** to **HDR through Single Strand Annealing (SSA)**. Progression was recorded at 127.0 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

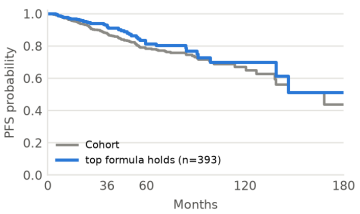


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0WT** amplified → **EIF4EBP1** interacts with → **ATM** in pathway → **HDR through Single Strand Annealing (SSA)**	67	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 158 carriers	**0.40**
2	**TCGA-B6-A0WT** amplified → **ADRB3** targeted by → **DL-Methylephedrine**	65	No learned formula fires on this path; its score is neural only. cohort: HR 0.99, n = 111 carriers	**0.39**
3	**TCGA-B6-A0WT** amplified → **ADAM32** interacts with → **PLXNA2** in pathway → **Other semaphorin interactions**	63	No learned formula fires on this path; its score is neural only. cohort: HR 1.83, n = 92 carriers	**0.38**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

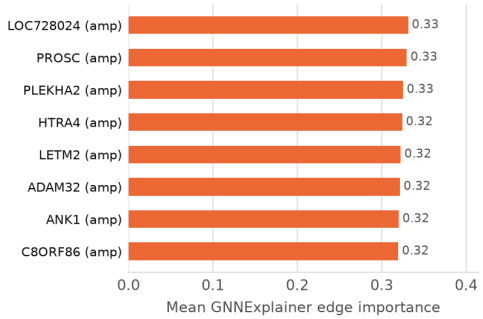
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
SF3B1 · driver	Mutation	p.K700E	4	0.31
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
PTEN · driver	Mutation	p.R130Q	22	0.31
LOC728024	High-level amplification	GISTIC ±2	0	0.33
PROSC	High-level amplification	GISTIC ±2	0	0.33
PLEKHA2	High-level amplification	GISTIC ±2	1	0.33
HTRA4	High-level amplification	GISTIC ±2	0	0.32
LETM2	High-level amplification	GISTIC ±2	0	0.32
ADAM32	High-level amplification	GISTIC ±2	0	0.32
ANK1	High-level amplification	GISTIC ±2	9	0.32
C8ORF86	High-level amplification	GISTIC ±2	0	0.32
ADAM2	High-level amplification	GISTIC ±2	1	0.32



34 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

SF3B1 → H3B-8800 (DB14017); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896) … and 47 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1PA	59	II	LumA	0.88	no progression, followed 18.9 mo
TCGA-E2-A1IE	61	II	LumA	0.88	progressed at 59.5 mo (group D)
TCGA-BH-A204	80	II	LumB	0.81	no progression, followed 83.3 mo
TCGA-WT-AB41	55	II	LumA	0.79	no progression, followed 53.0 mo
TCGA-A2-A0ES	52	II	LumA	0.77	no progression, followed 72.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**69**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

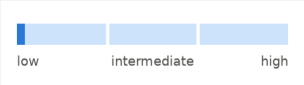
Sample: **TCGA-AR-A2LE-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **8.63** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

3rd

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 12%

Model-estimated progression time

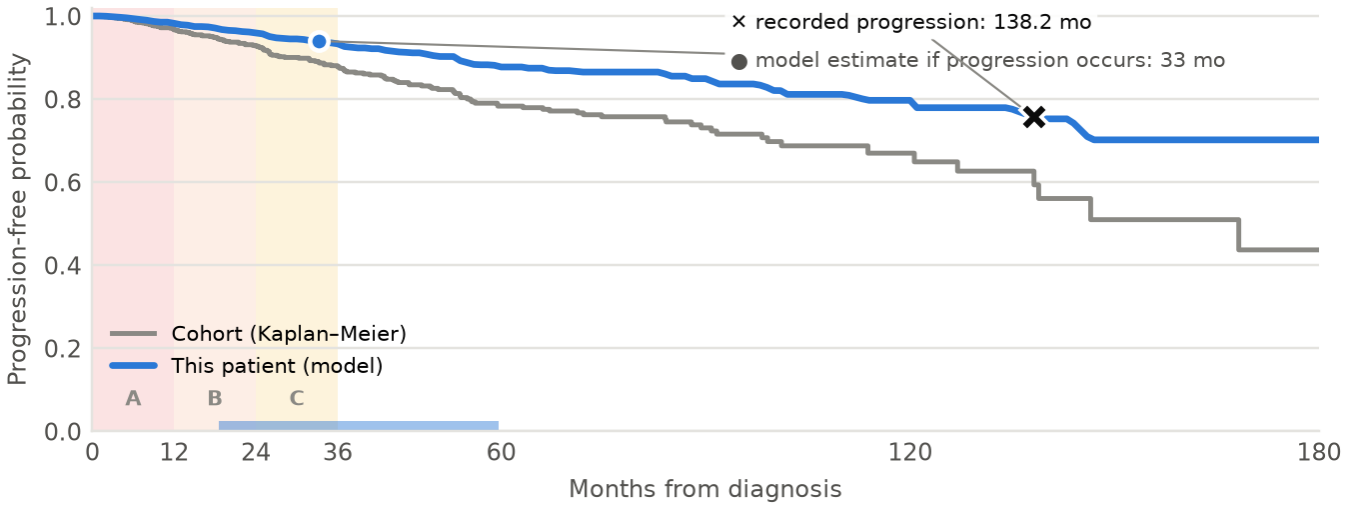
≈ 33 months

Window 24–36 months · IQR 19–60 mo
if progression occurs

Recorded progression

138.2 months

Group D · Late progression
✗ **recorded window differs** · |error| 105 mo



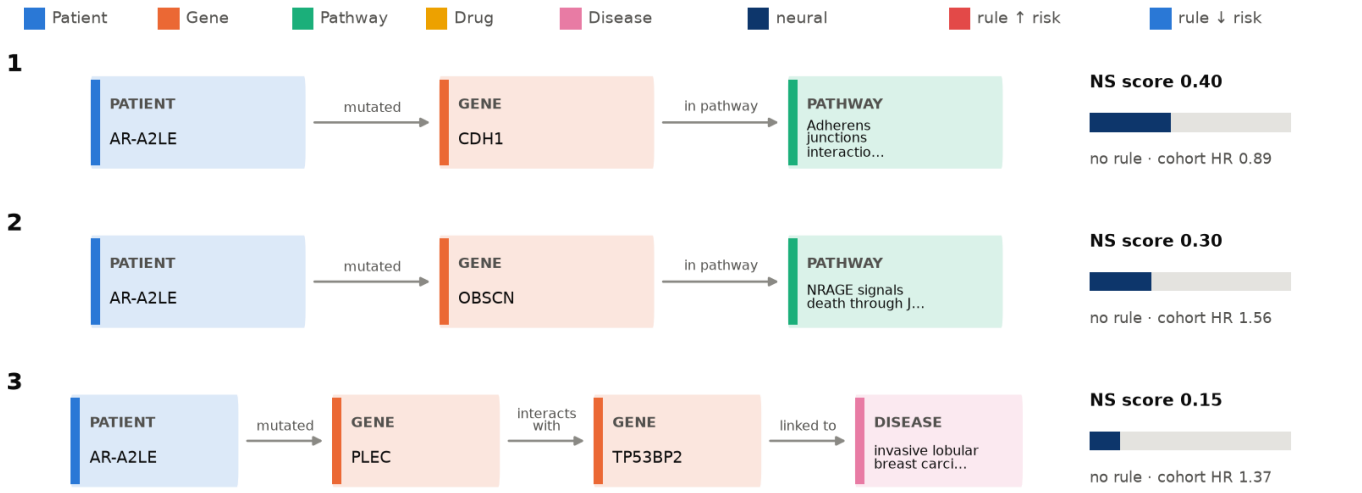
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 69-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **3rd risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.095 per SD). The highest-scoring knowledge-graph path runs from **CDH1** to **Adherens junctions interactions**. Progression was recorded at 138.2 months; the model's estimate given progression was 33 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

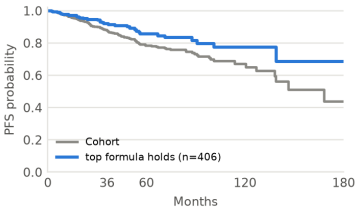


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A2LE** mutated → **CDH1** in pathway → **Adherens junctions interactions**	67	No learned formula fires on this path; its score is neural only. cohort: HR 0.89, n = 207 carriers	**0.40**
2	**TCGA-AR-A2LE** mutated → **OBSCN** in pathway → **NRAGE signals death through JNK**	51	No learned formula fires on this path; its score is neural only. cohort: HR 1.56, n = 119 carriers	**0.30**
3	**TCGA-AR-A2LE** mutated → **PLEC** interacts with → **TP53BP2** linked to → **invasive lobular breast carcinoma**	25	No learned formula fires on this path; its score is neural only. cohort: HR 1.37, n = 189 carriers	**0.15**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

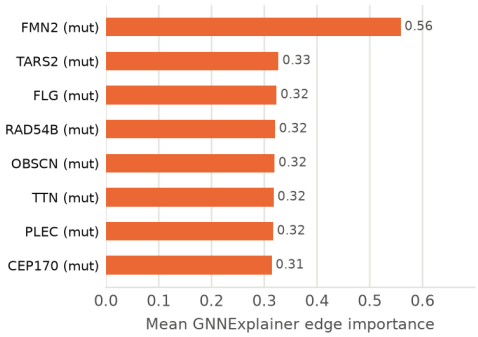
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.095 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	397 pts
-0.055 ↓ risk	Stage(p, I) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	81 pts
-0.051 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	118 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.Q255*	17	0.31
ARID1A · driver	Mutation	p.S711*	10	0.30
FMN2	Mutation	p.E134K	0	0.56
TARS2	Mutation	p.S649C	2	0.33
FLG	Mutation	p.R903T	1	0.32
RAD54B	Mutation	p.R658*	0	0.32
OBSCN	Mutation	p.Q1151H	8	0.32
TTN	Mutation	p.N26888Kfs*9, p.S13089C	2	0.32
PLEC	Mutation	p.X610_splice	3	0.32
CEP170	Mutation	p.E558K	0	0.31
GNPAT	Mutation	p.R163Q	3	0.31
MUC16	Mutation	p.Q14194H	8	0.31



14 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TARS2 → Threonine (DB00156); MUC16 → Oregovomab (DB04964); ATP6V1C1 → Tiludronic acid (DB01133); ATP6V1C1 → D-tartaric acid (DB01694); ATP6V1C1 → Thonzonium (DB09552). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1IJ	57	I	LumA	0.77	no progression, followed 28.4 mo
TCGA-A2-A0EP	56	I	LumA	0.74	no progression, followed 118.5 mo
TCGA-BH-A8FY	87	I	LumA	0.69	no progression, followed 9.7 mo
TCGA-E2-A576	69	I	LumA	0.67	no progression, followed 34.3 mo
TCGA-B6-A0IP	74	I	LumA	0.66	no progression, followed 129.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

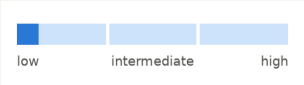
Sample: **TCGA-B6-A2IU-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.23** · MSIsensor score: **0.03** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

8th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 14%

Model-estimated progression time

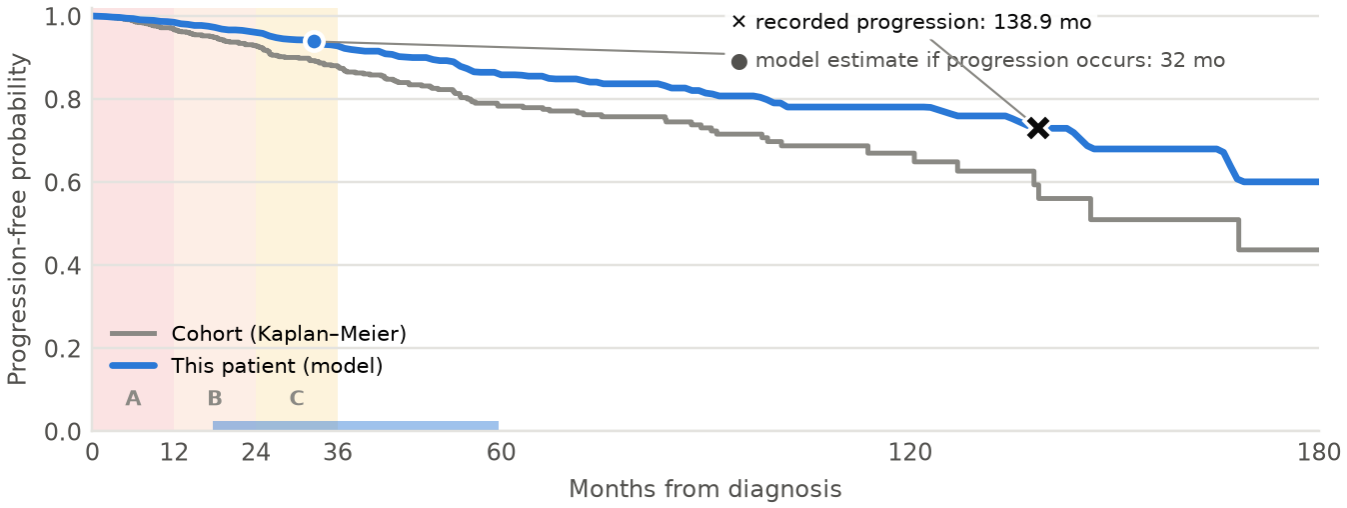
≈ 32 months

Window 24–36 months · IQR 18–60 mo
if progression occurs

Recorded progression

138.9 months

Group D · Late progression
✗ recorded window differs · |error| 106 mo



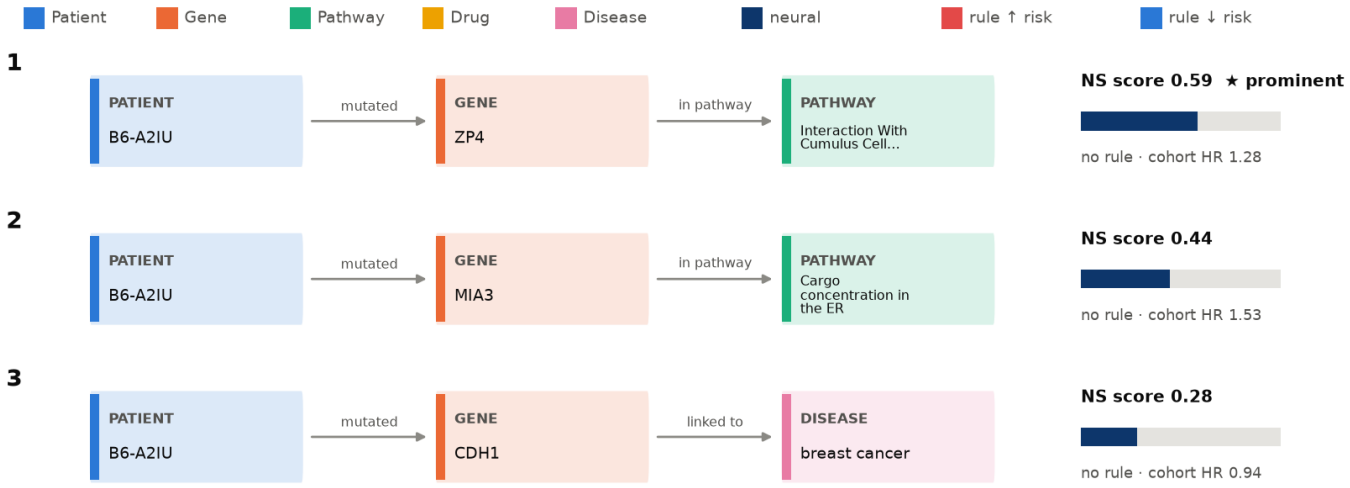
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **8th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **ZP4** to **Interaction With Cumulus Cells And The Zona Pellucida**. Progression was recorded at 138.9 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

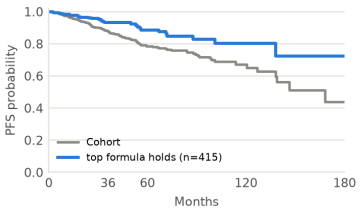


Only 1 of the 21 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A2IU** mutated → **ZP4** in pathway → **Interaction With Cumulus Cells And The Zona Pellucida**	98	No learned formula fires on this path; its score is neural only. cohort: HR 1.28, n = 183 carriers	**0.59**
2	**TCGA-B6-A2IU** mutated → **MIA3** in pathway → **Cargo concentration in the ER**	74	No learned formula fires on this path; its score is neural only. cohort: HR 1.53, n = 92 carriers	**0.44**
3	**TCGA-B6-A2IU** mutated → **CDH1** linked to → **breast cancer**	47	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 792 carriers	**0.28**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

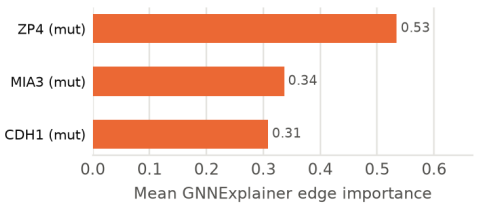
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
-0.032 ↓ risk	NStage(p, N0) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	194 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.Q195*	17	0.31
ZP4	Mutation	p.P484=	1	0.53
MIA3	Mutation	p.I387V	3	0.34



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A2LN	65	II	LumA	0.90	no progression, followed 38.2 mo
TCGA-AR-A5QM	62	II	LumA	0.89	progressed at 31.8 mo (group C)
TCGA-PE-A5DD	64	II	LumA	0.89	no progression, followed 64.2 mo
TCGA-E2-A10F	47	II	LumA	0.87	no progression, followed 28.9 mo
TCGA-A8-A091	61	II	LumA	0.86	progressed at 8.0 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

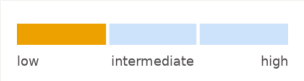
Sample: **TCGA-B6-A0IG-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.23** · MSIsensor score: **0.24** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

34th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

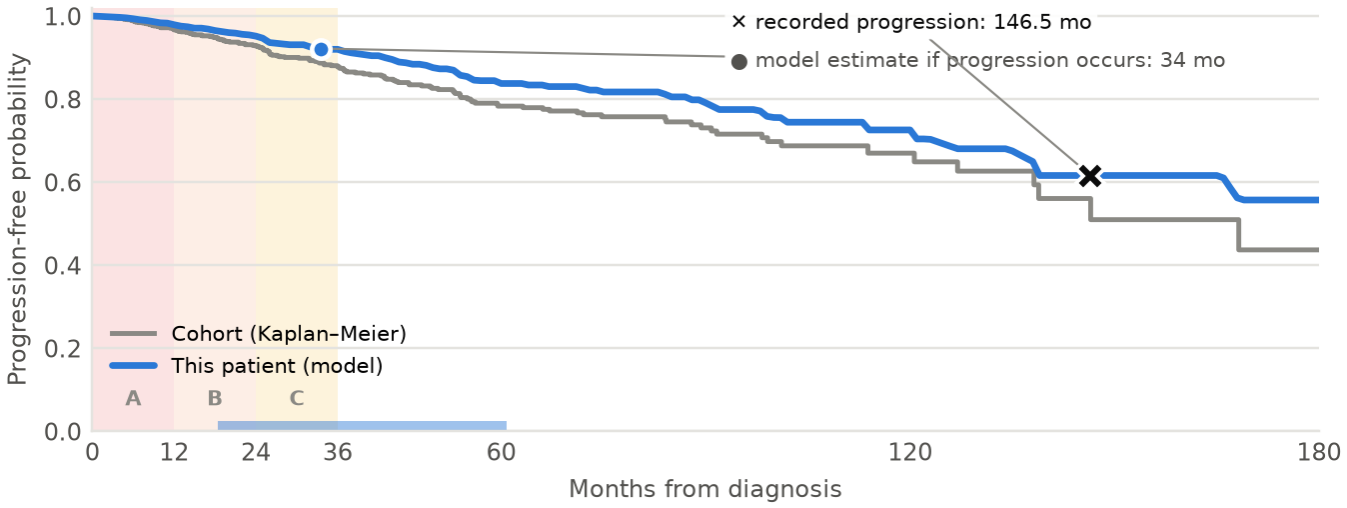
≈ 34 months

Window 24–36 months · IQR 18–61 mo
if progression occurs

Recorded progression

146.5 months

Group D · Late progression
✗ recorded window differs · |error| 113 mo



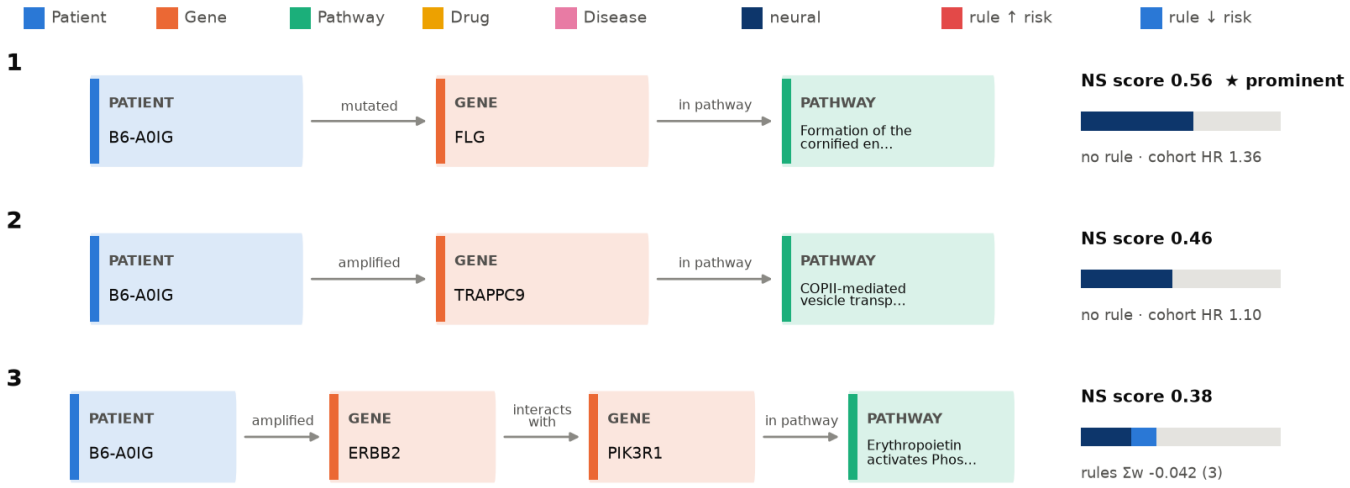
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **34th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.025 per SD). The highest-scoring knowledge-graph path runs from **FLG** to **Formation of the cornified envelope**. Progression was recorded at 146.5 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

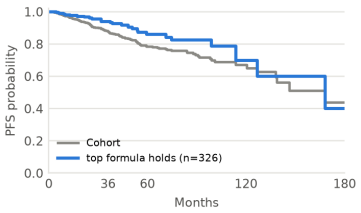


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-B6-A0IG** mutated → **FLG** in pathway → **Formation of the cornified envelope**	94	No learned formula fires on this path; its score is neural only. cohort: HR 1.36, n = 158 carriers	**0.56**
2	**TCGA-B6-A0IG** amplified → **TRAPPC9** in pathway → **COPII-mediated vesicle transport**	77	No learned formula fires on this path; its score is neural only. cohort: HR 1.10, n = 109 carriers	**0.46**
3	**TCGA-B6-A0IG** amplified → **ERBB2** interacts with → **PIK3R1** in pathway → **Erythropoietin activates Phosphoinositide-3-kinase (PI3K)**	42	↓ **slower progression** · Σw -0.042 over 3 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.76, n = 334 carriers	**0.38**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

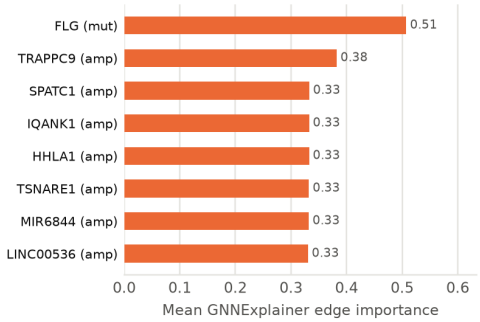
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.011 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.006 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR4') ⇒ Progress(p) [KG]	mined pair	233 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
BRCA1 · driver	High-level amplification	GISTIC ±2	29	0.32
RB1 · driver	Mutation	p.K873E	25	0.31
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
FLG	Mutation	p.R461W	1	0.51
TRAPPC9	High-level amplification	GISTIC ±2	2	0.38
SPATC1	High-level amplification	GISTIC ±2	0	0.33
IQANK1	High-level amplification	GISTIC ±2	0	0.33
HHLA1	High-level amplification	GISTIC ±2	0	0.33
TSNARE1	High-level amplification	GISTIC ±2	0	0.33
MIR6844	High-level amplification	GISTIC ±2	0	0.33



274 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 115 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A24V	52	II	LumA	0.84	no progression, followed 105.3 mo
TCGA-E2-A150	48	II	Basal	0.81	no progression, followed 63.6 mo
TCGA-C8-A12L	67	II	Her2	0.79	no progression, followed 11.9 mo
TCGA-A8-A097	65	II	LumA	0.79	no progression, followed 12.0 mo
TCGA-LL-A5YN	46	II	LumA	0.78	no progression, followed 14.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE II**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

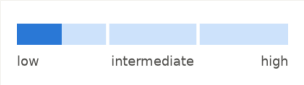
Sample: **TCGA-B6-A0WZ-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.87** · MSIsensor score: **0.04** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

17th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

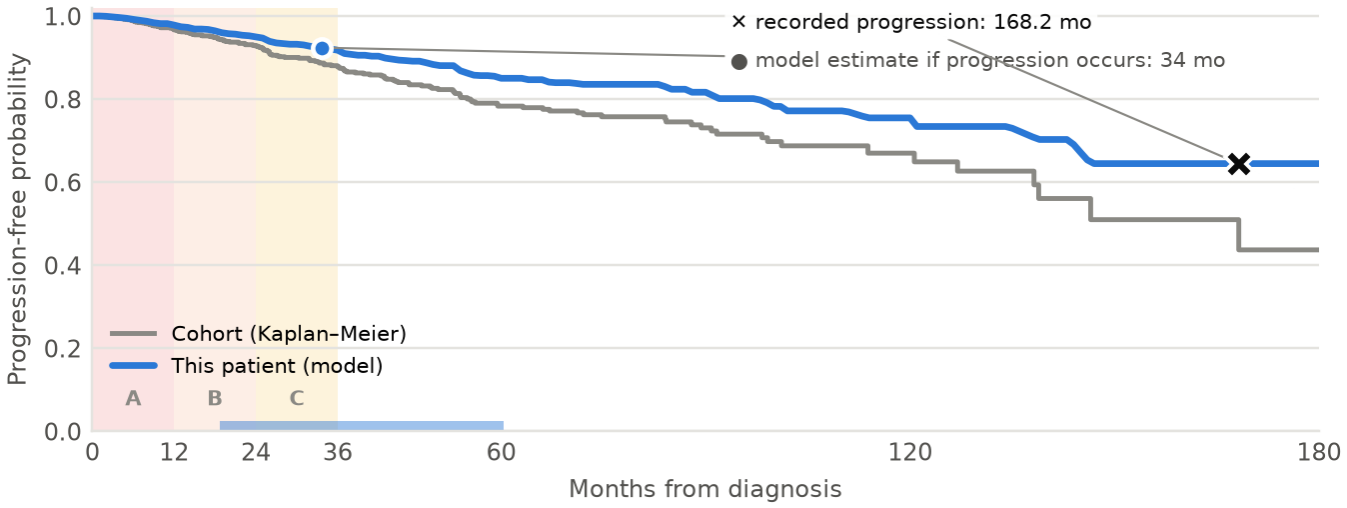
≈ 34 months

Window 24–36 months · IQR 19–60 mo
if progression occurs

Recorded progression

168.2 months

Group D · Late progression
✗ recorded window differs · |error| 135 mo



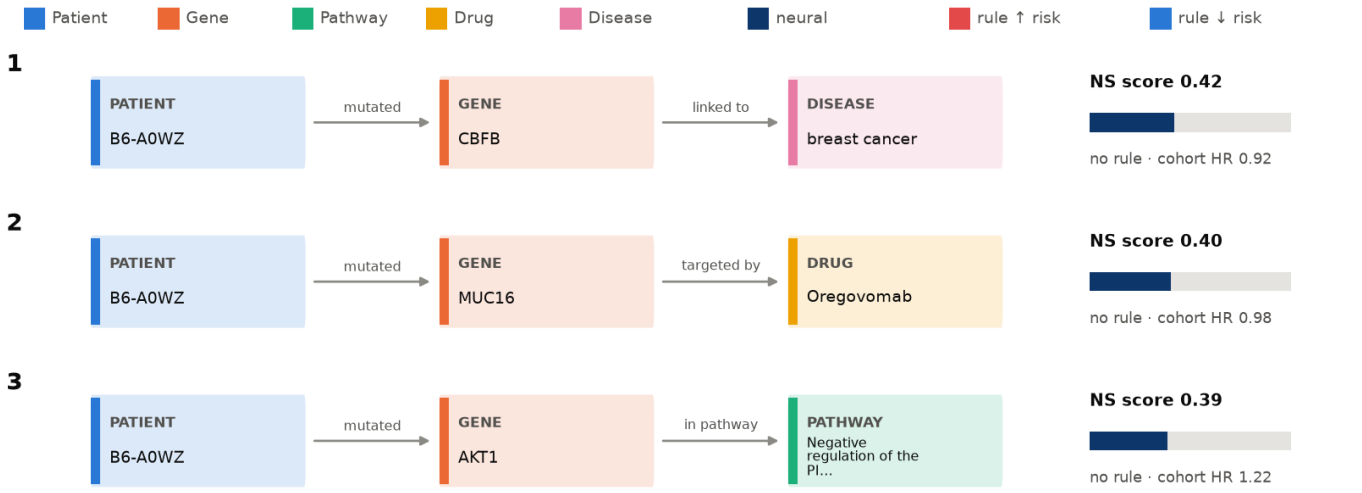
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **17th risk percentile** (low tier). The strongest learned formula that holds for them (*Subtype(p, LumA) ⇒ Progress(p)*) lowers the progression hazard (weight -0.010 per SD). The highest-scoring knowledge-graph path runs from **CBFB** to **breast cancer**. Progression was recorded at 168.2 months; the model's estimate given progression was 34 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

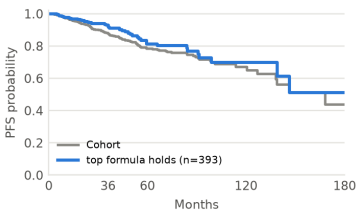


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0WZ** mutated → **CBFB** linked to → **breast cancer**	70	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 793 carriers	**0.42**
2	**TCGA-B6-A0WZ** mutated → **MUC16** targeted by → **Oregovomab**	67	No learned formula fires on this path; its score is neural only. cohort: HR 0.98, n = 122 carriers	**0.40**
3	**TCGA-B6-A0WZ** mutated → **AKT1** in pathway → **Negative regulation of the PI3K/AKT network**	65	No learned formula fires on this path; its score is neural only. cohort: HR 1.22, n = 193 carriers	**0.39**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

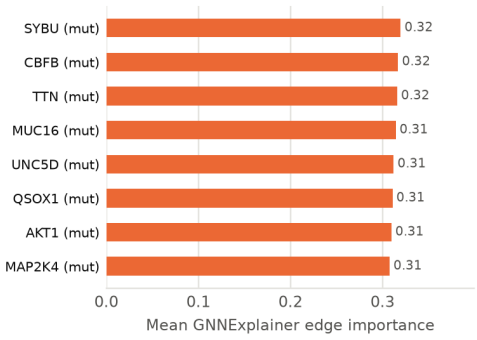
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CBFB · driver	Mutation	p.V86=	29	0.32
AKT1 · driver	Mutation	p.E17K	54	0.31
MAP2K4 · driver	Mutation	p.S184L	12	0.31
SYBU	Mutation	p.E444K	0	0.32
TTN	Mutation	p.L19843=	2	0.32
MUC16	Mutation	p.I8478=	8	0.31
UNC5D	Mutation	p.E736K	1	0.31
QSOX1	Mutation	p.L190=	4	0.31



8 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

AKT1 → ATP (DB00171); AKT1 → Arsenic trioxide (DB01169); AKT1 → Genistein (DB01645); AKT1 → Inositol 1,3,4,5-Tetrakisphosphate (DB01863); AKT1 → Resveratrol (DB02709); AKT1 → Archexin (DB05971); AKT1 → Enzastaurin (DB06486); AKT1 → Perifosine (DB06641); AKT1 → N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine (DB07584); AKT1 → 5-(5-chloro-7H-pyrrolo[2,3-d]pyrimidin-4-yl)-4,5,6,7-tetrahydro-1H-imidazo[4,5-c]pyridine (DB07585) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1L9	40	II	LumA	0.77	no progression, followed 19.7 mo
TCGA-E9-A22H	42	II	LumB	0.76	no progression, followed 40.5 mo
TCGA-BH-A0W4	46	II	LumA	0.76	no progression, followed 25.0 mo
TCGA-E2-A15I	44	II	LumA	0.76	no progression, followed 55.6 mo
TCGA-D8-A141	40	missing	LumA	0.75	no progression, followed 20.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.